DATA VISUALIZATION (CONT.)

Reading:
Chapter 6 from Skiena.

PRINCIPLES OF DATA VISUALIZATION

‘*Graphical excellence begins with telling the truth about the data*’ E. Tufte

Slides will make heavy use of ‘The Visual Display of Quantitative Information’ by Edward Tufte.

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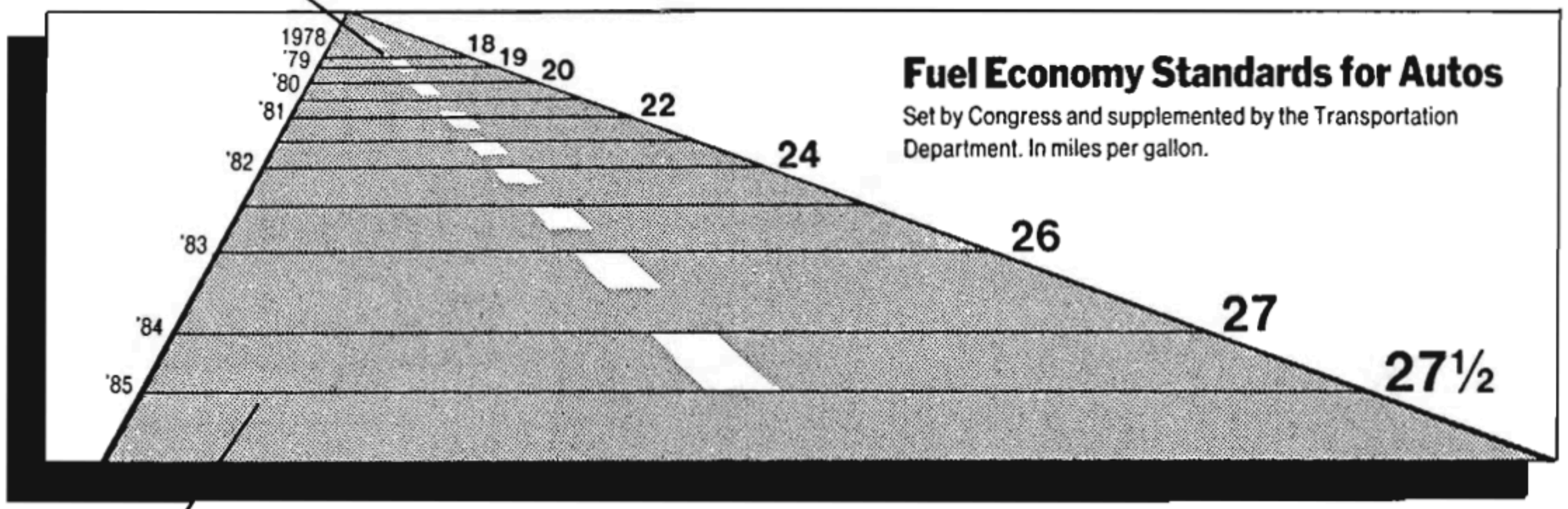
Lie Factor = size of effect shown in graphic / size of effect in data

PRINCIPLES OF DATA VISUALIZATION

Graphical Integrity: Design and Data Variation

Is this a good visualization?

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

E. Tufte

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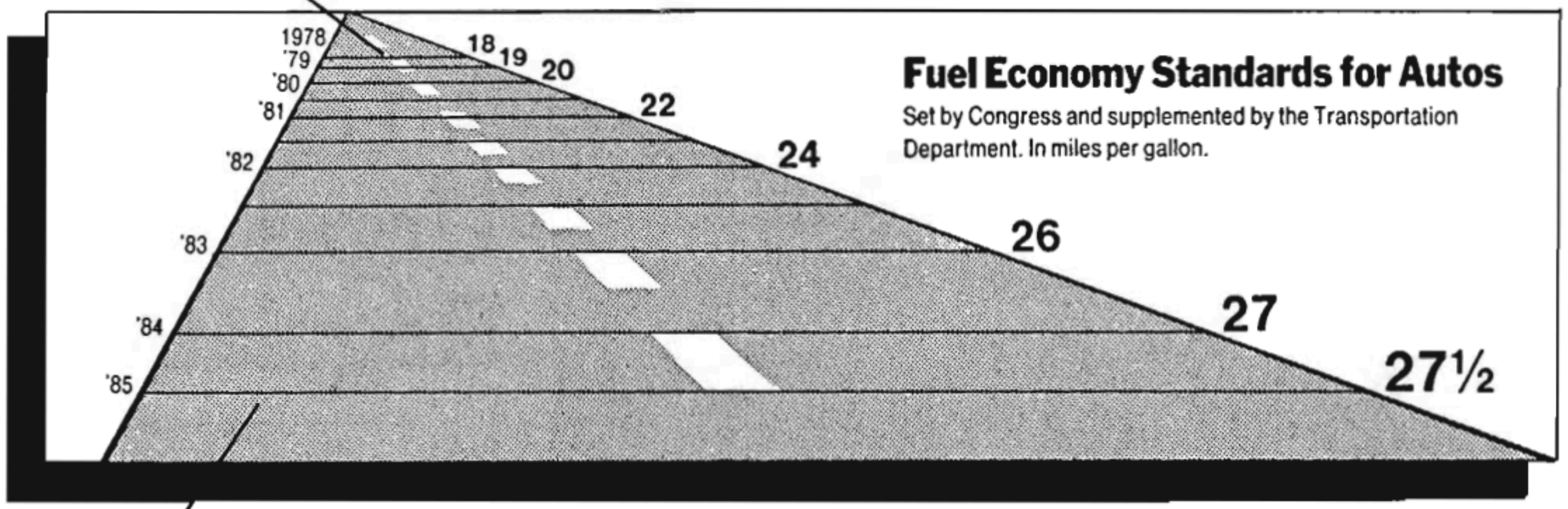

This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

E. Tufte

- Which direction is the future?
- Perspective on left numbers
- Road width: perspective or change in values

Biggest mistake: Lie factor

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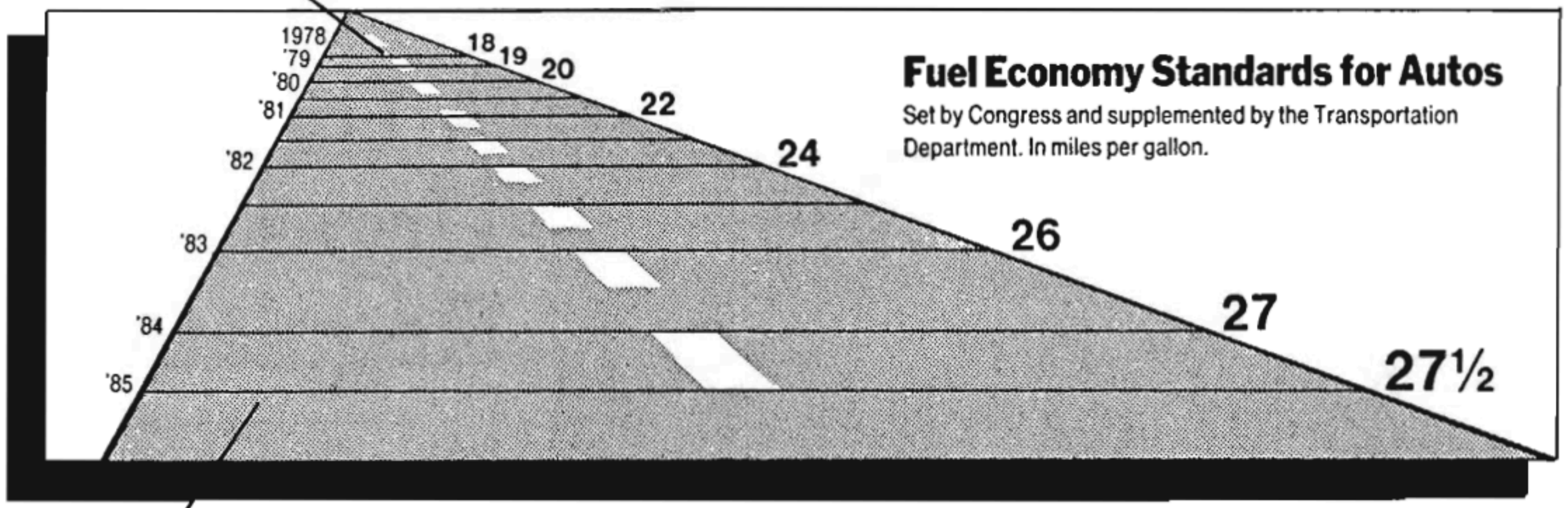

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E. Tufte

Actual data:

$$\frac{27.5 - 18.0}{18.0} = 53\,\%$$

In graphic:

$$\frac{5.3 - 0.6}{0.6} = 783\,\%$$

Lie factor $= 783/53 = 14.8$

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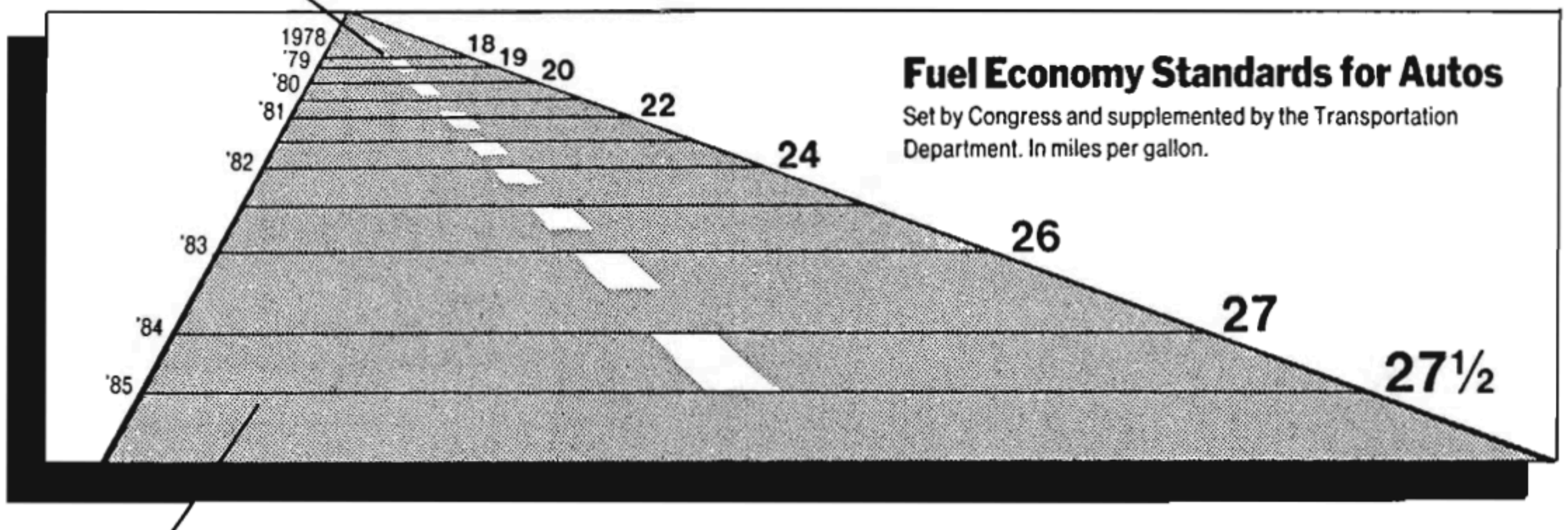

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Problem with the lie factor:

Trying to capture both design variation and data variation.

Need to show data variation not design variation!

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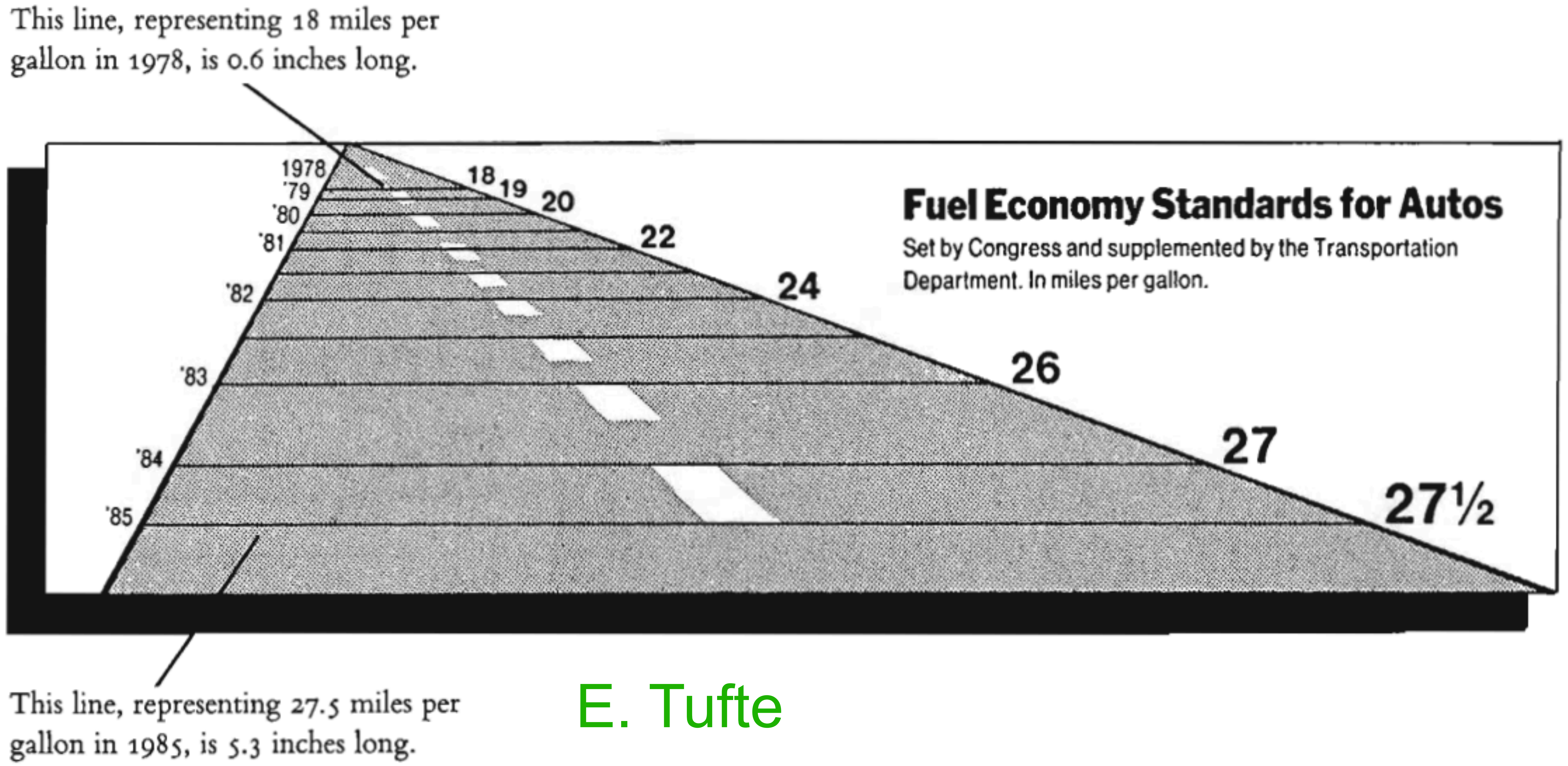
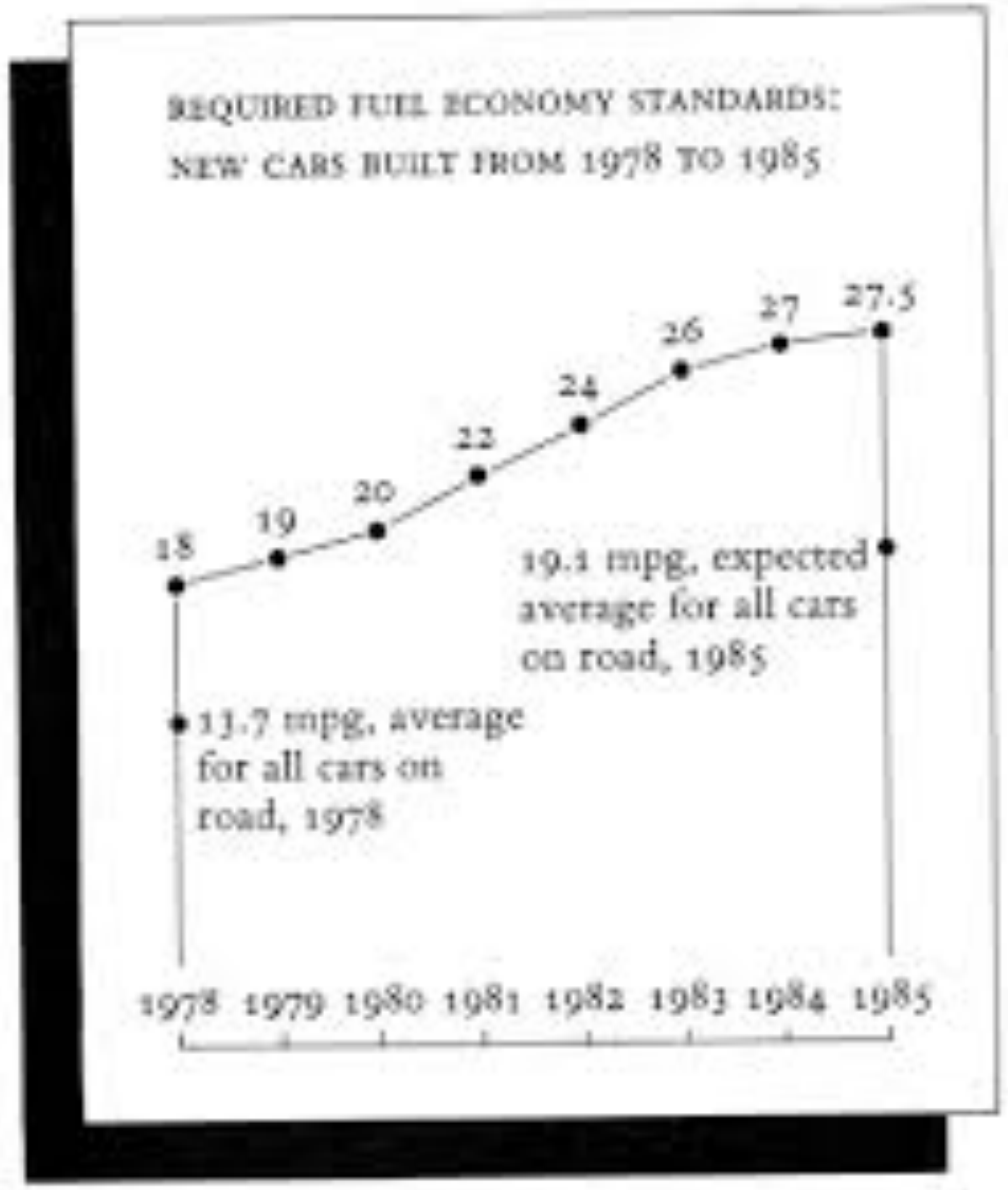


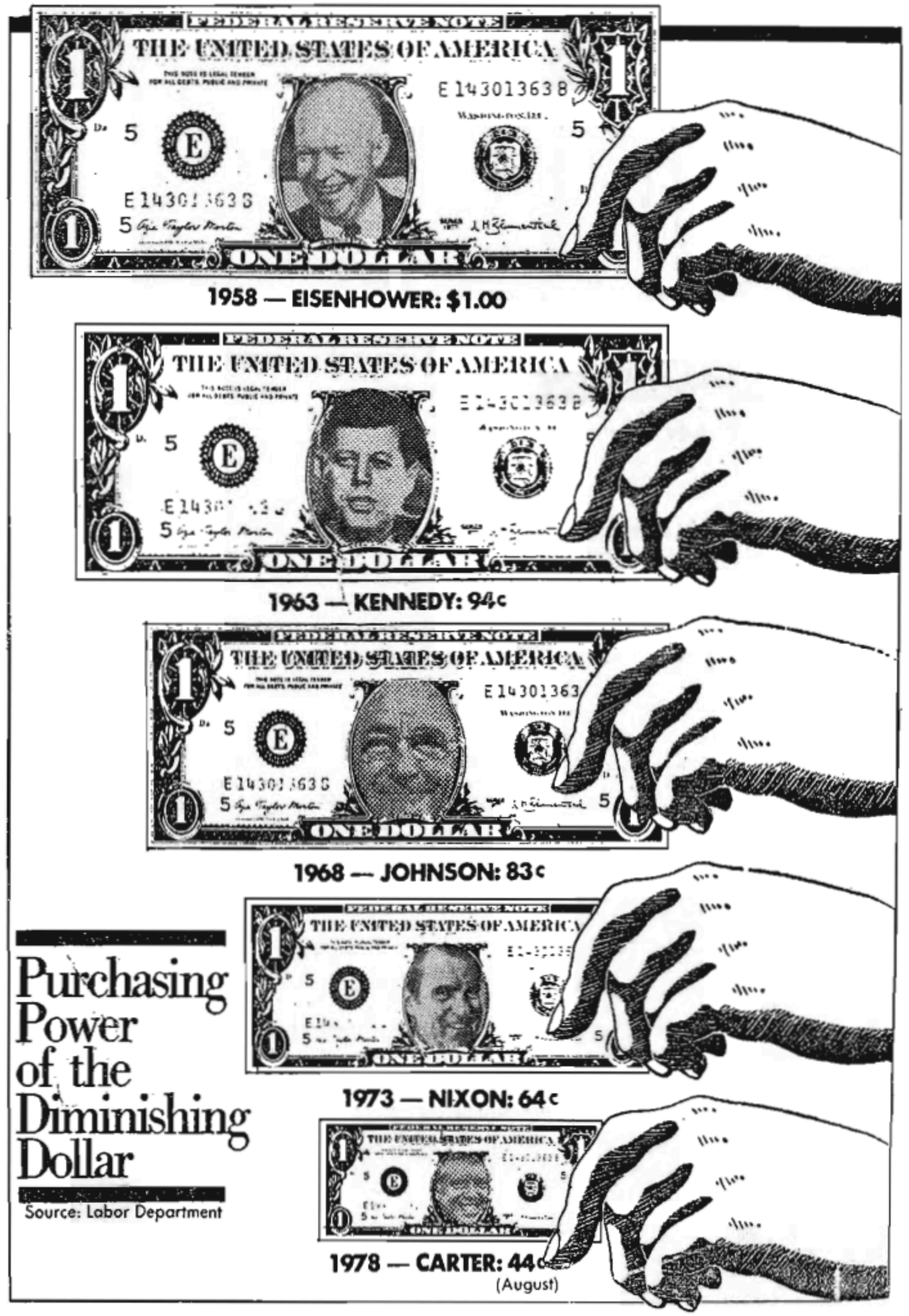
E. Tufte

Visualization matching data



PRINCIPLES OF DATA VISUALIZATION

Graphical Integrity: Visual Area and Numerical Measure



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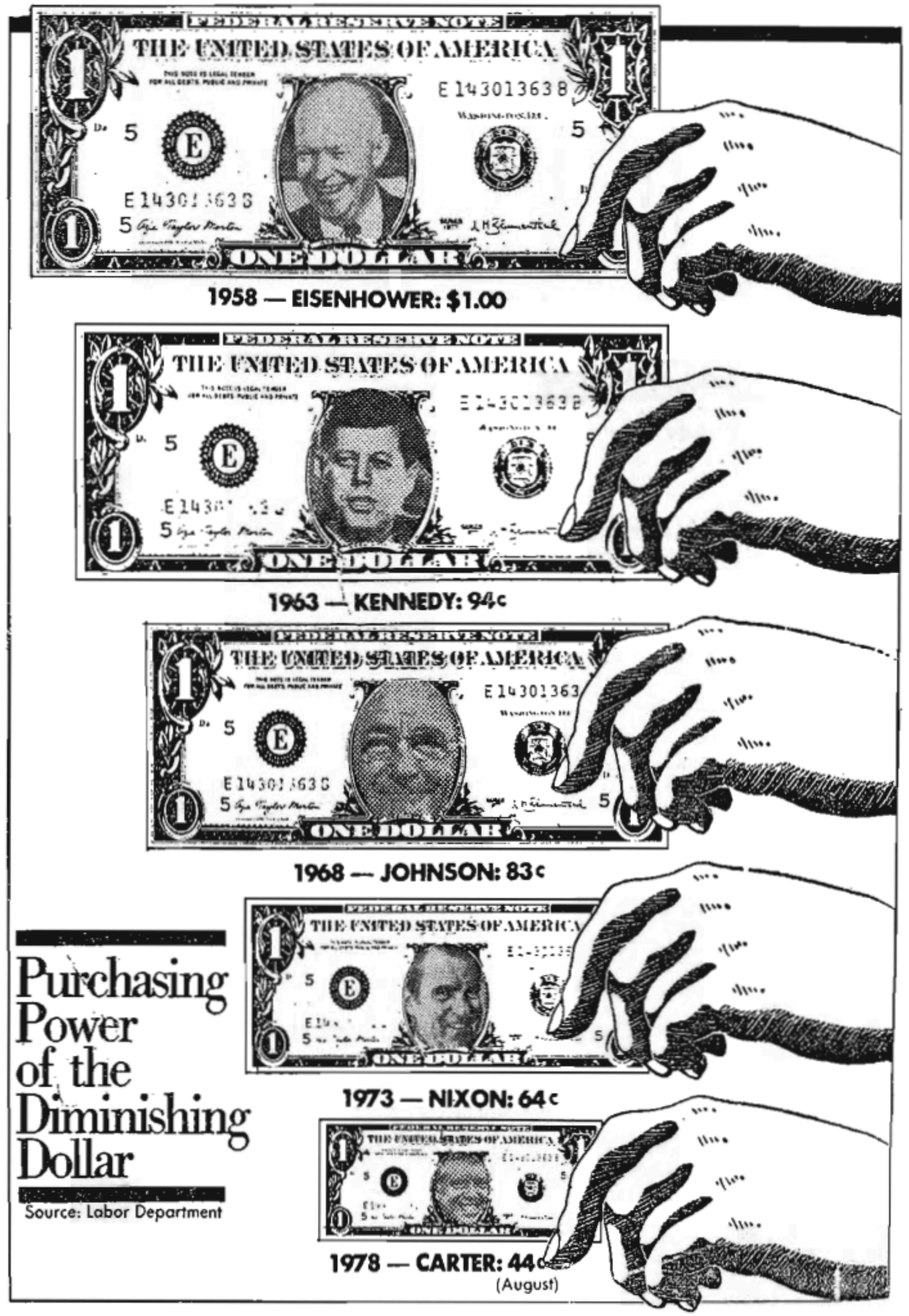

Not a good idea to use area/volume to show one-dimensional data.

Bottom dollar’s area vs top dollar’s area?

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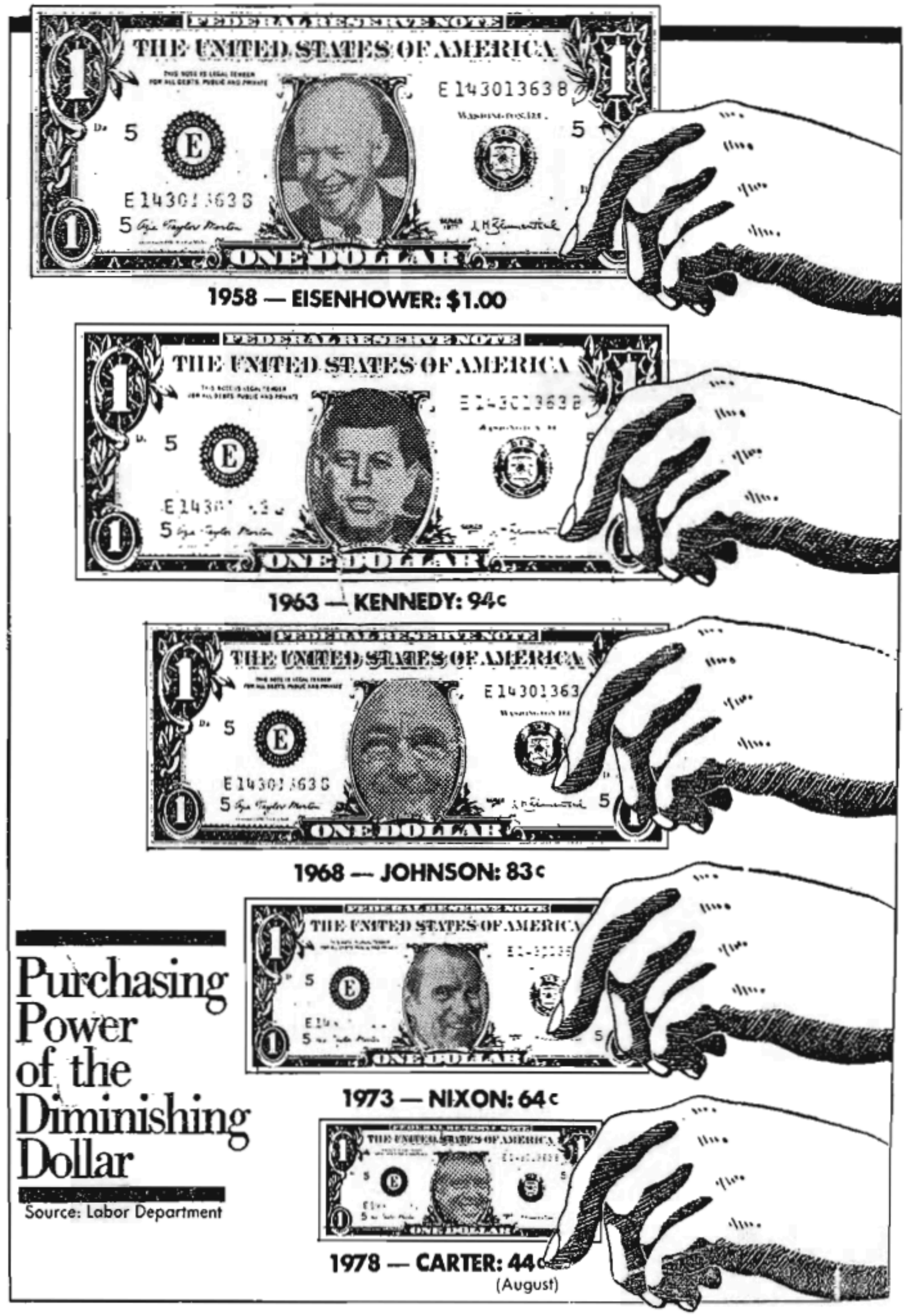

Not a good idea to use area/volume to show one-dimensional data.

Bottom dollar’s area vs top dollar’s area?

Bottom dollar’s value 44, top dollar’s value 100

Bottom dollar had to be twice in size.

PRINCIPLES OF DATA VISUALIZATION

Graphical Integrity: Offset Distortion

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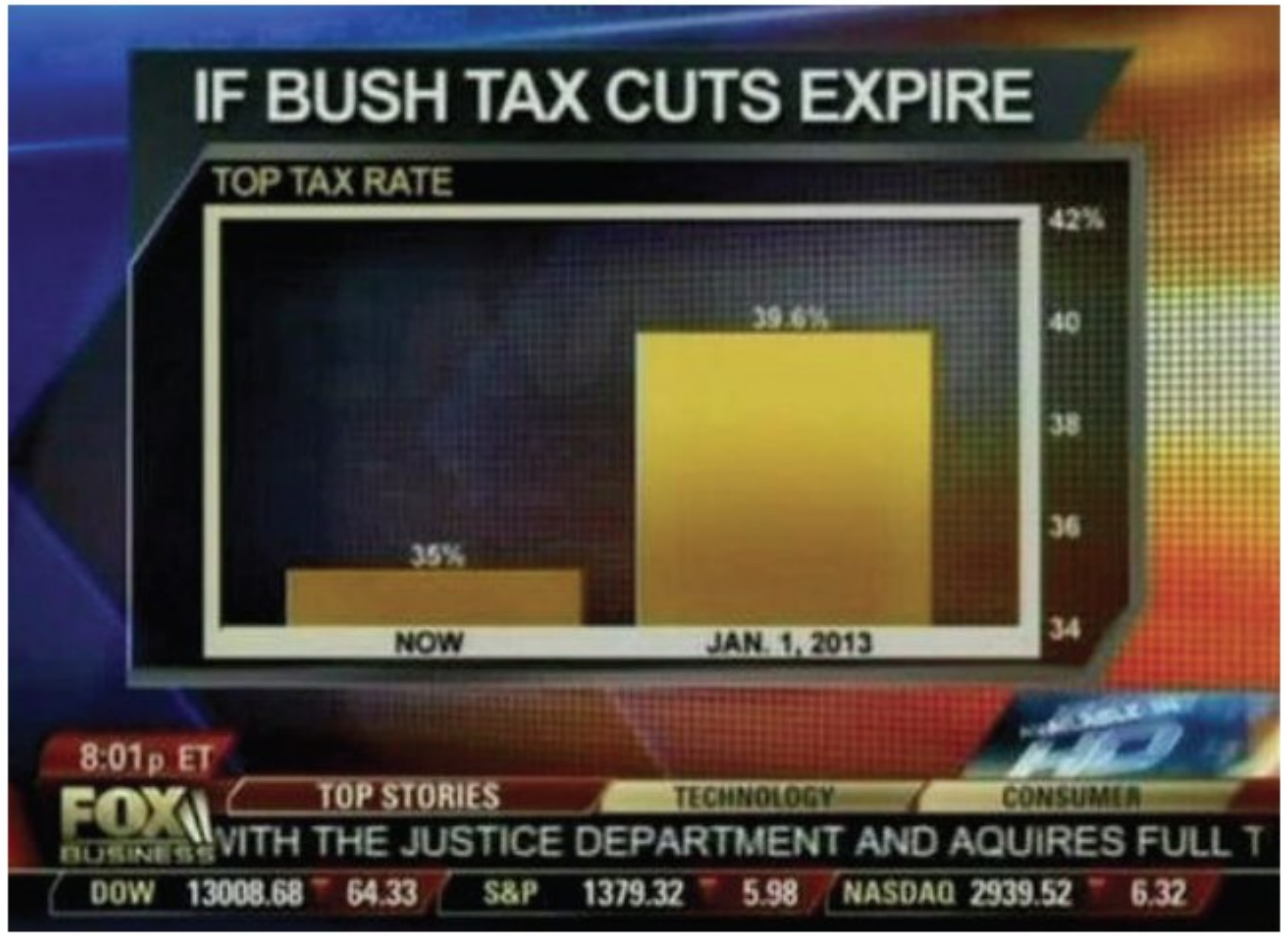

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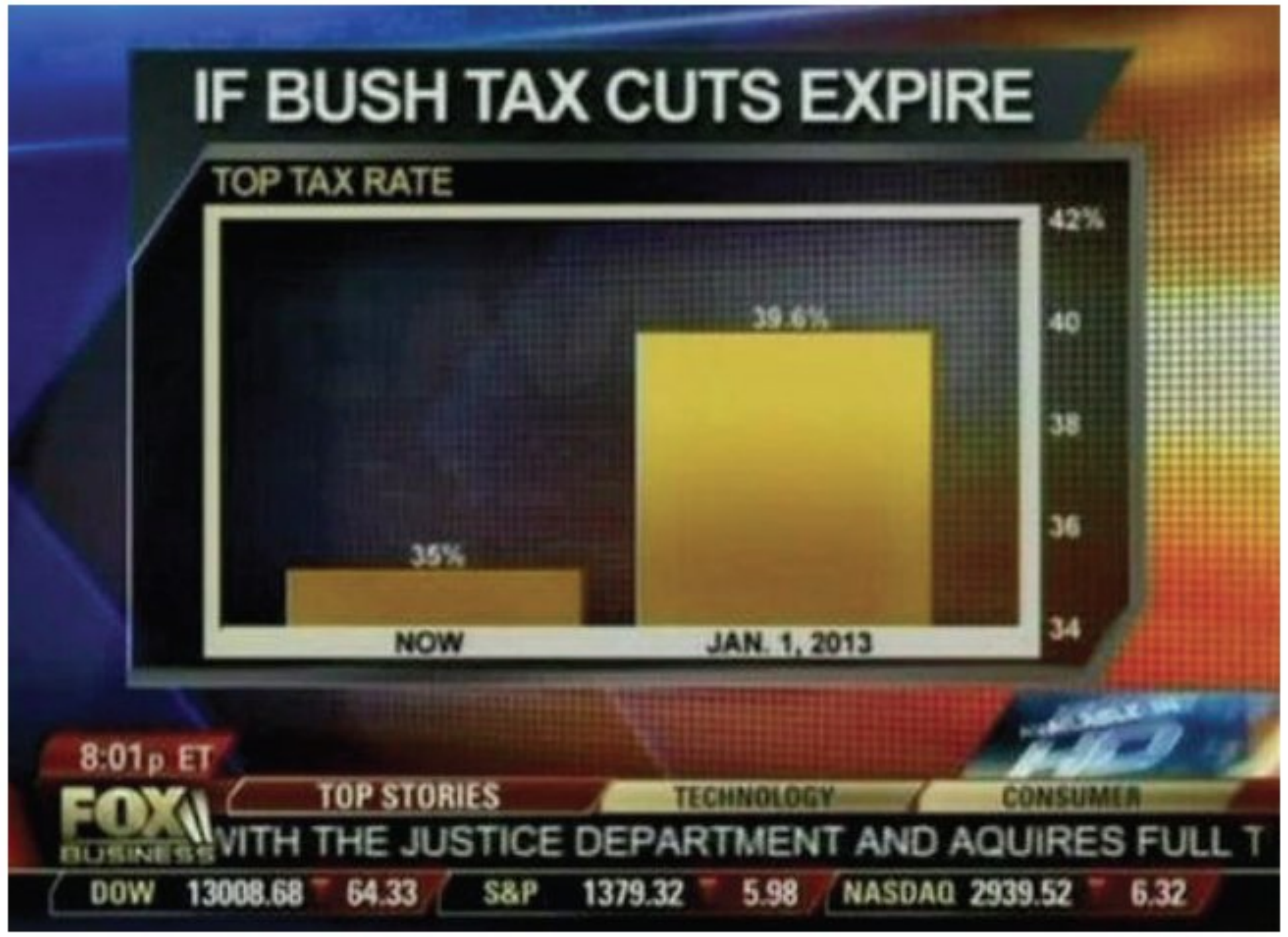

What a difference!

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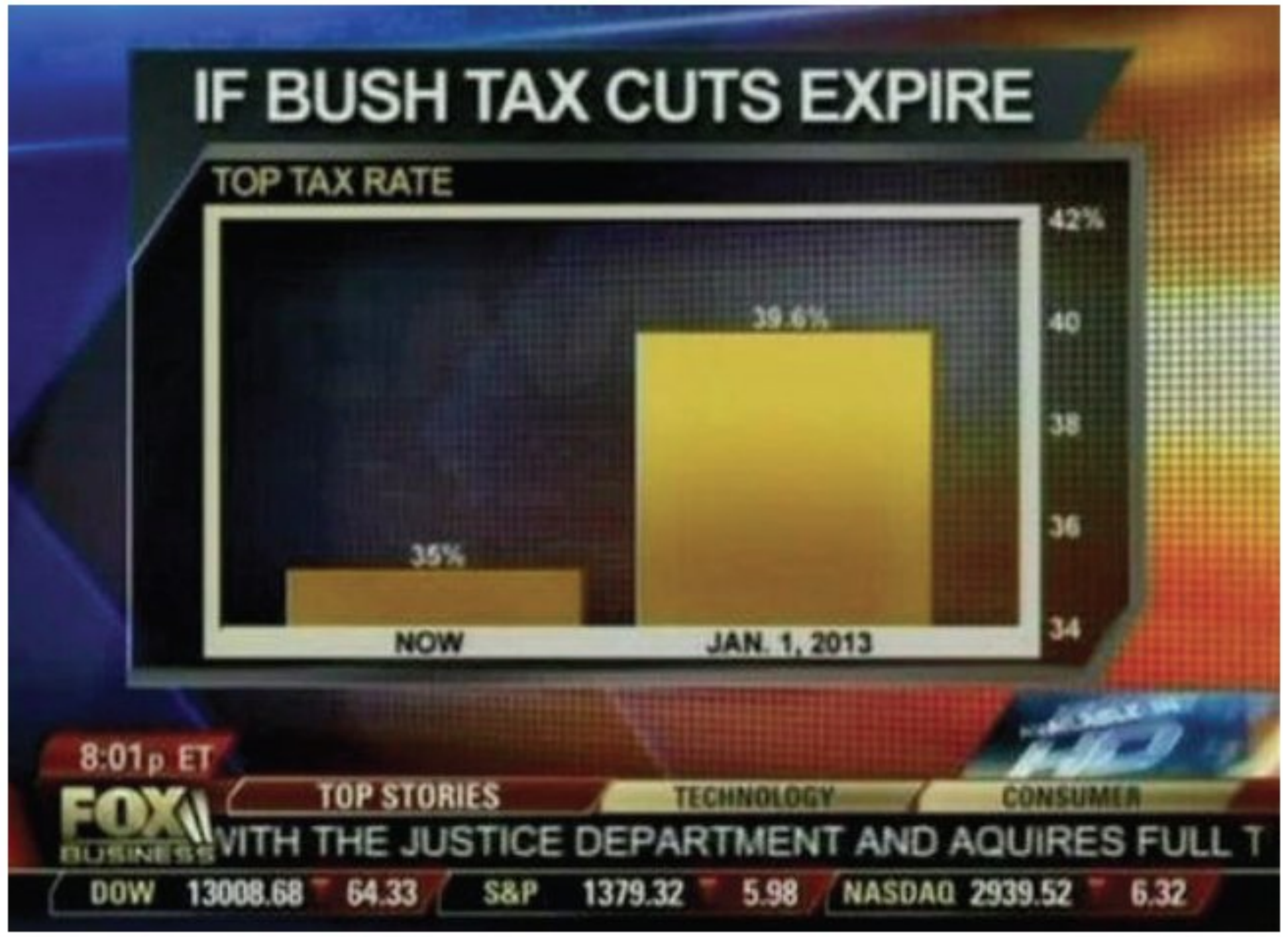
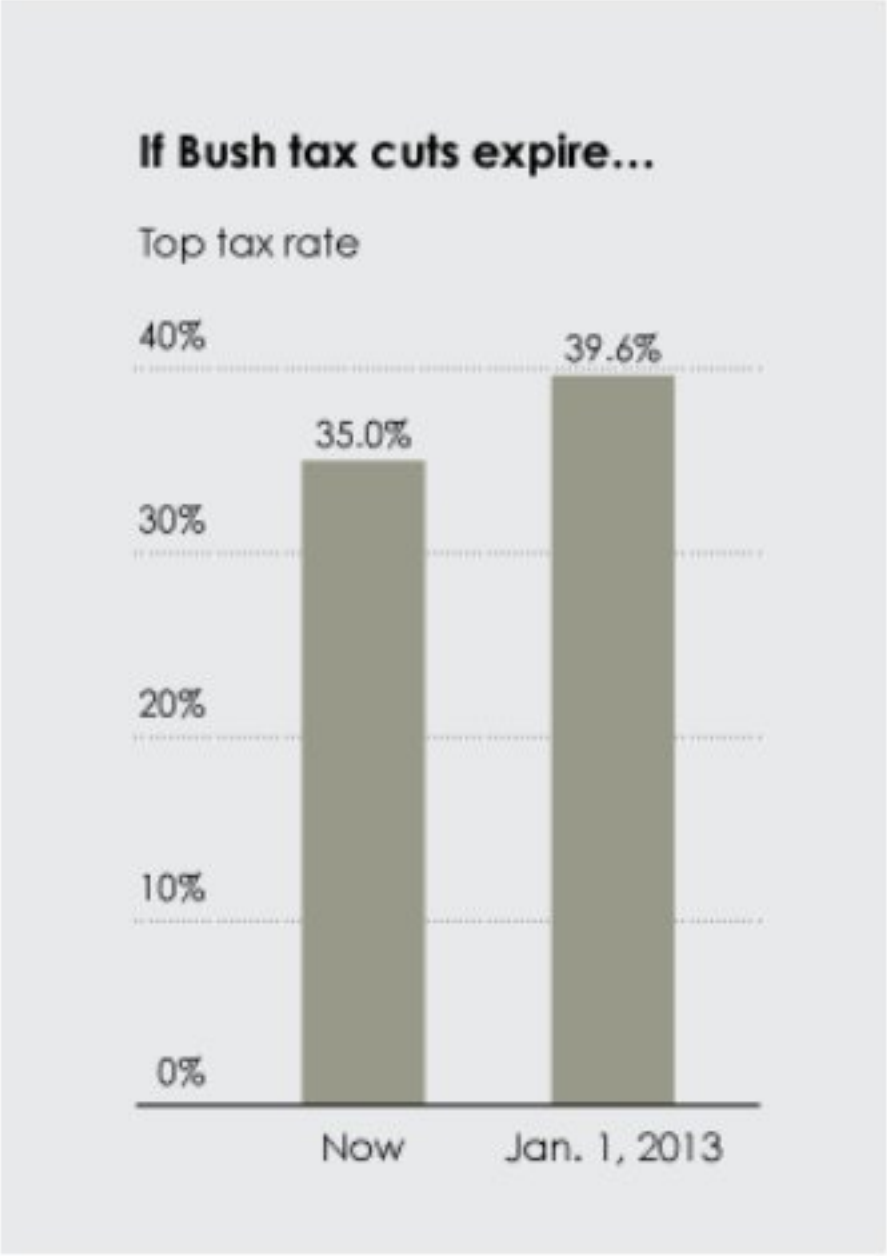




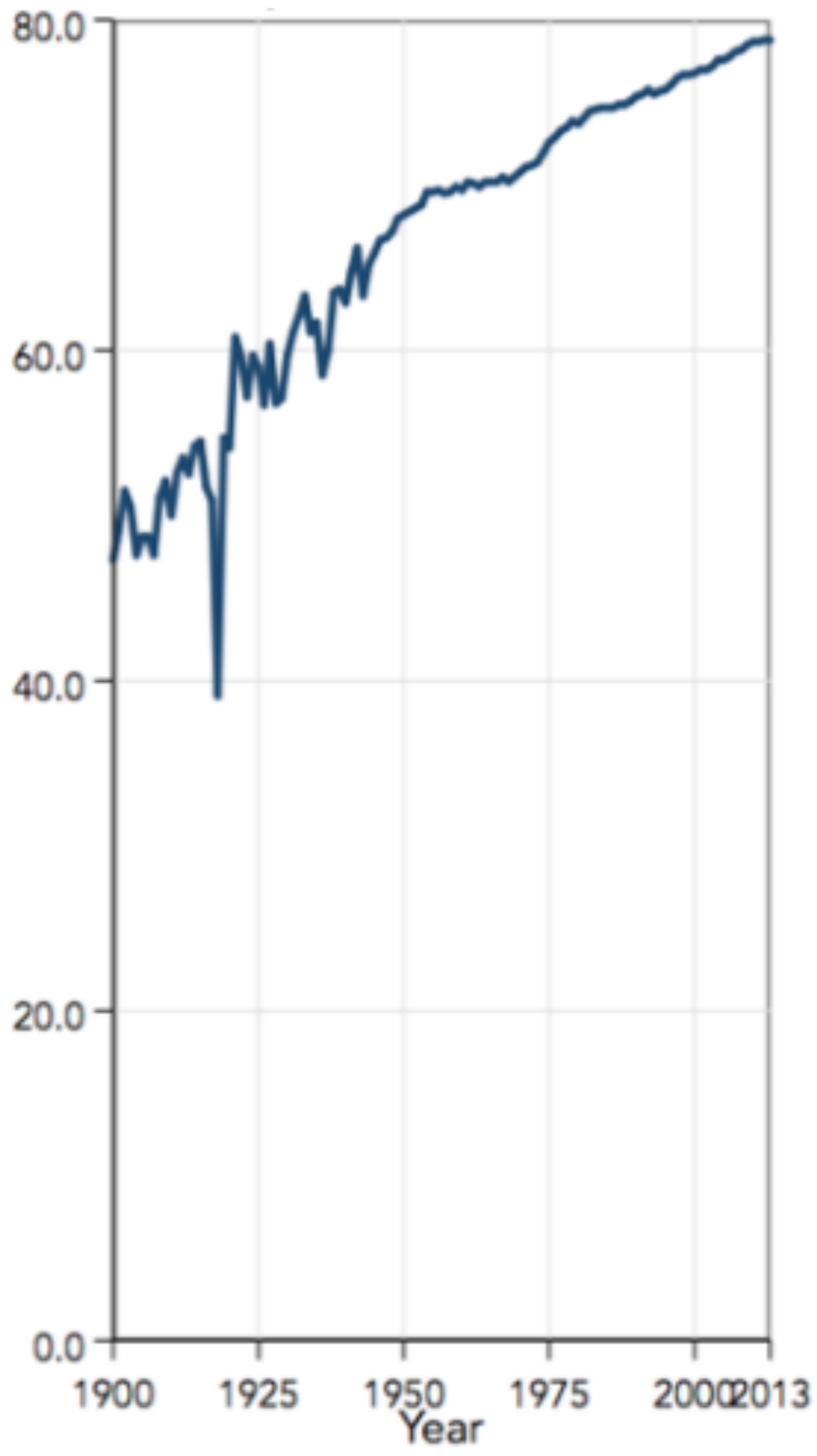
What a difference!

Actual difference not so spectacular!

Start bar graphs at 0.

PRINCIPLES OF DATA VISUALIZATION

Graphical Integrity: Scale Distortion



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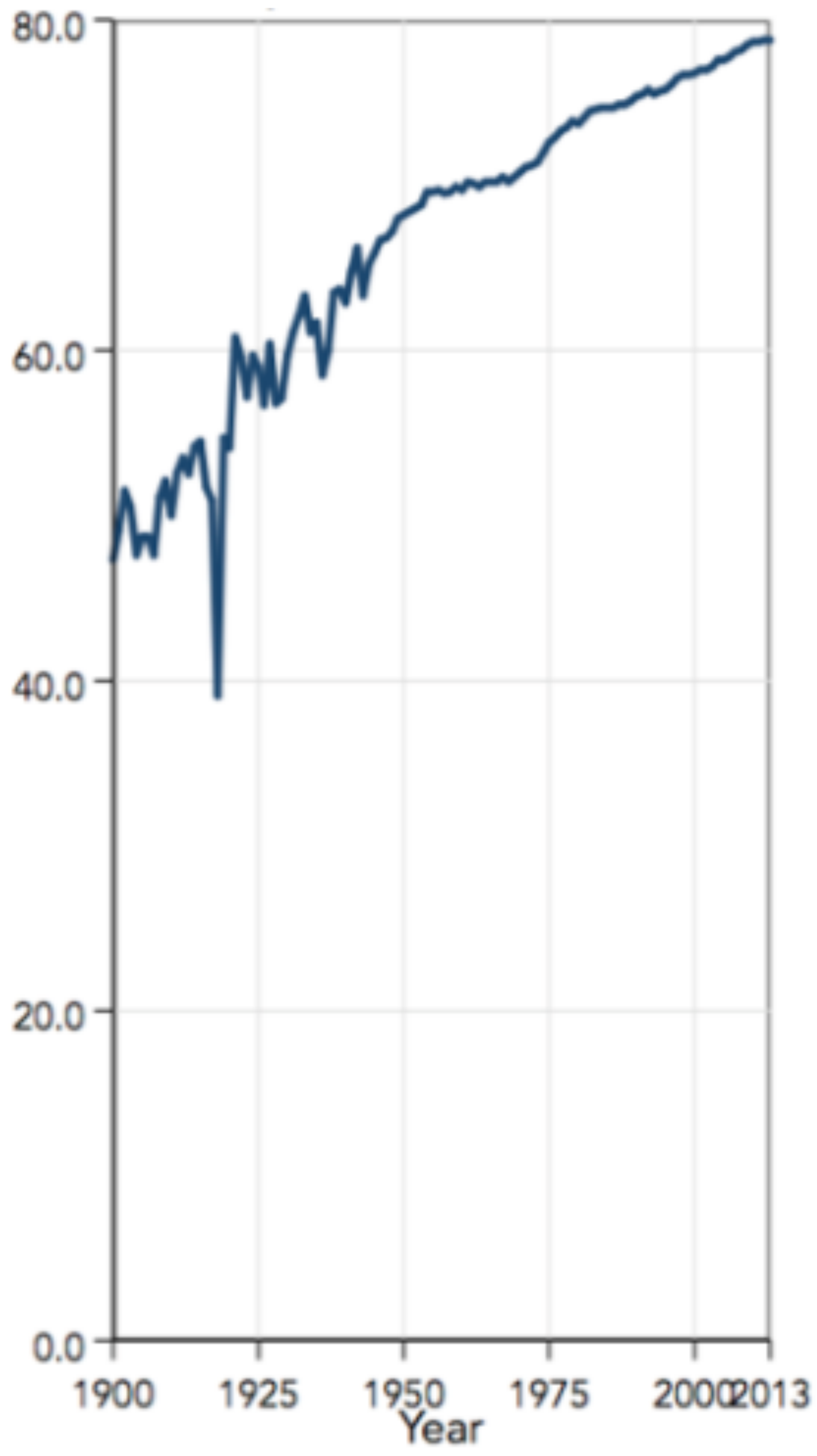

Something is steeply increasing.
How serious is the problem?

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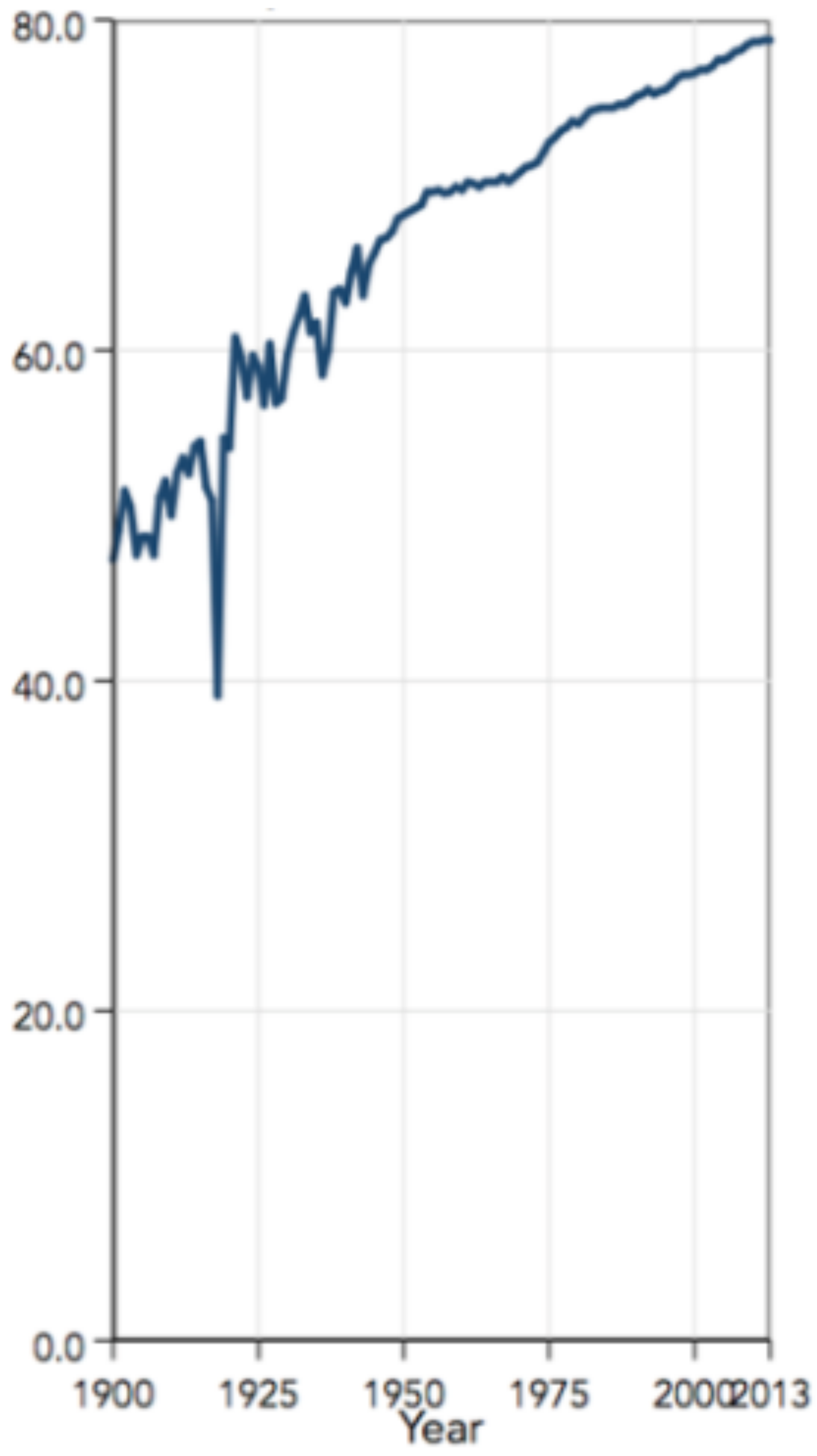

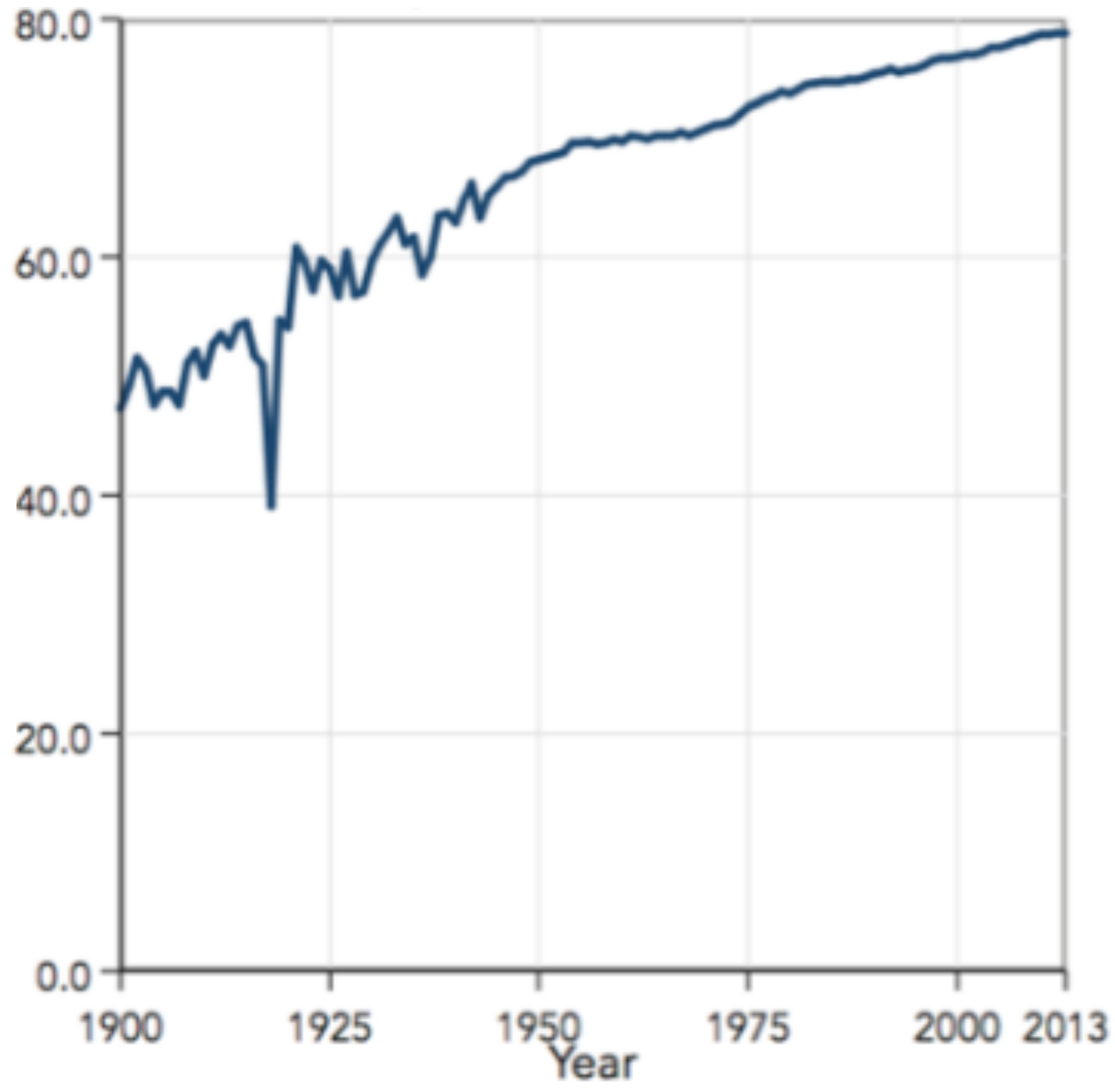

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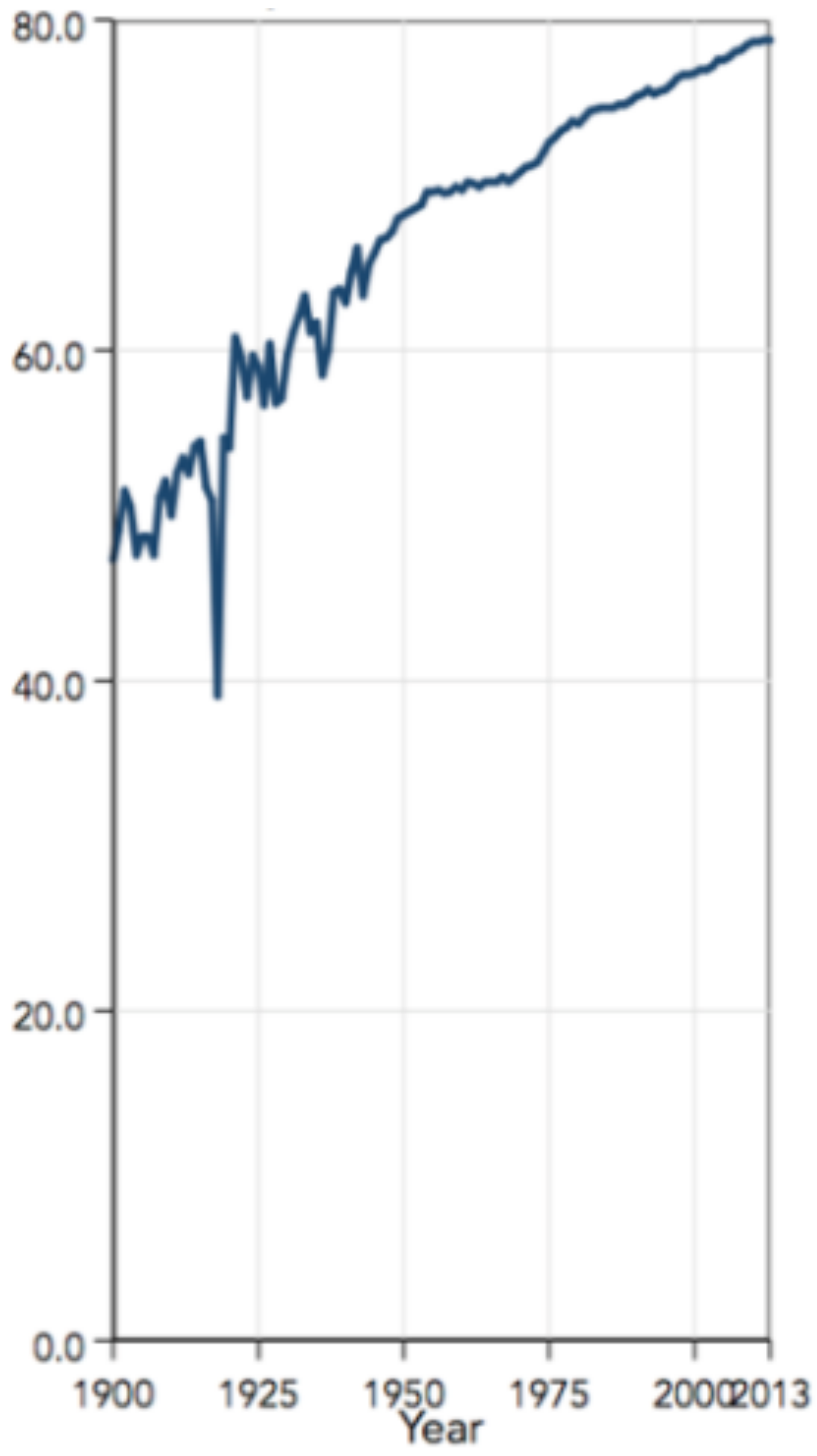

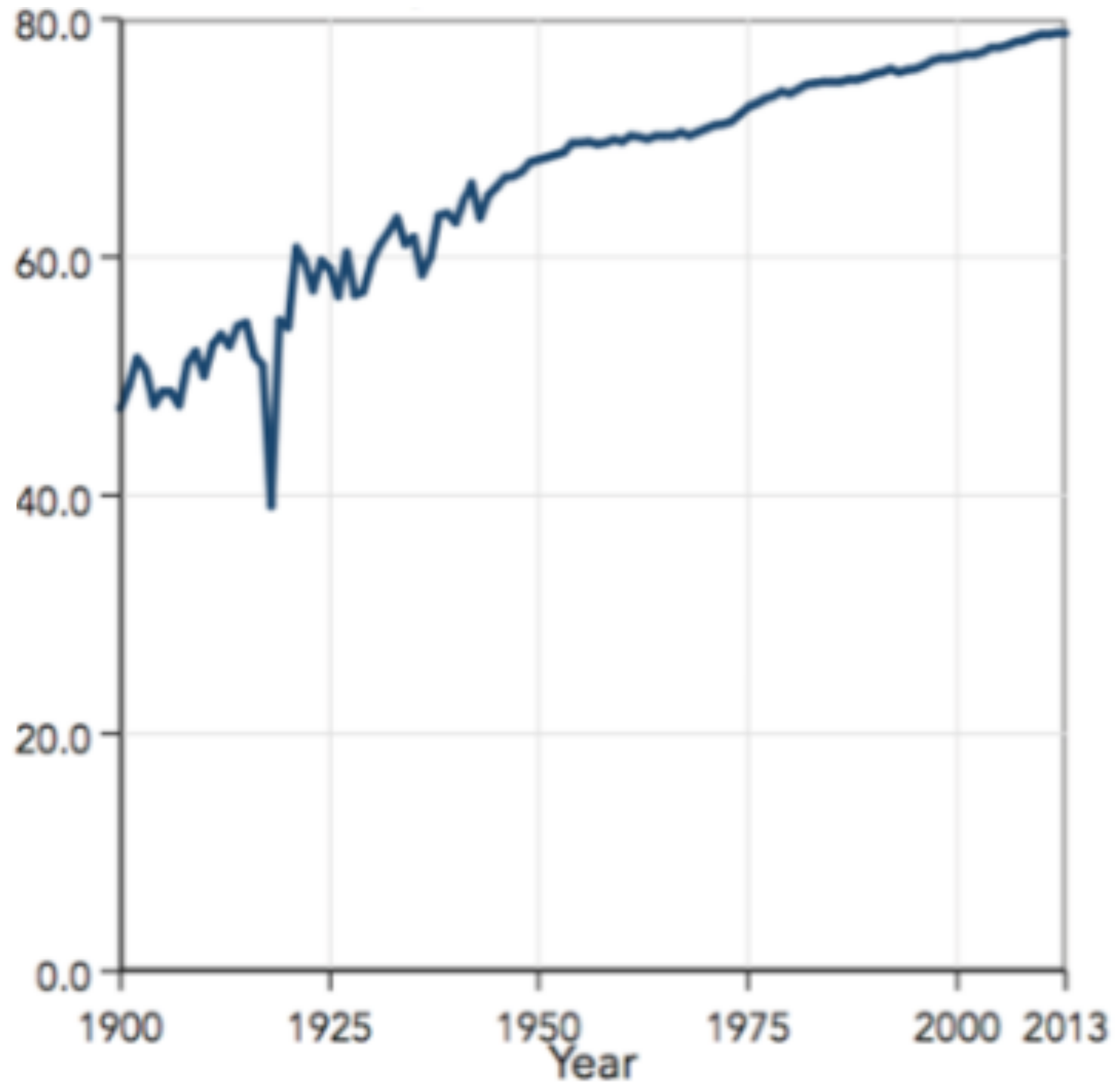

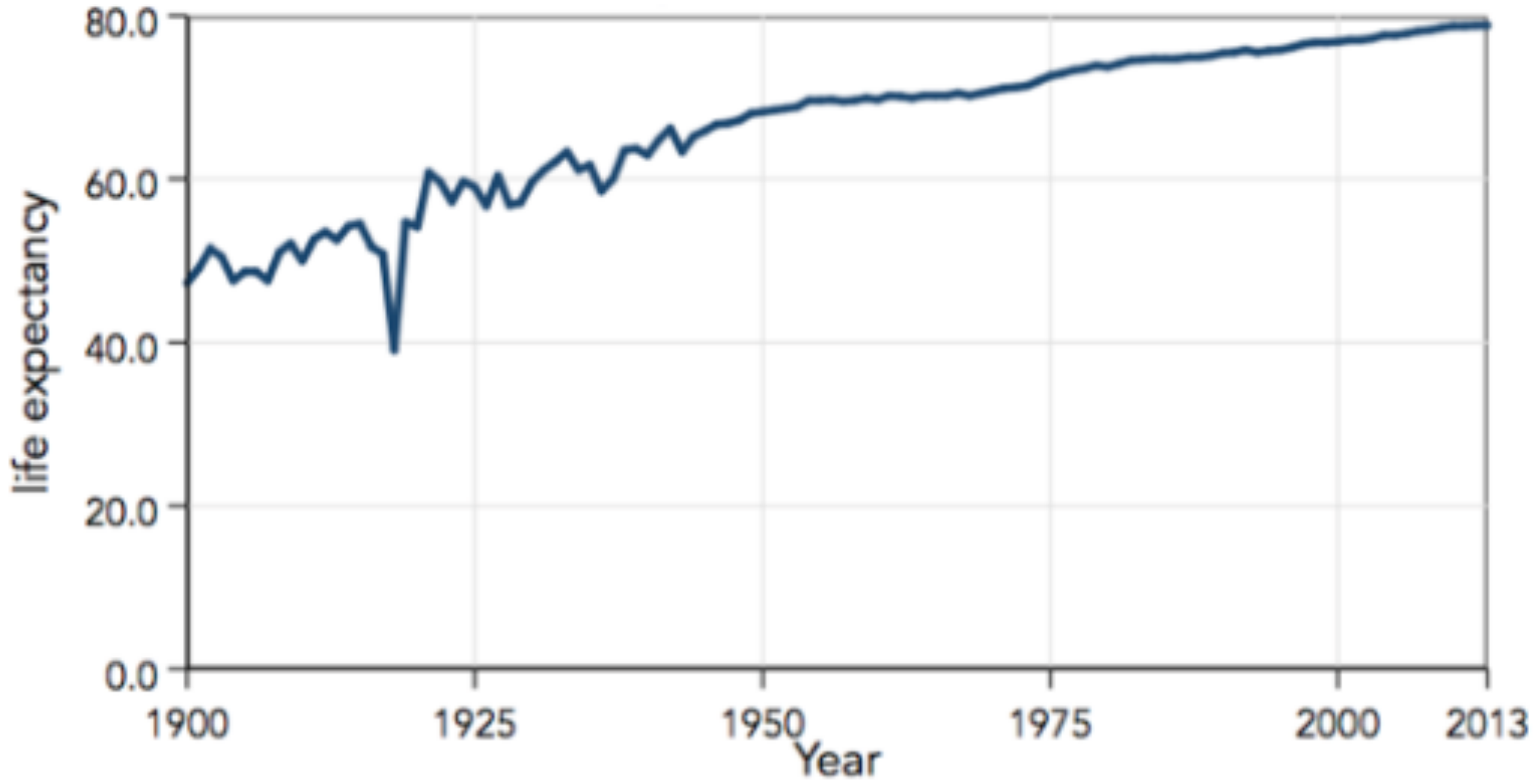

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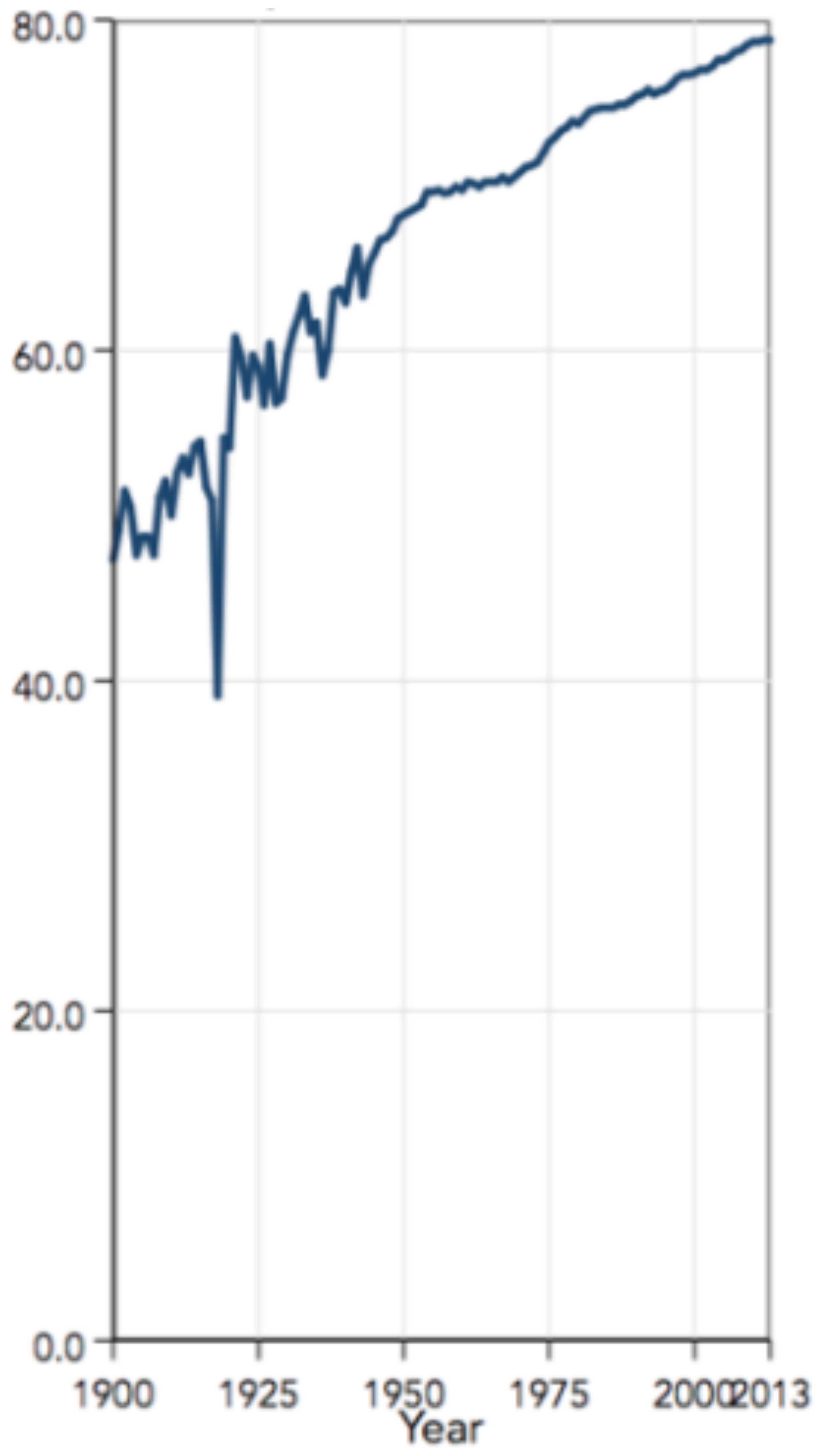
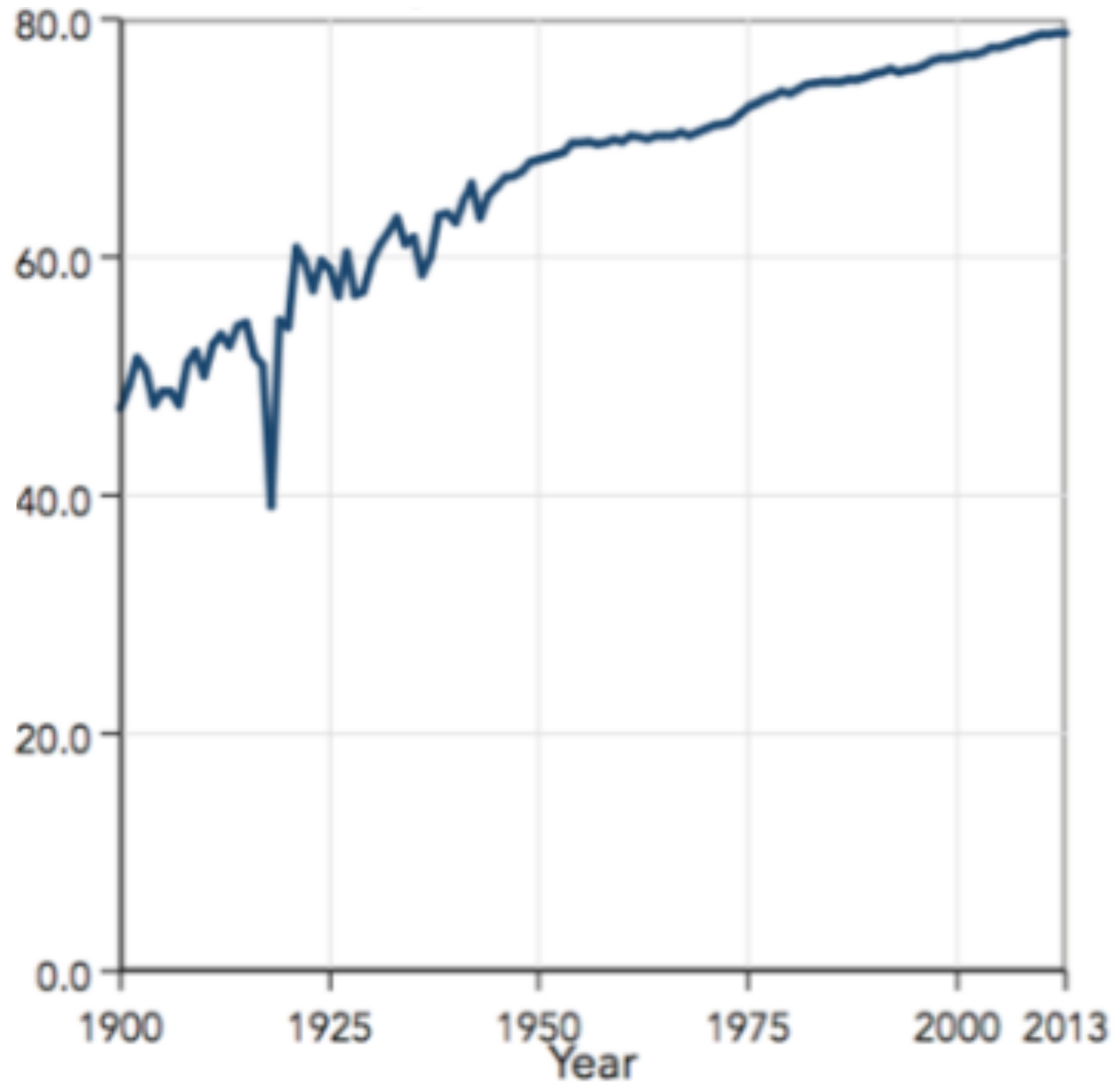
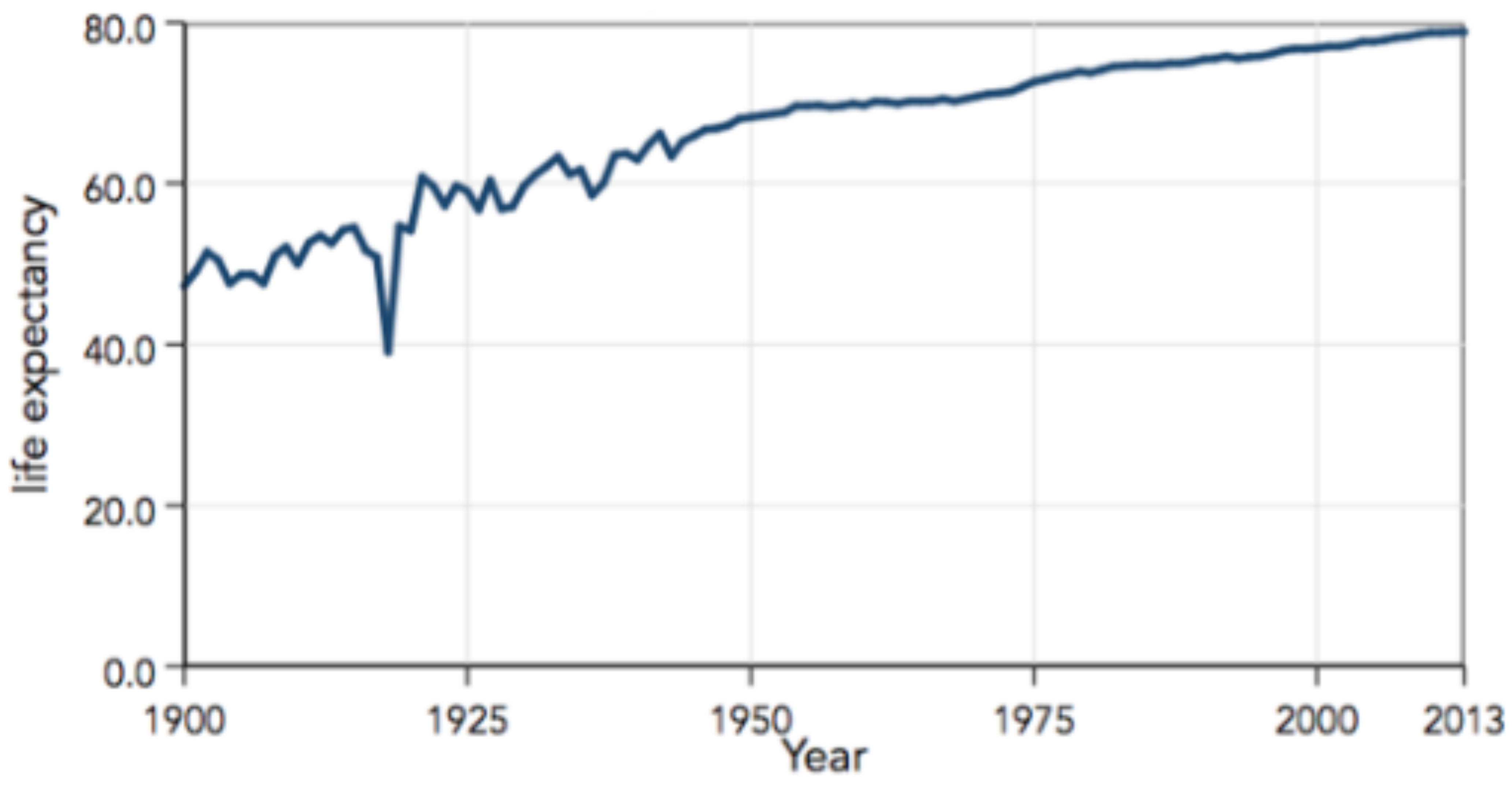




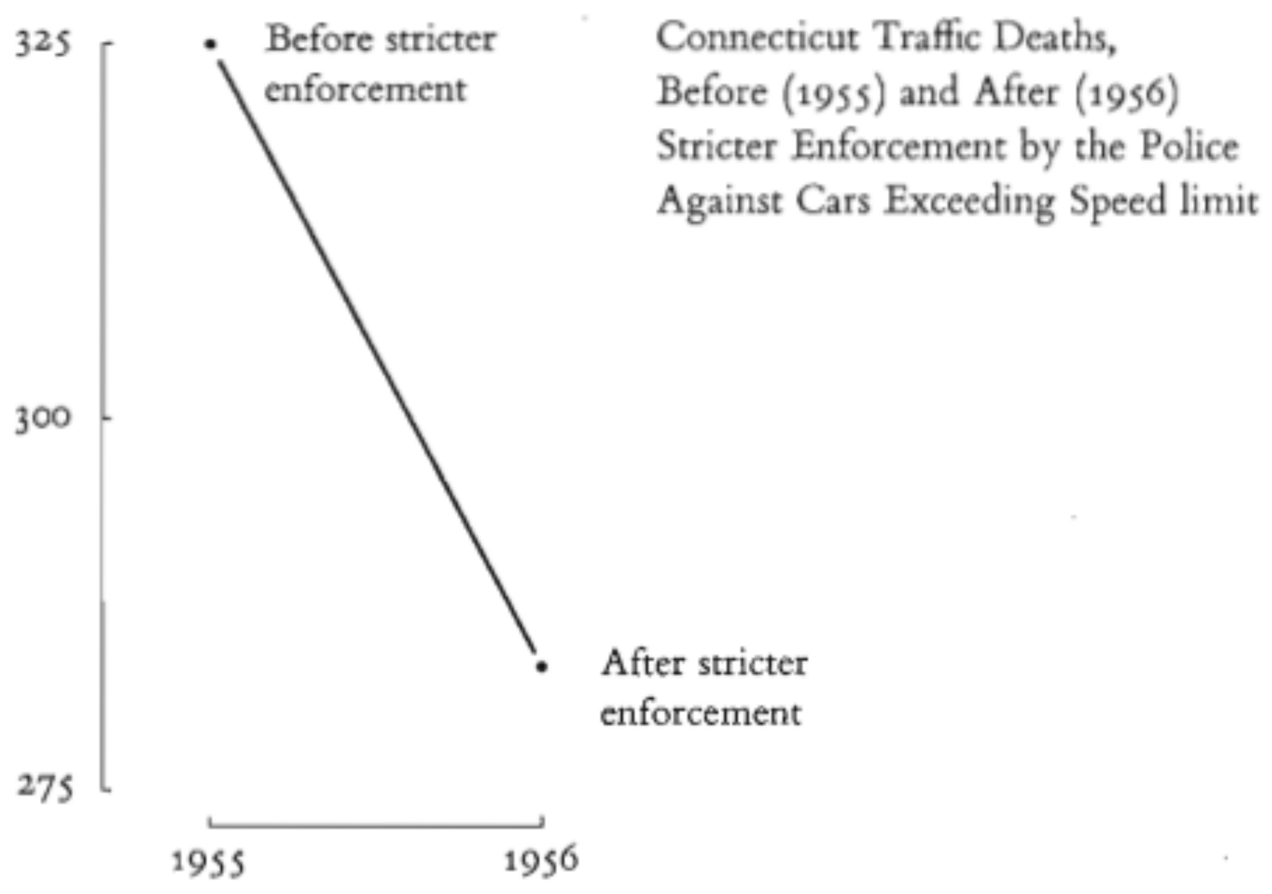
Aspect ratio when visualizing variables with different scales?

Depends. One possible rule:
Golden ratio (W=1.6 x H)



PRINCIPLES OF DATA VISUALIZATION

Graphical Integrity: Omitting Context



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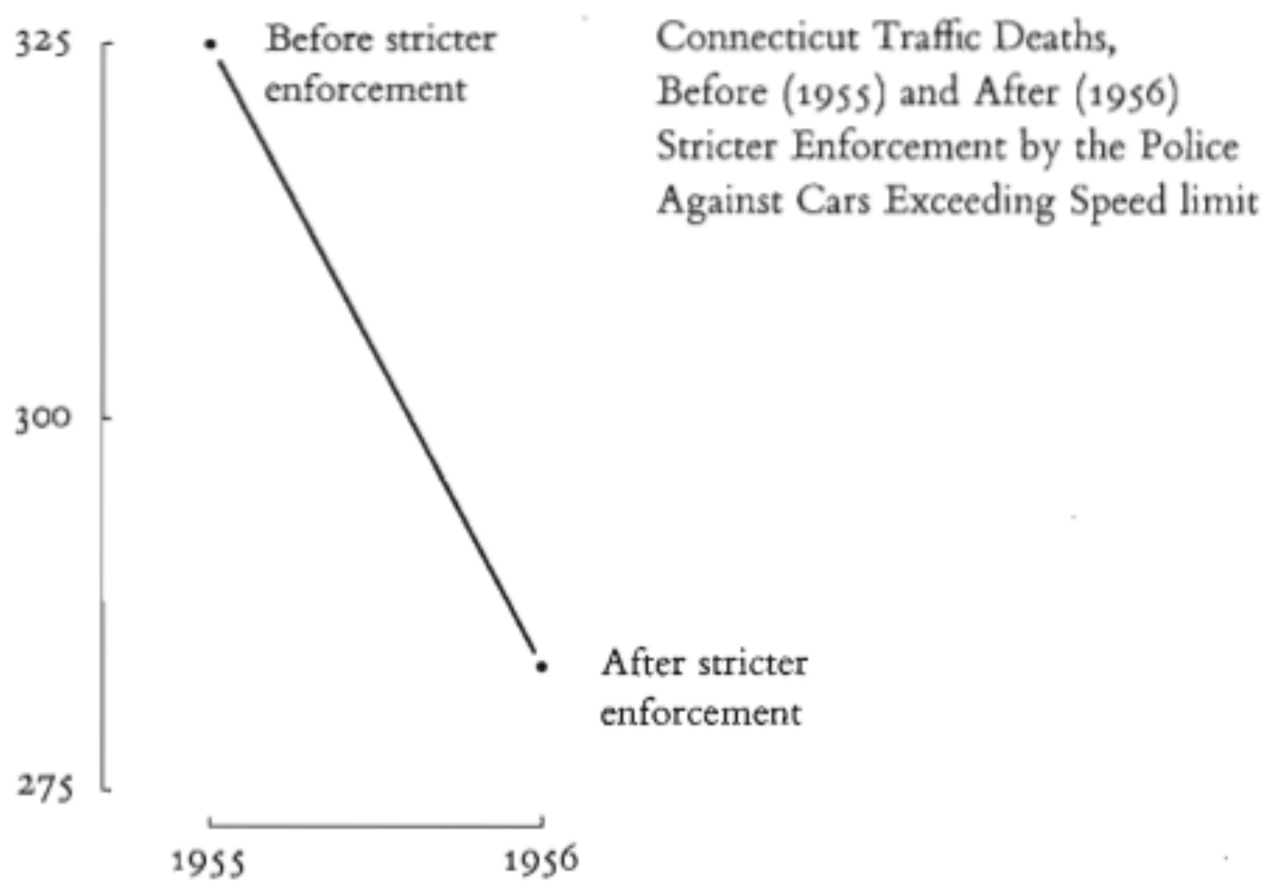

A large decline in traffic deaths in Connecticut: Due to stricter enforcement?

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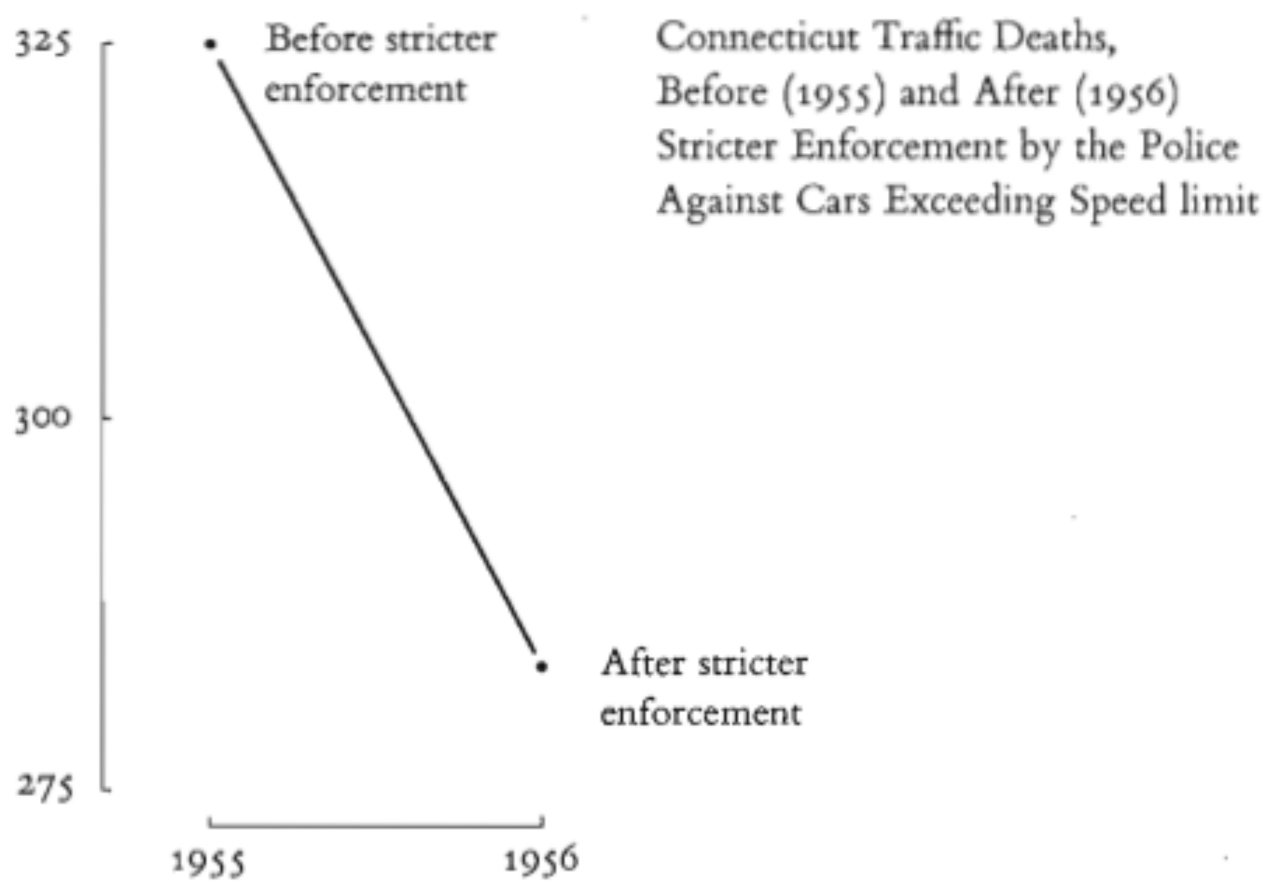

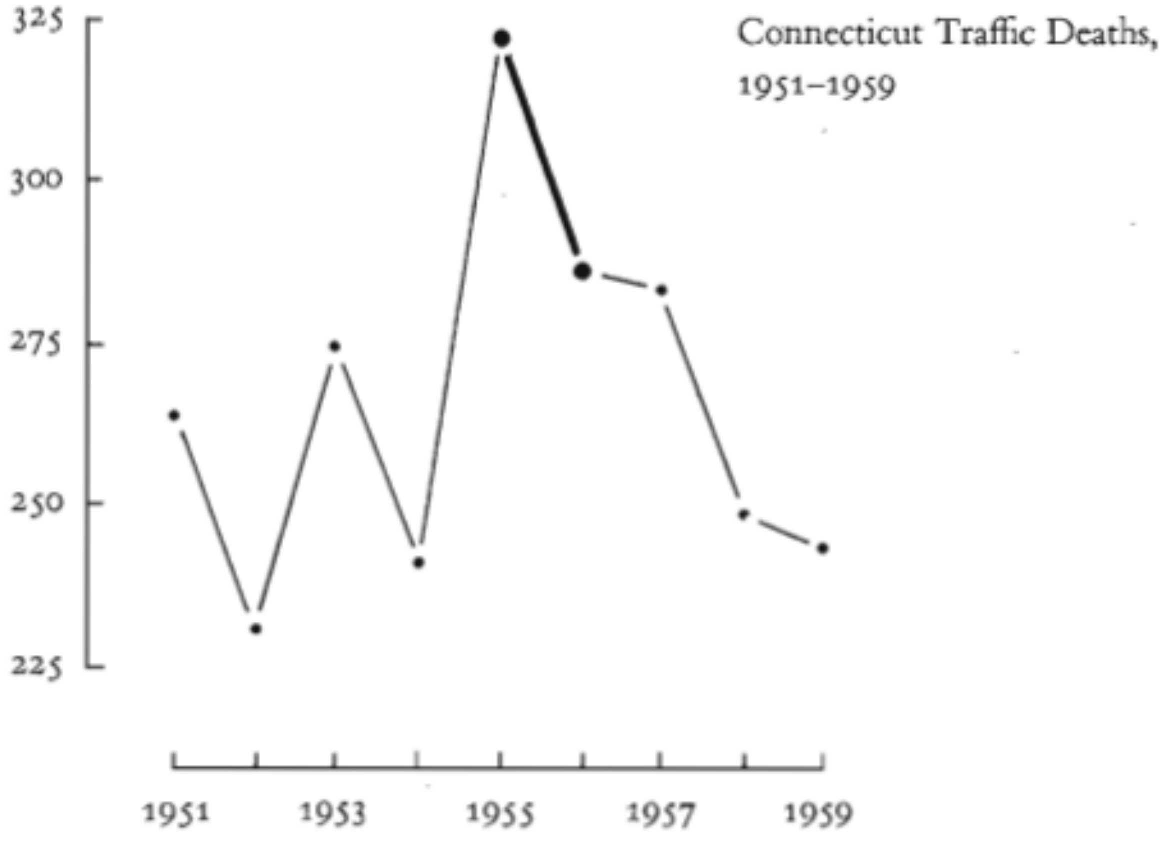

A large decline in traffic deaths in Connecticut: Due to stricter enforcement?

Adding context changes the story

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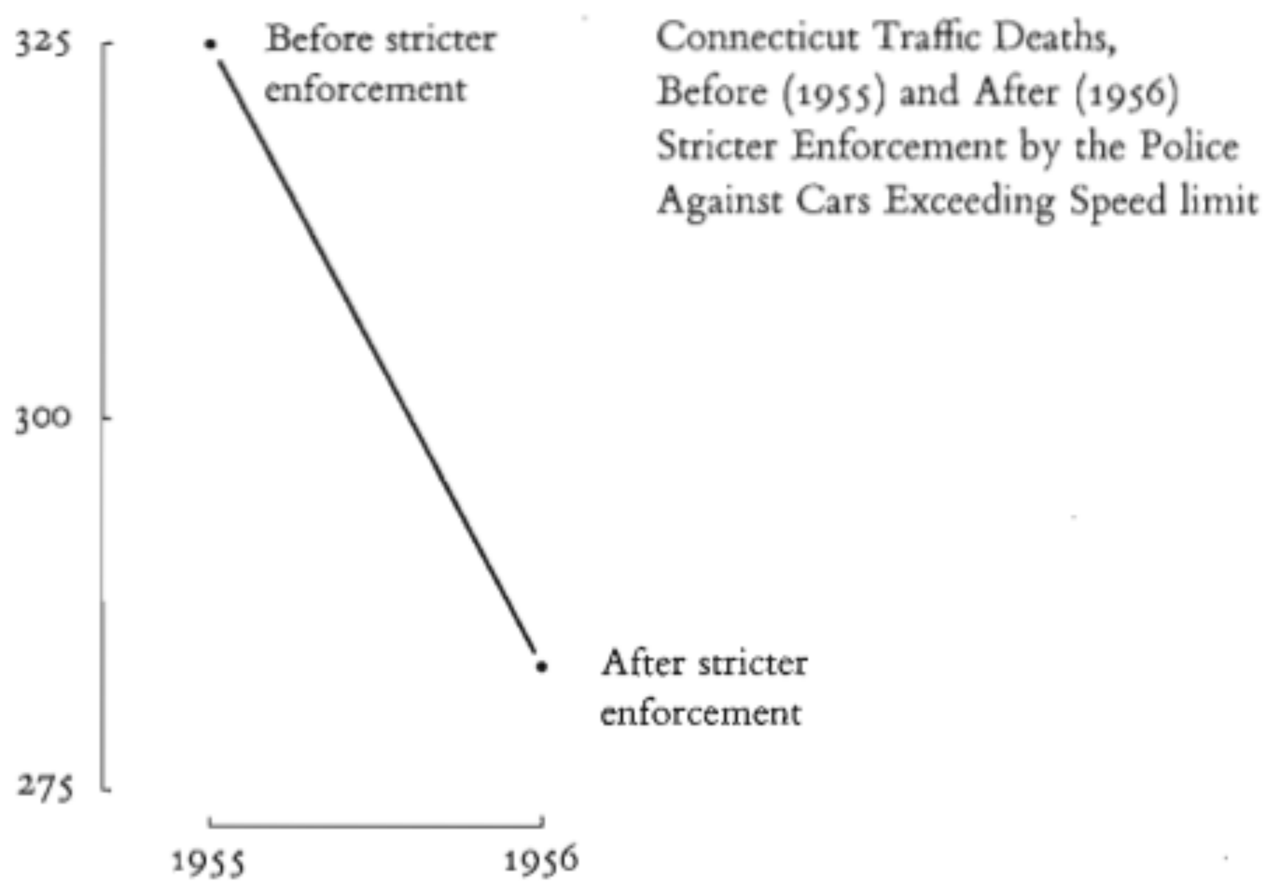

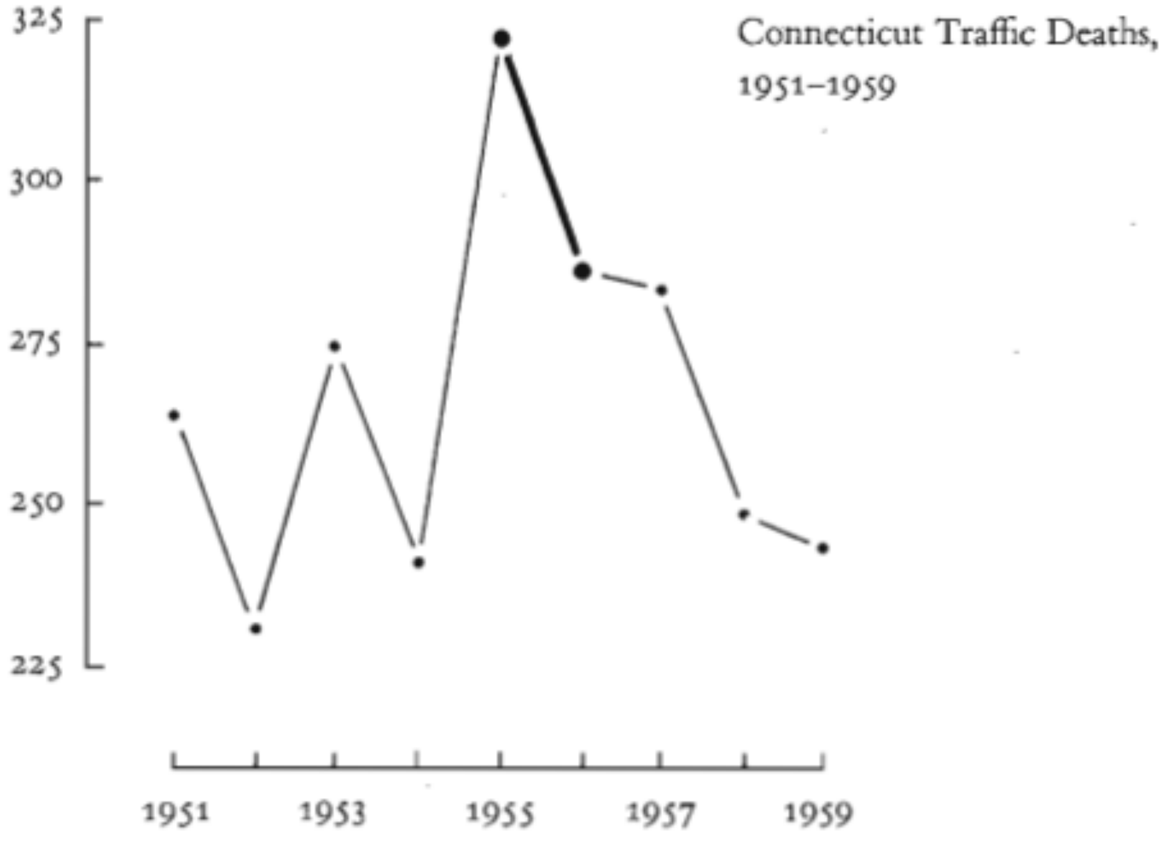

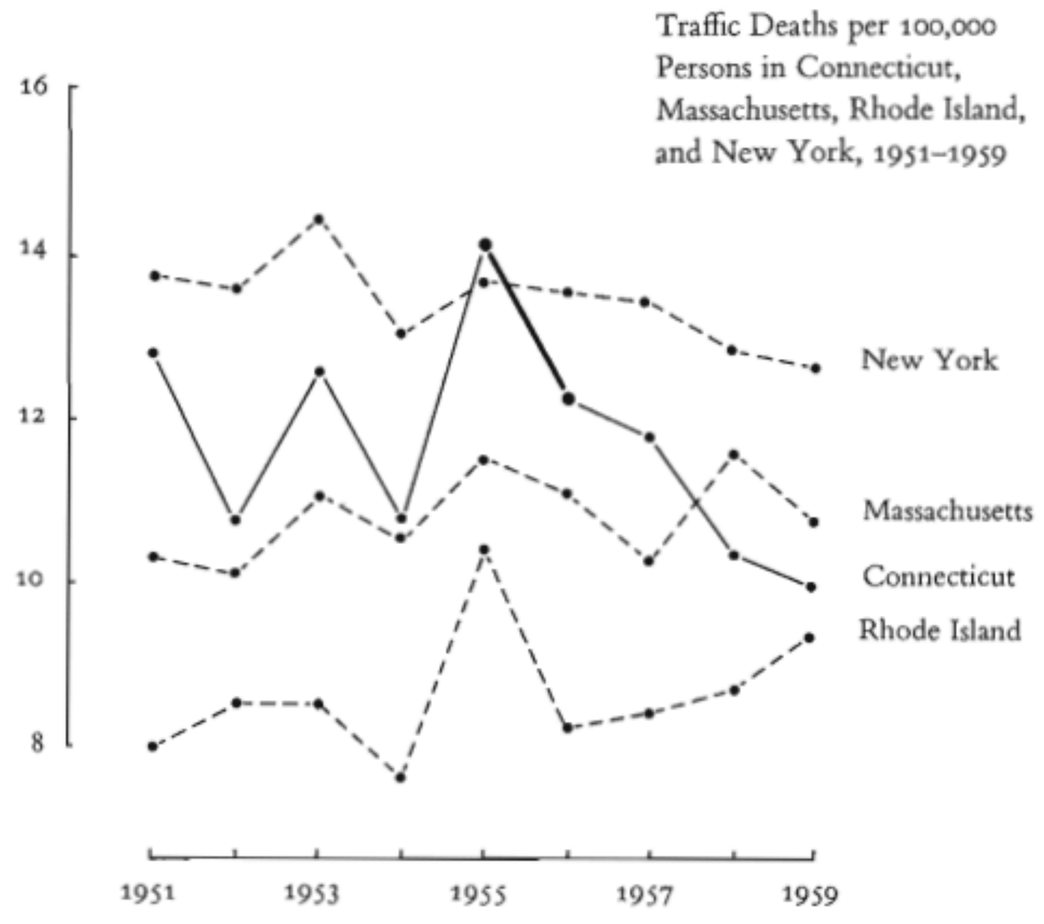

A large decline in traffic deaths in Connecticut: Due to stricter enforcement?

Adding context changes the story

Adding further context: not only Connecticut had a decline.

Graphics must not quote data out of context: There should always be a ‘Compared to what?’

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Data-ink ratio = data-ink / total ink used for the graphic

Goal: Maximize the ratio.

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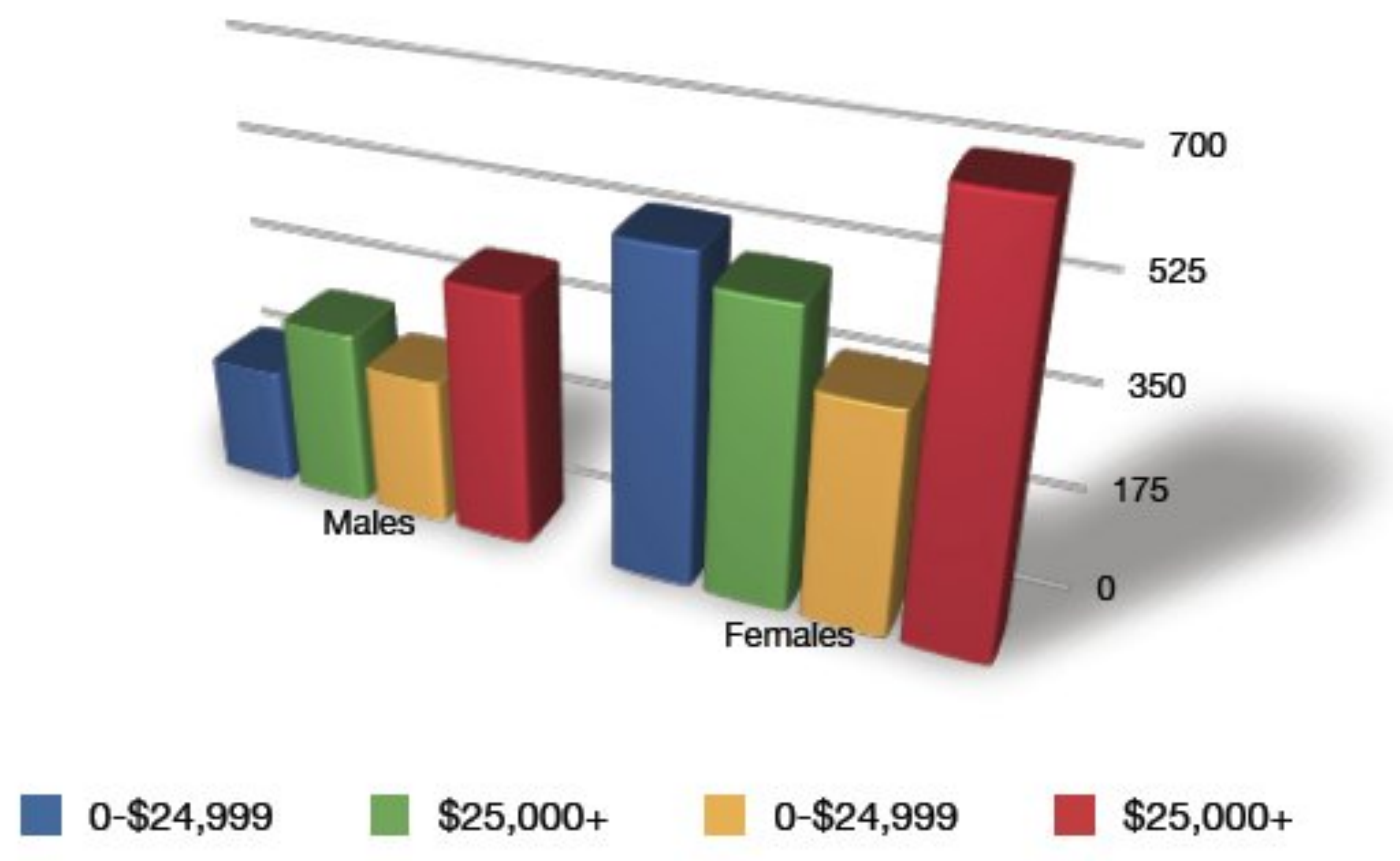

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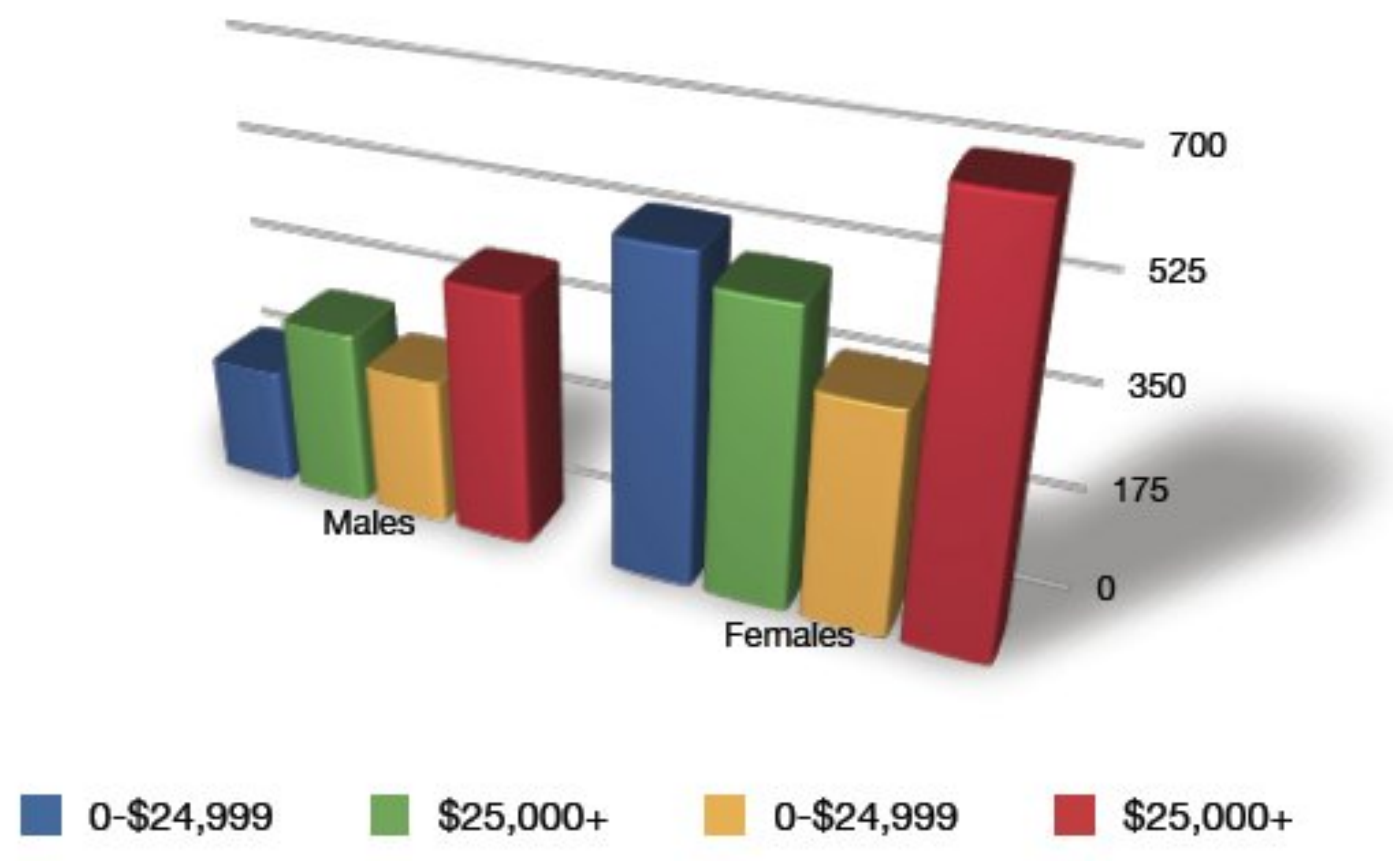

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Much better when most ink is
used to present the data.

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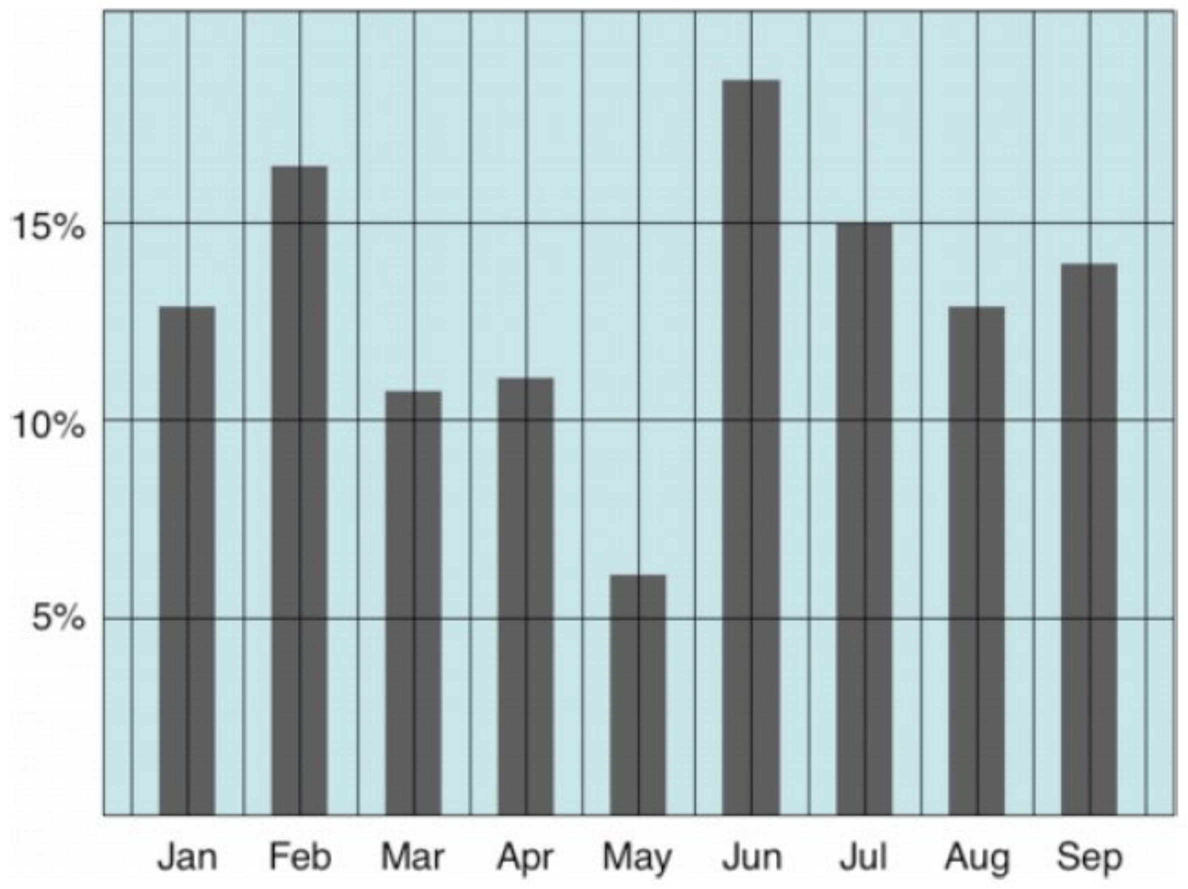

Do we need grid lines?

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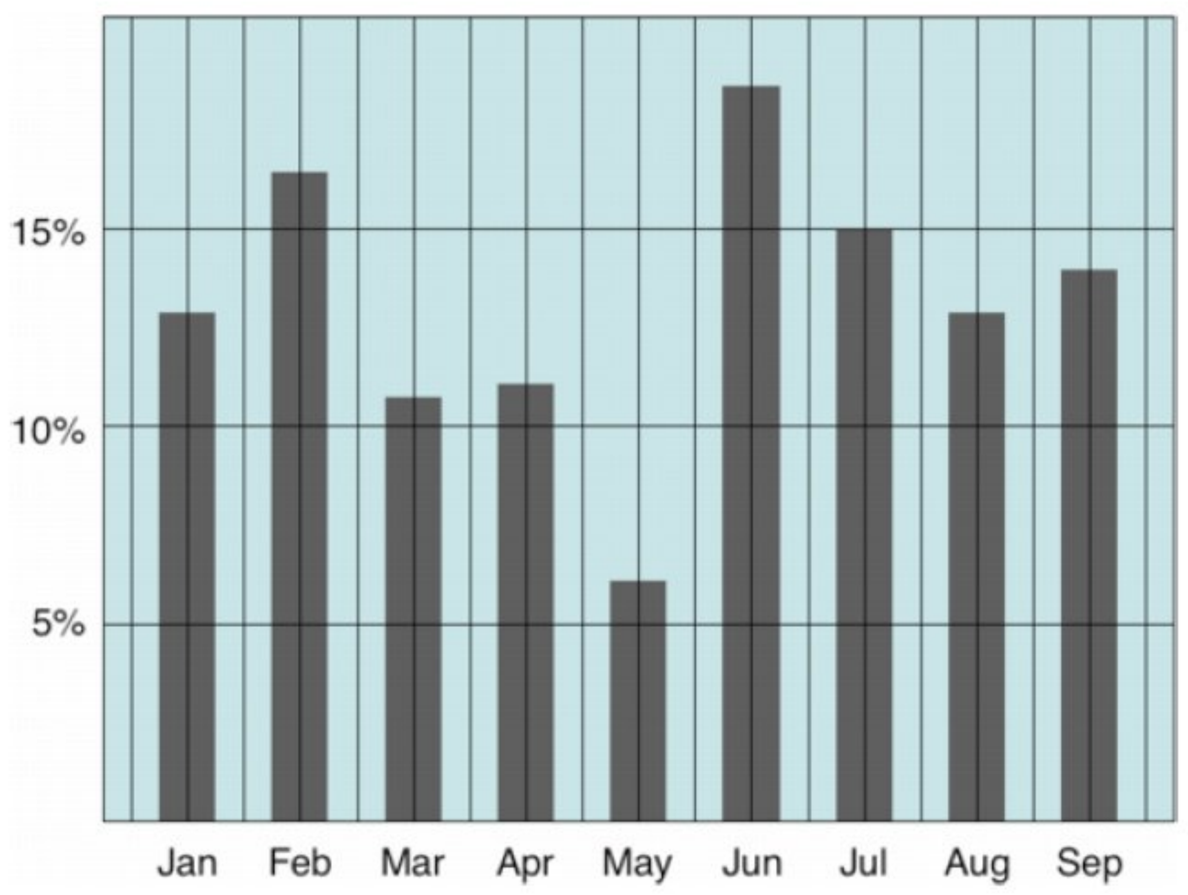

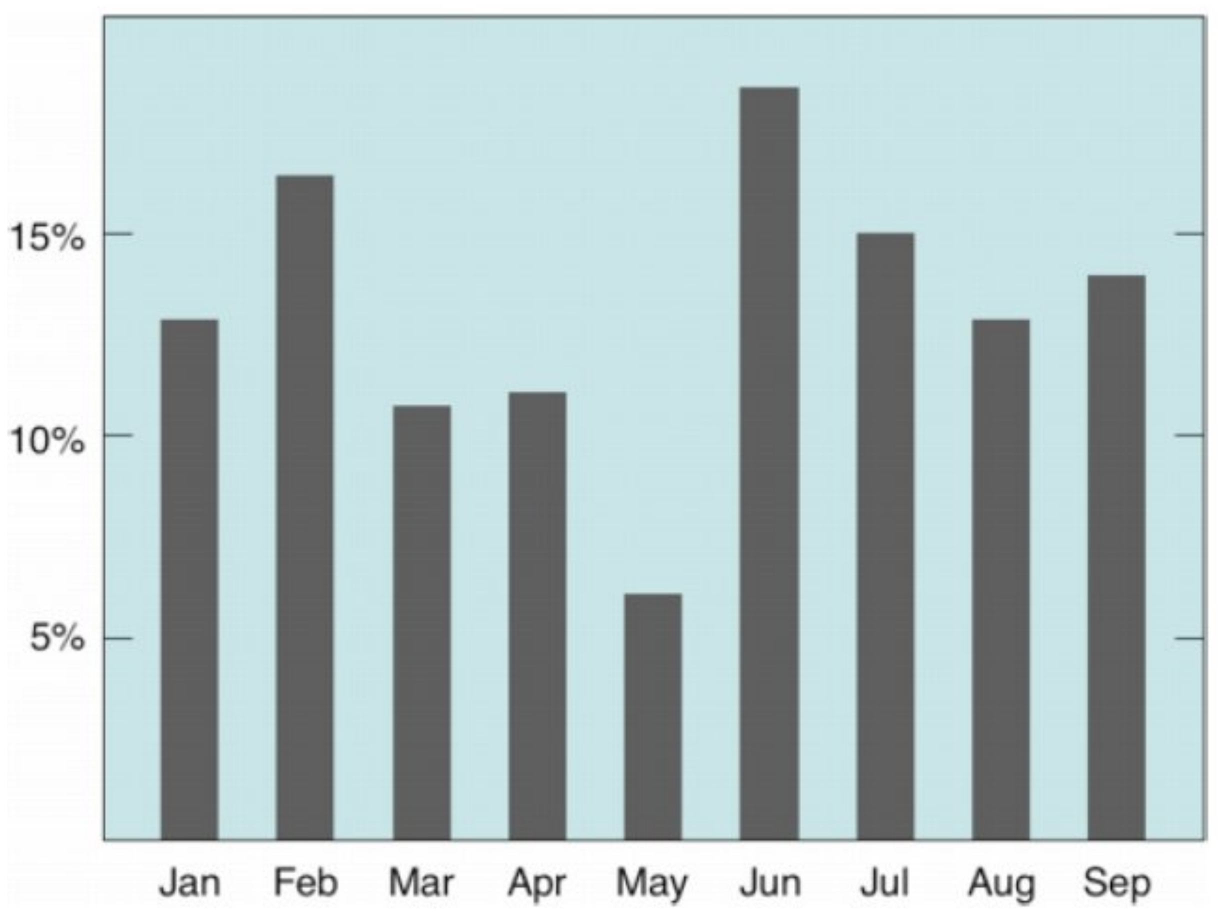

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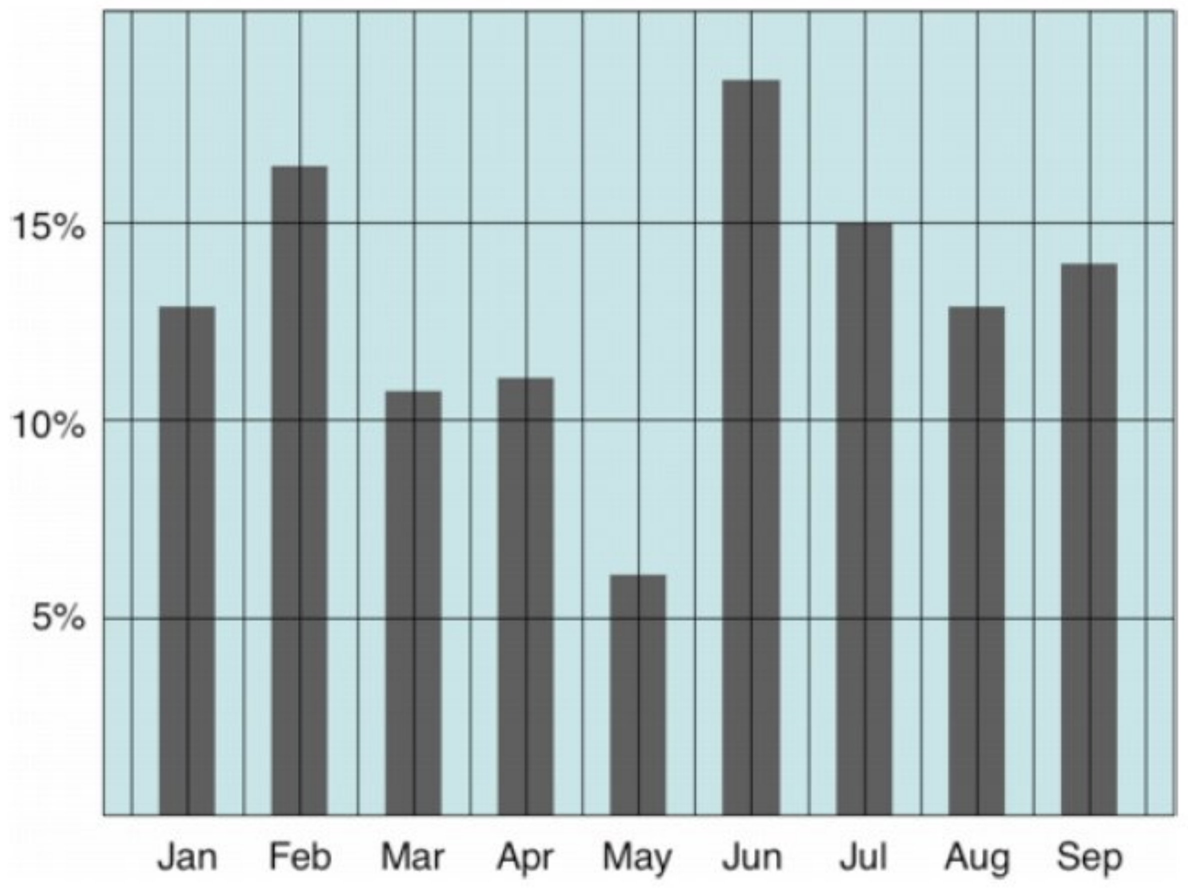

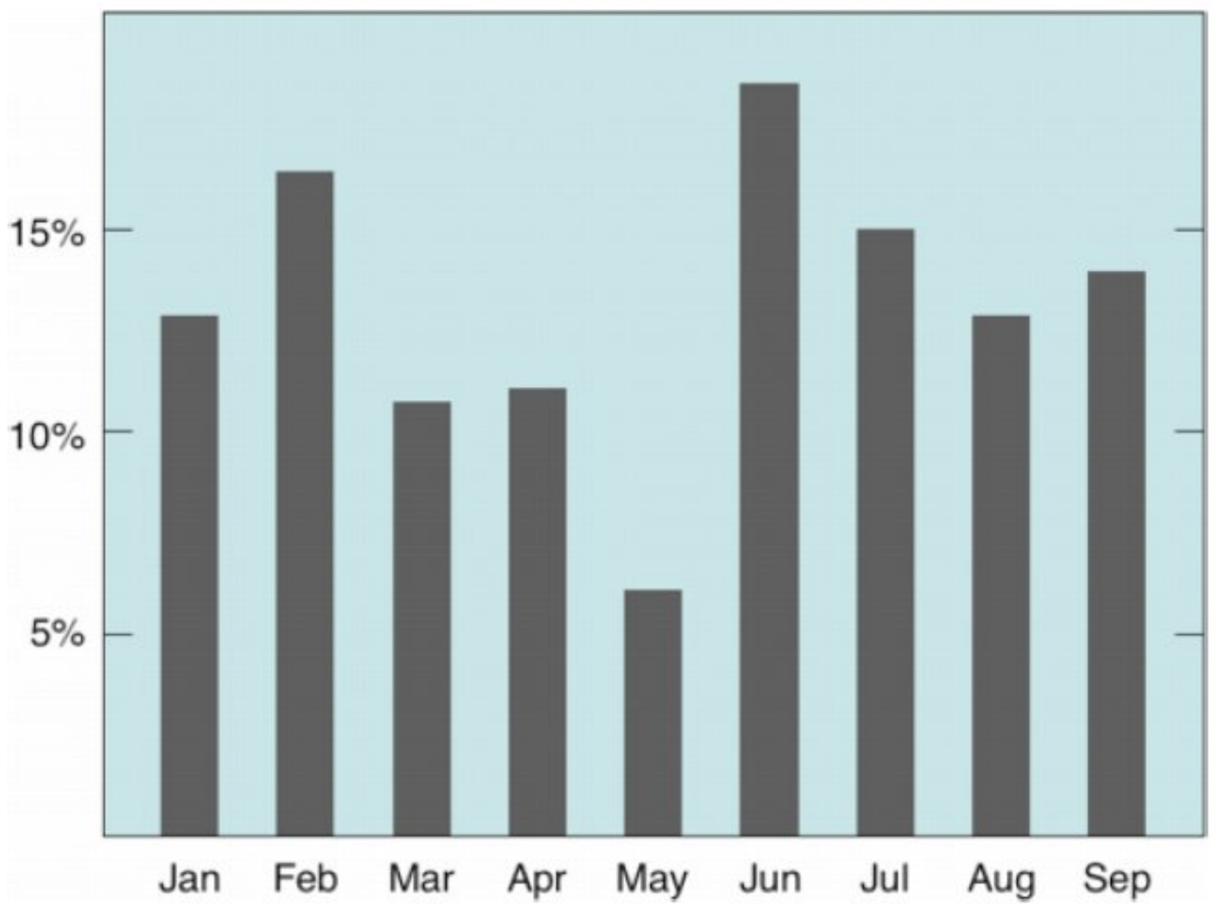

Do we need color?

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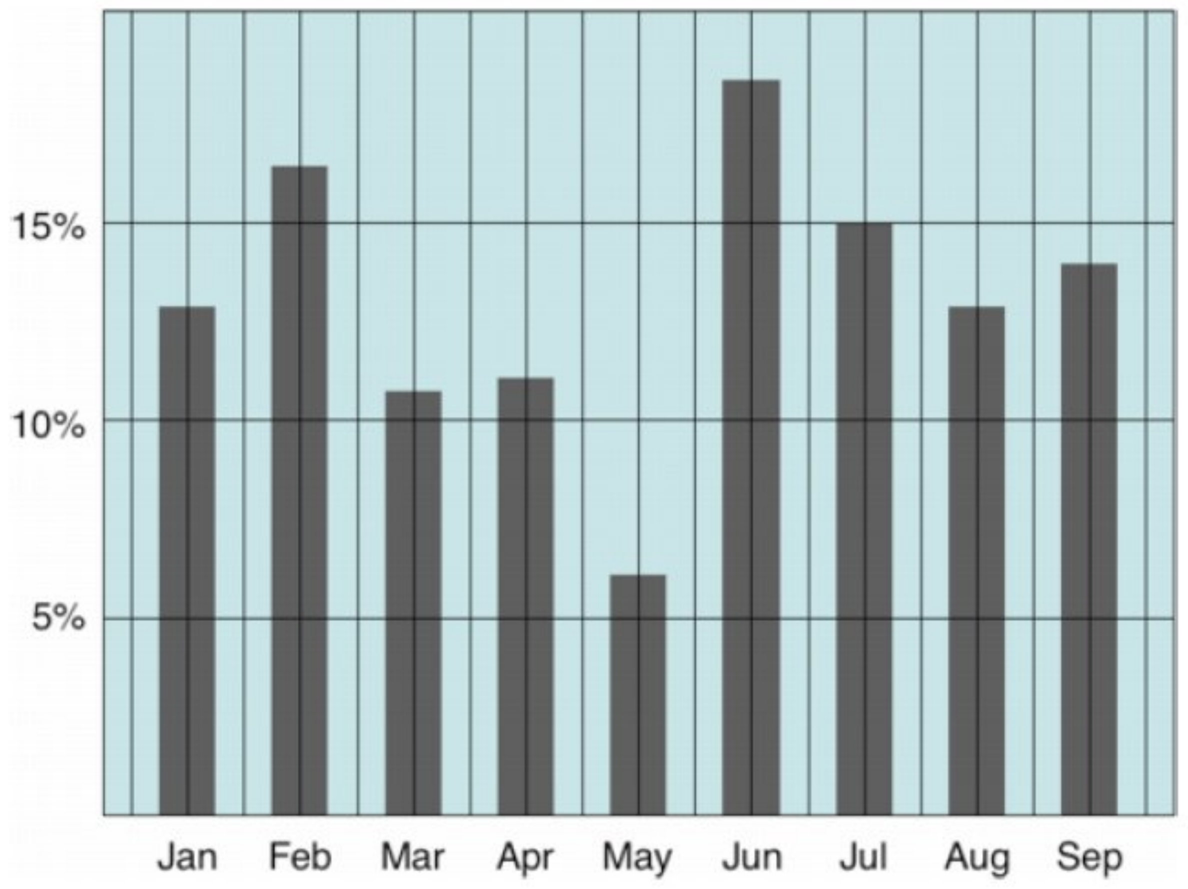

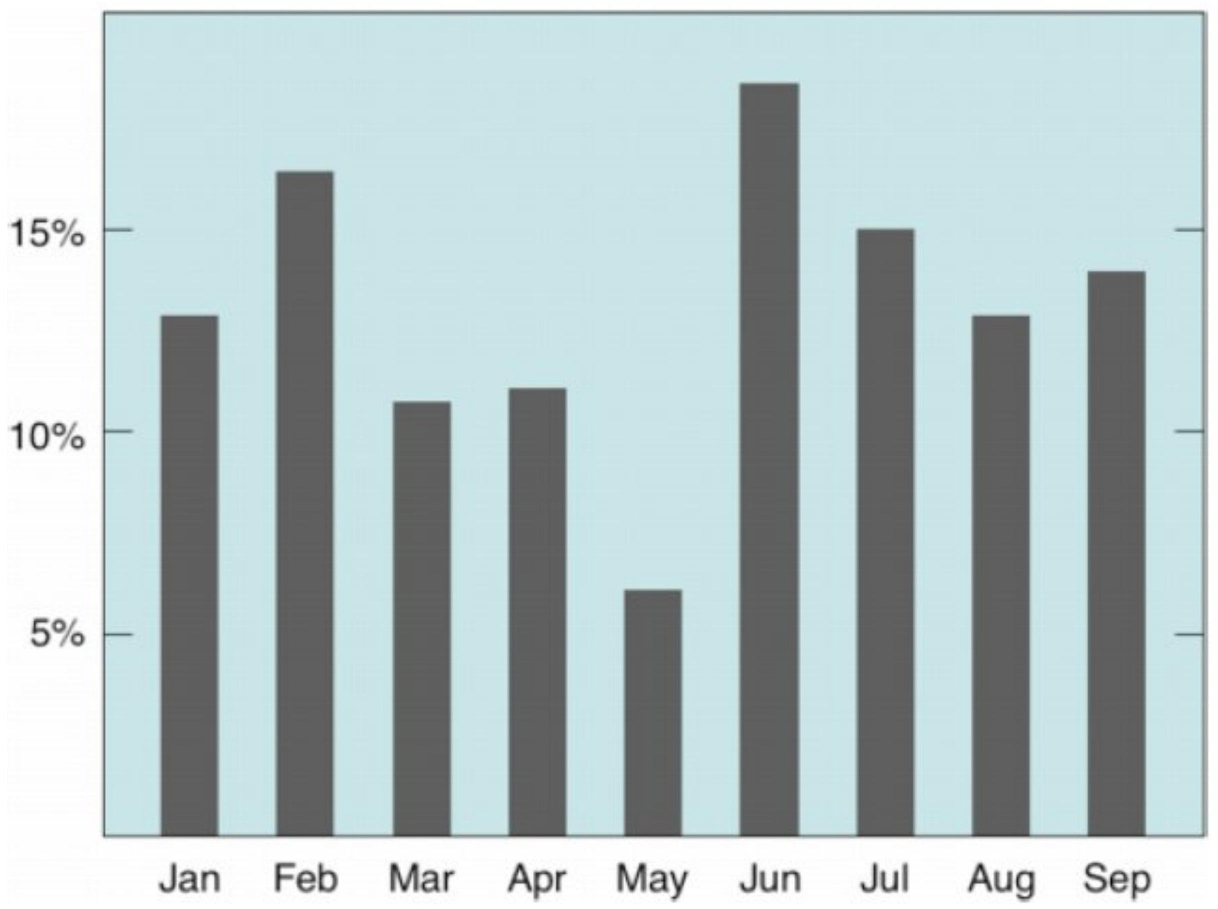

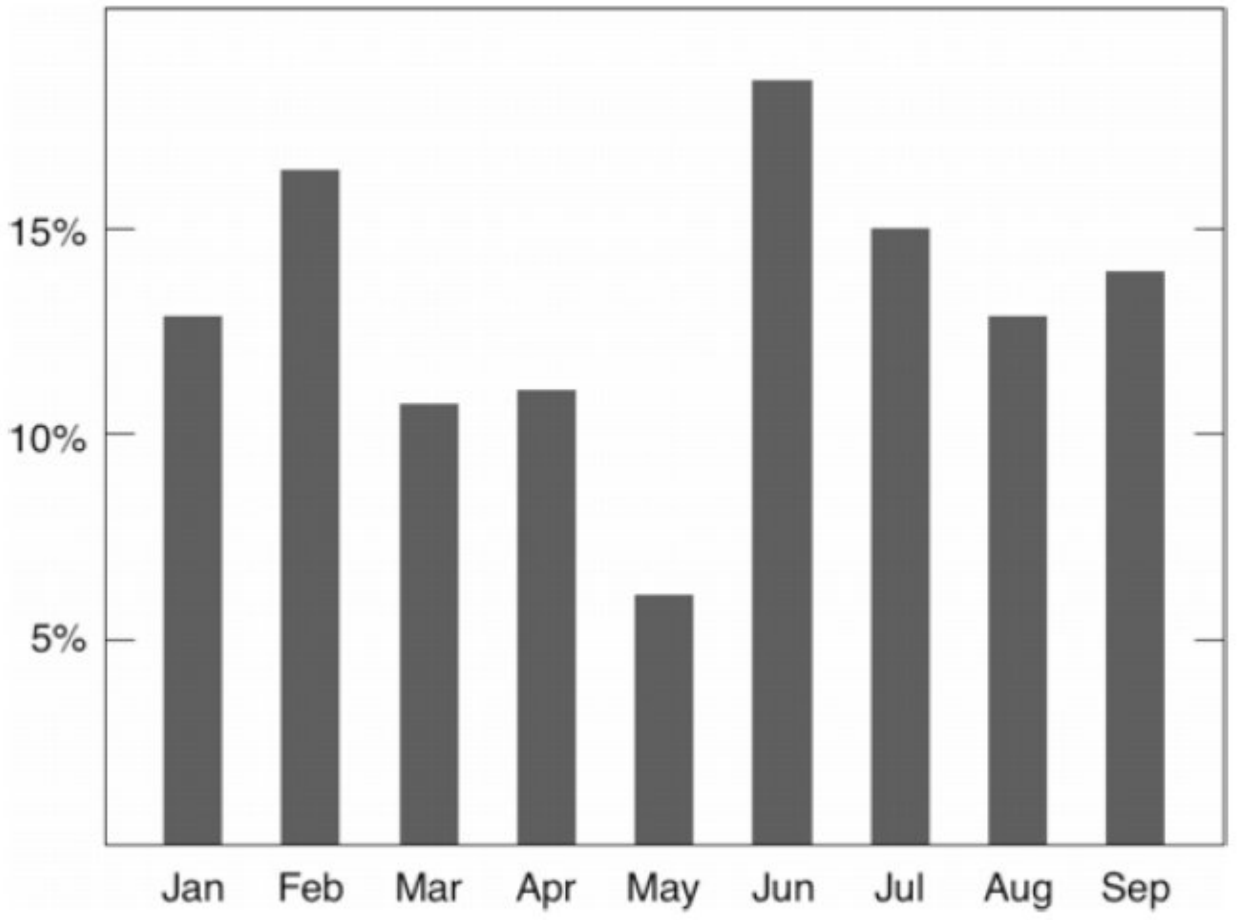

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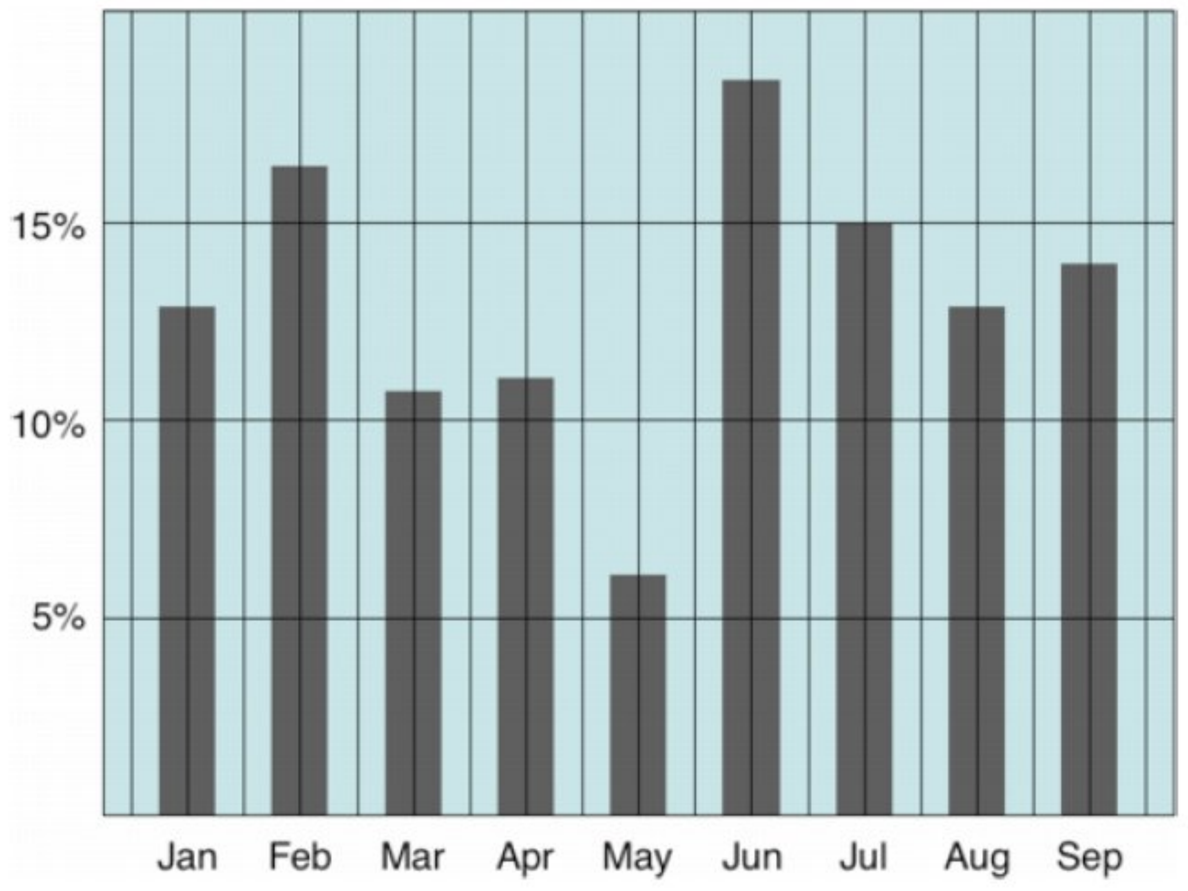
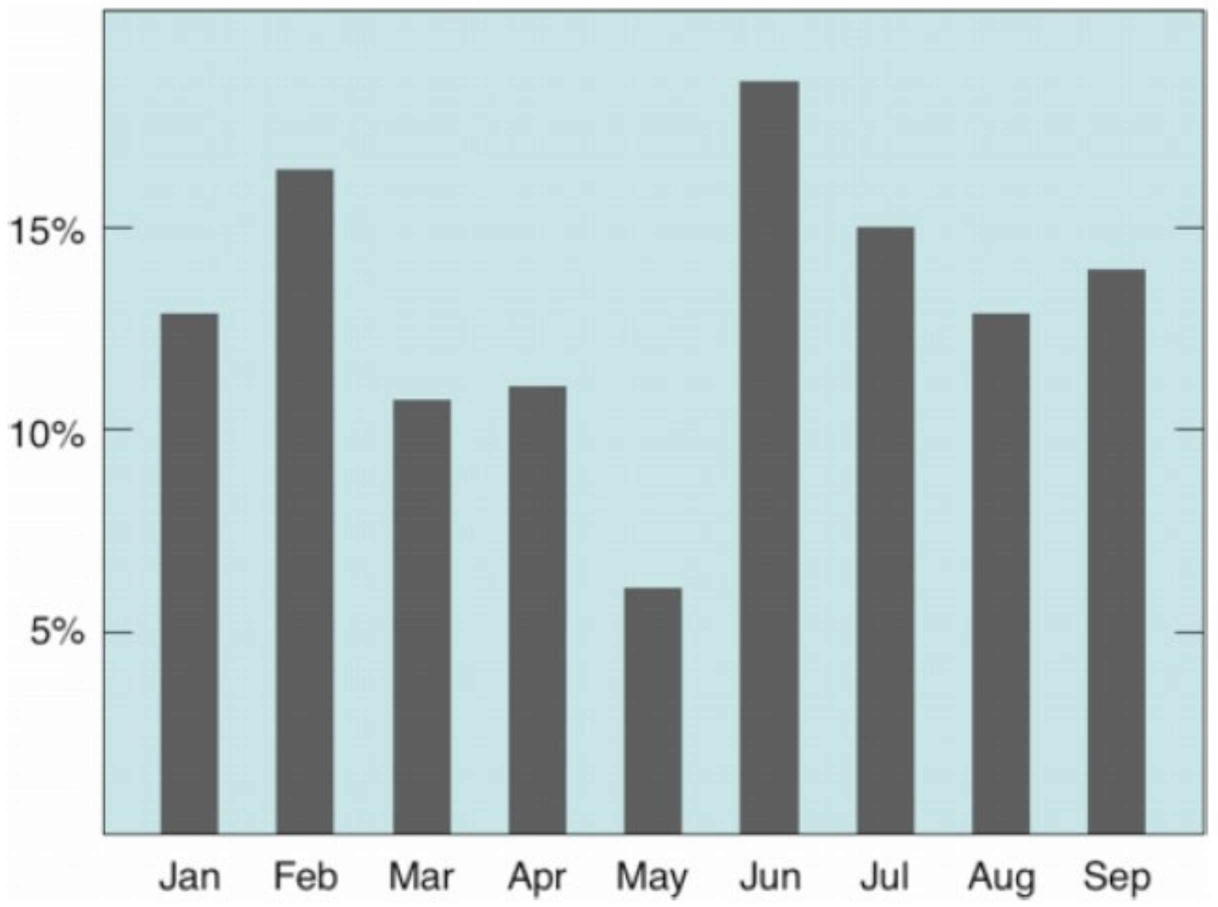
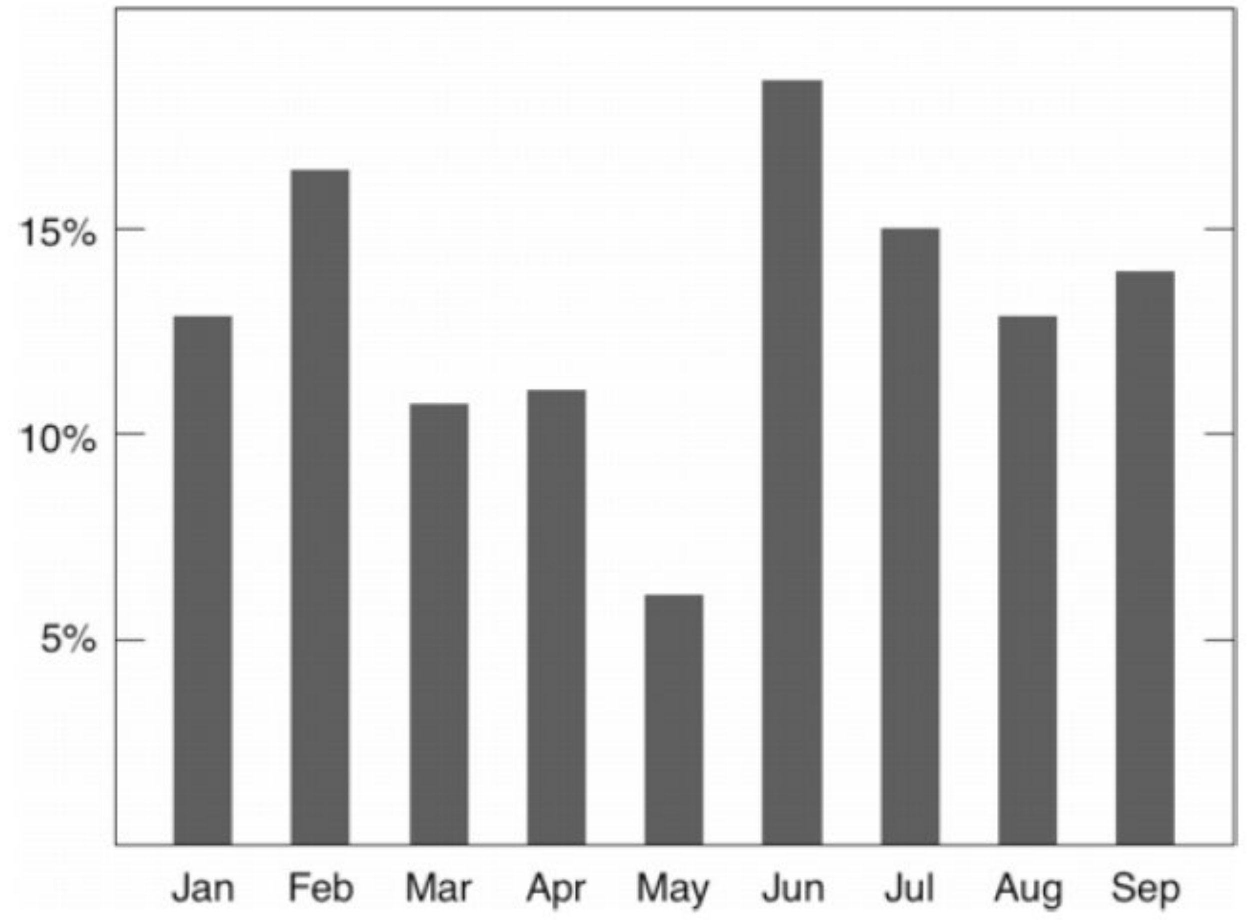




Do we need the full box?



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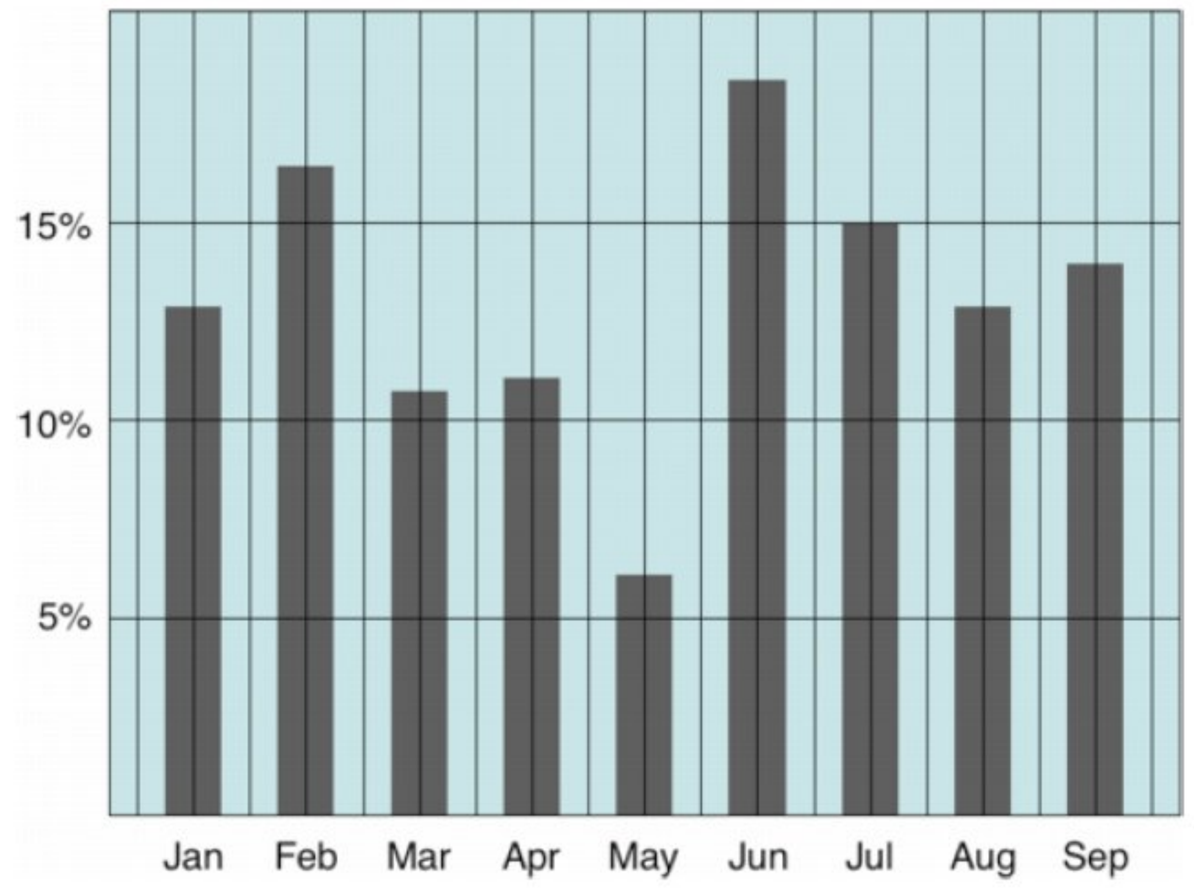

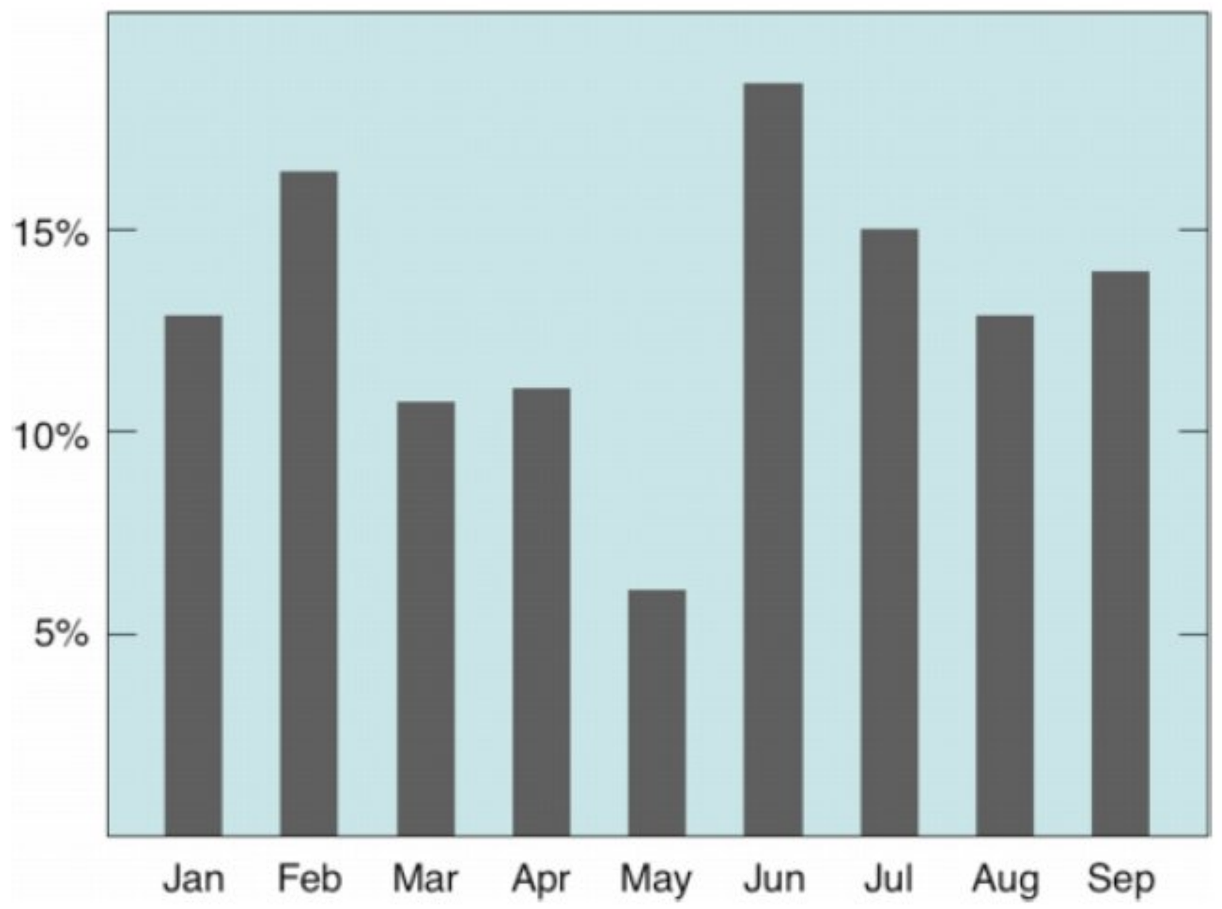

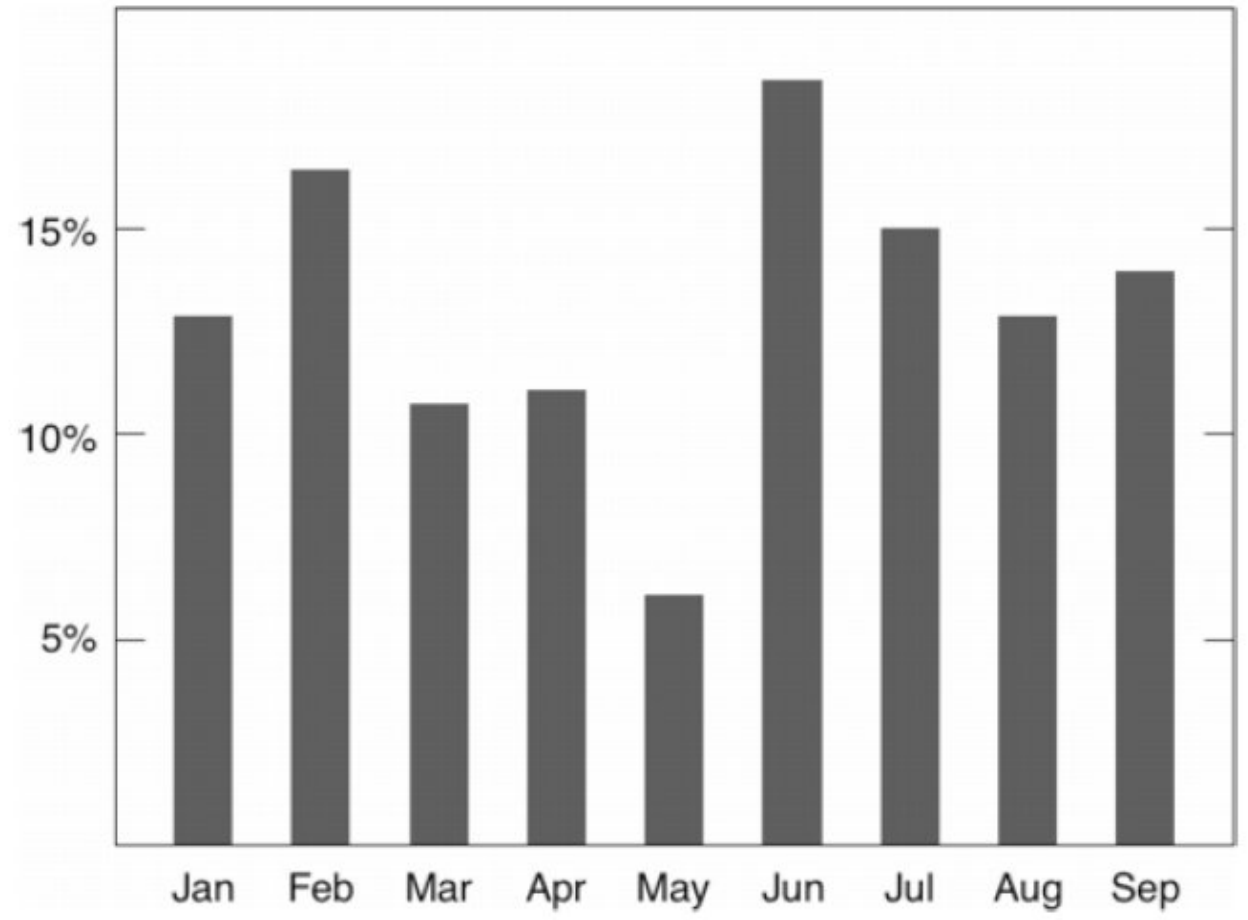

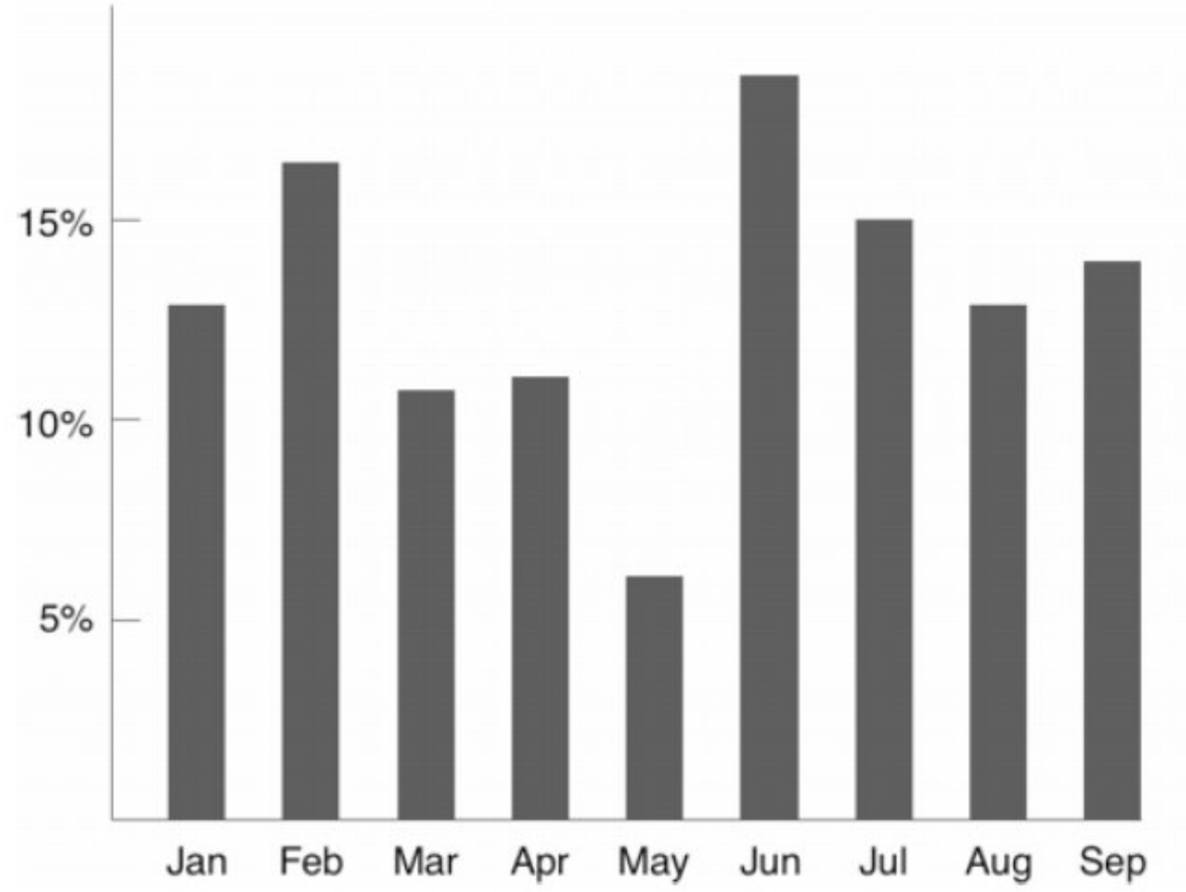

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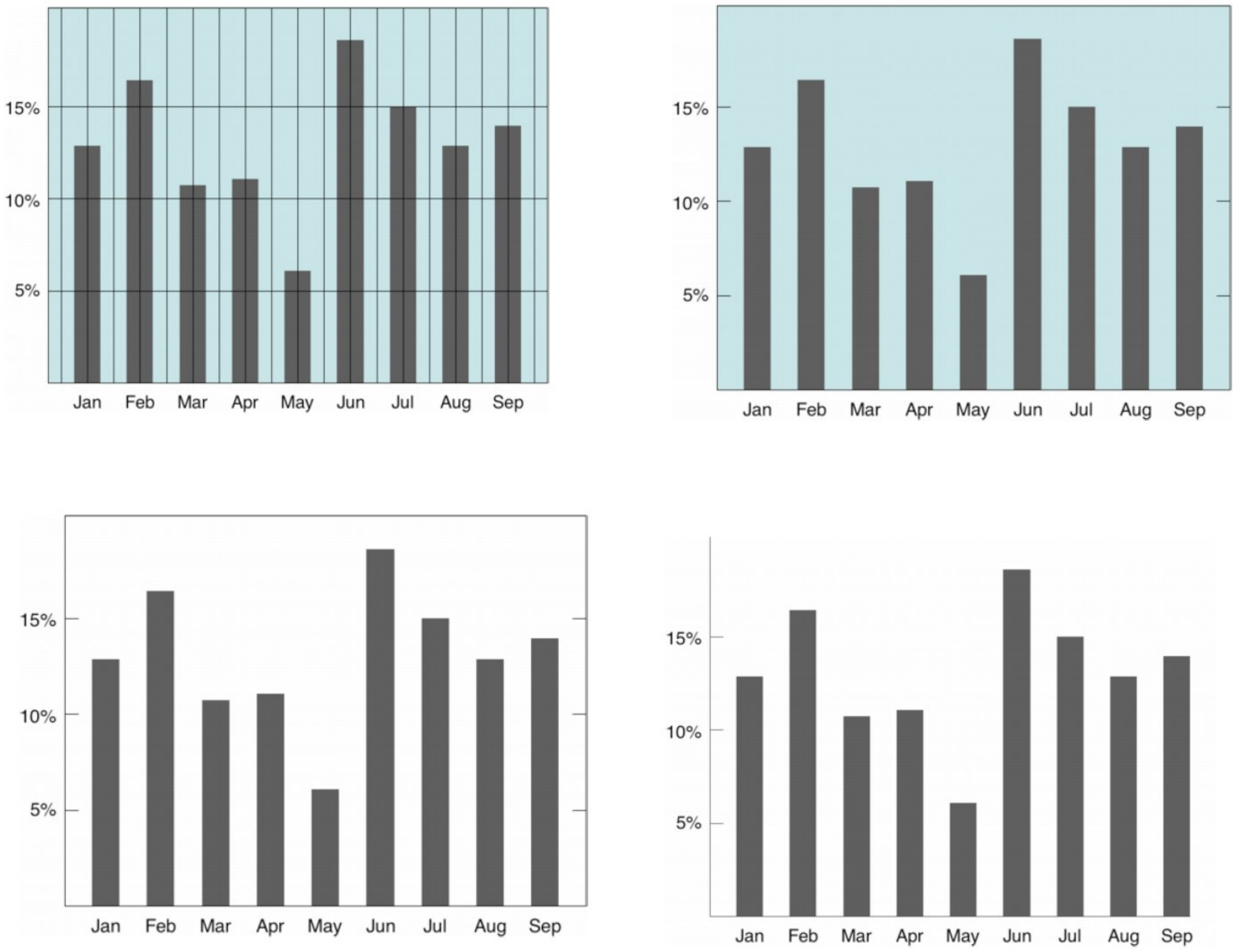

Can less be more?

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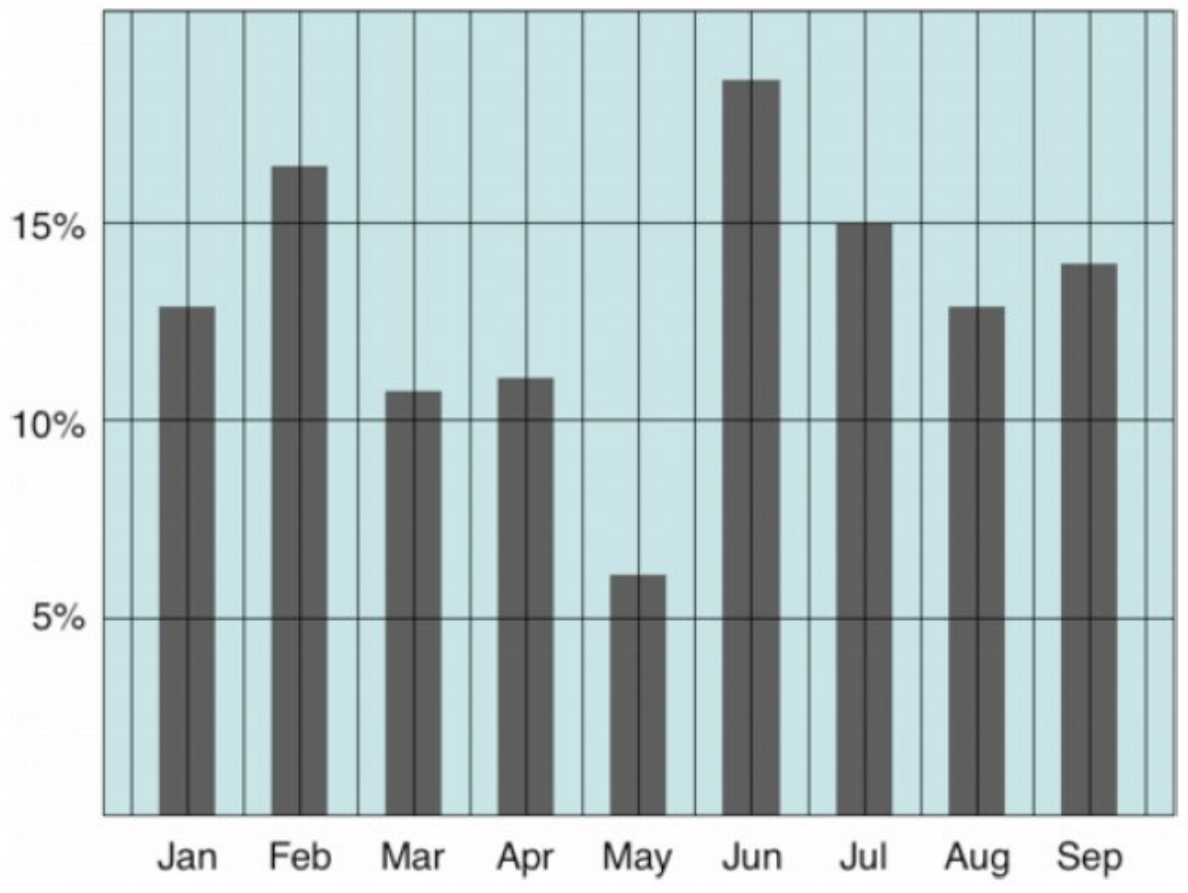

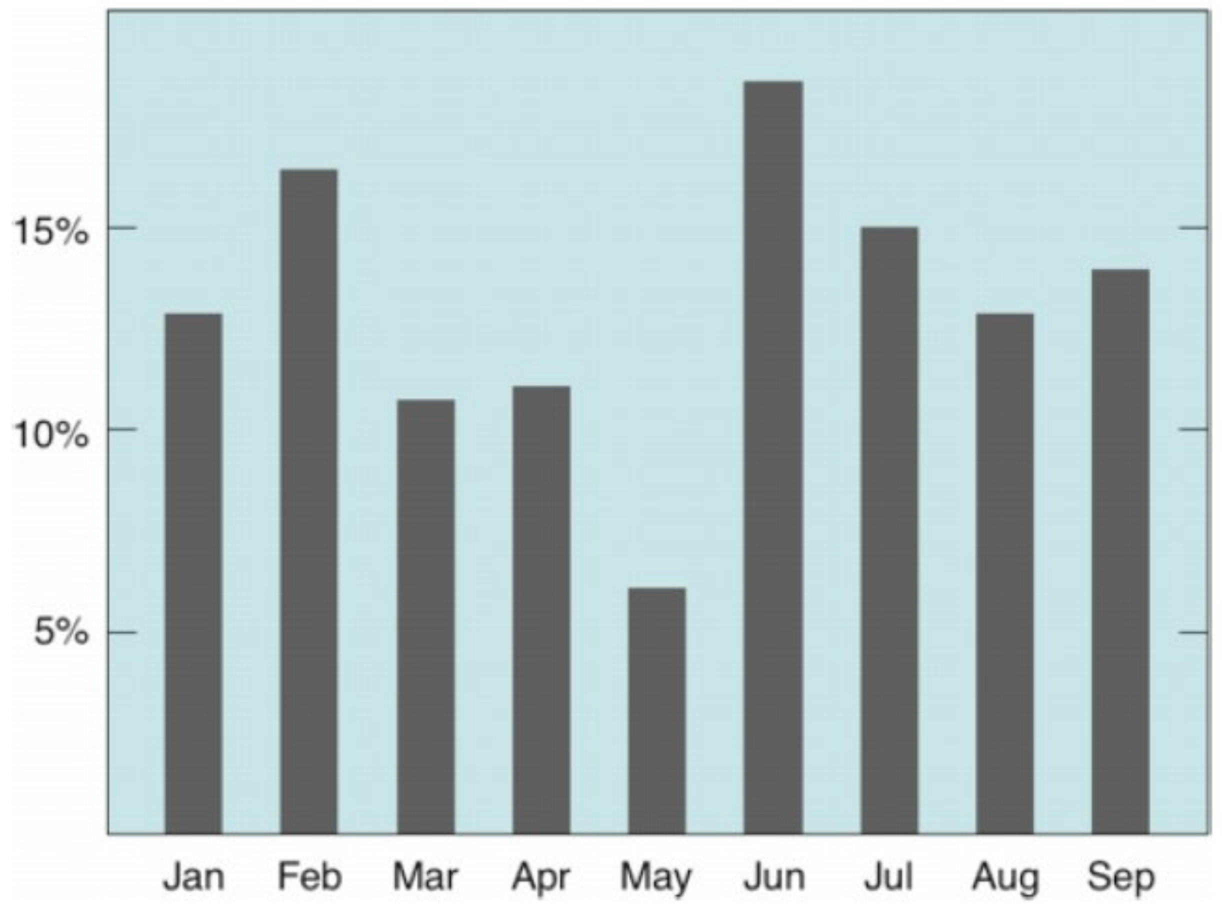

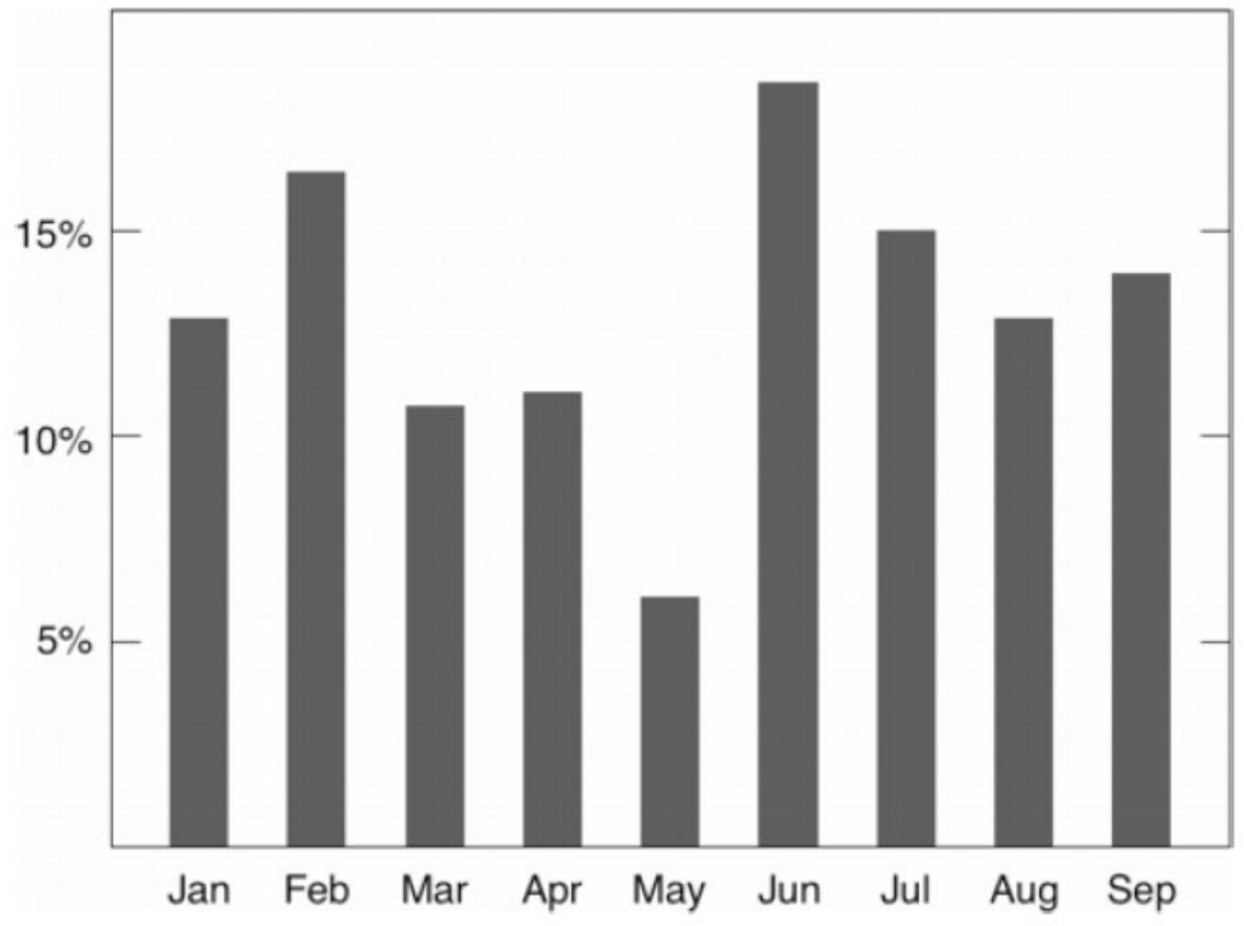

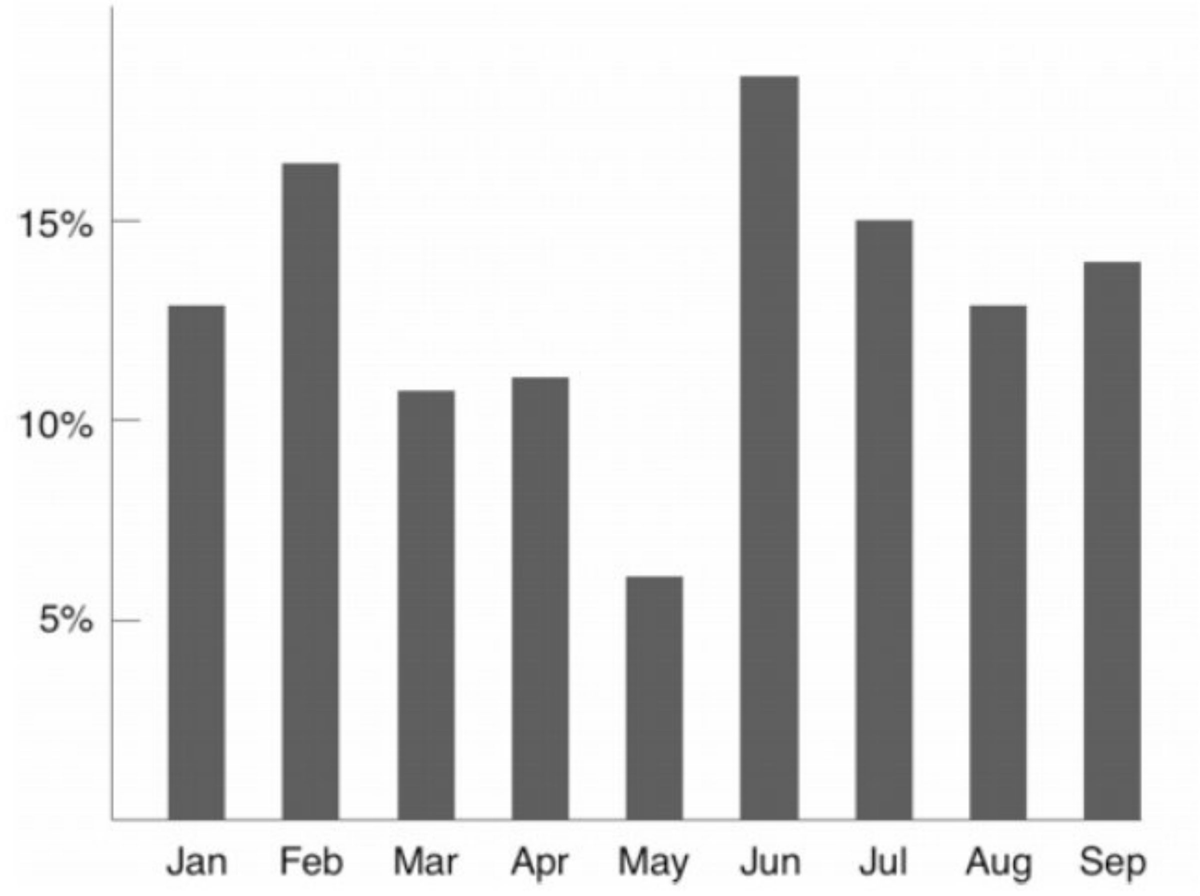

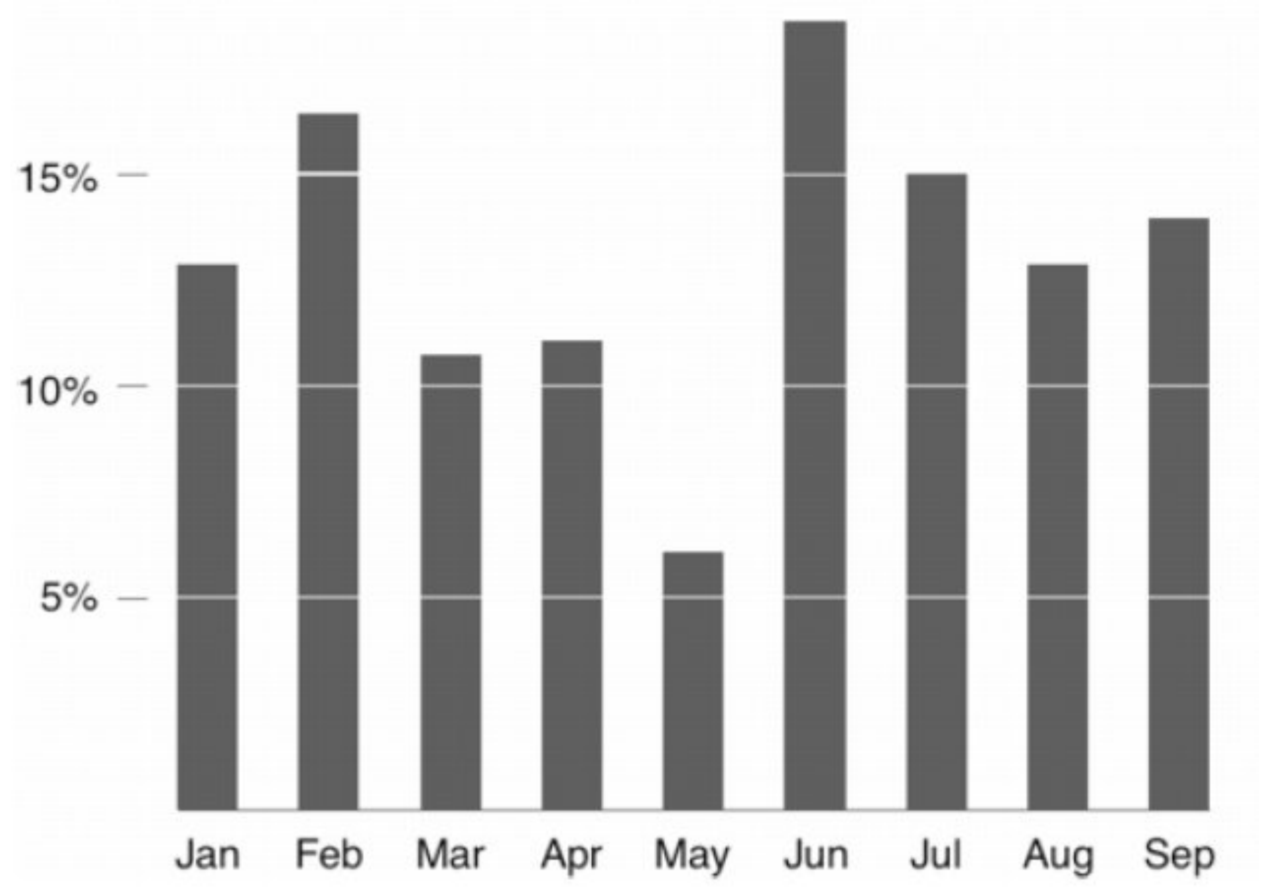

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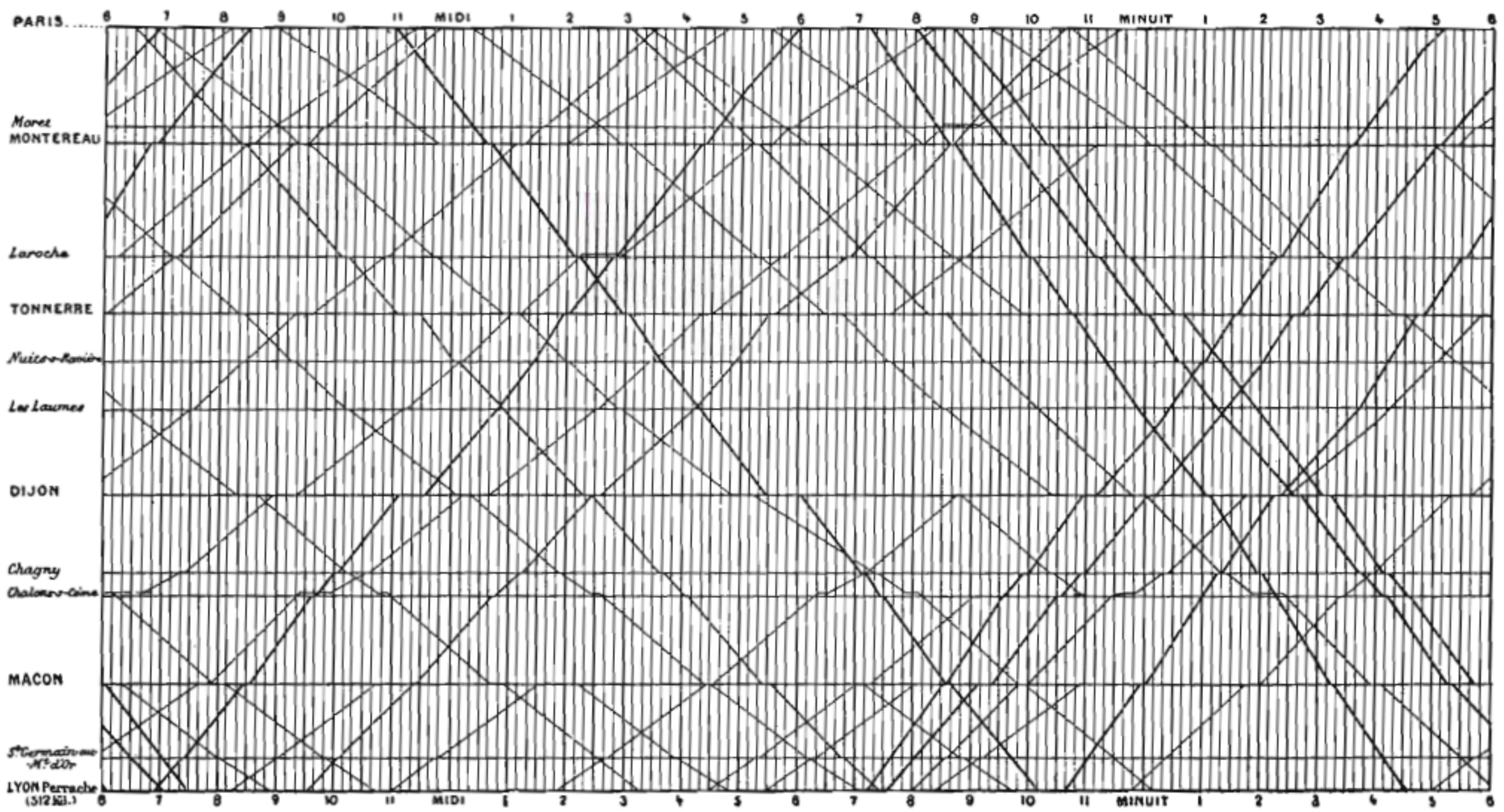
Further examples: Marey train schedule



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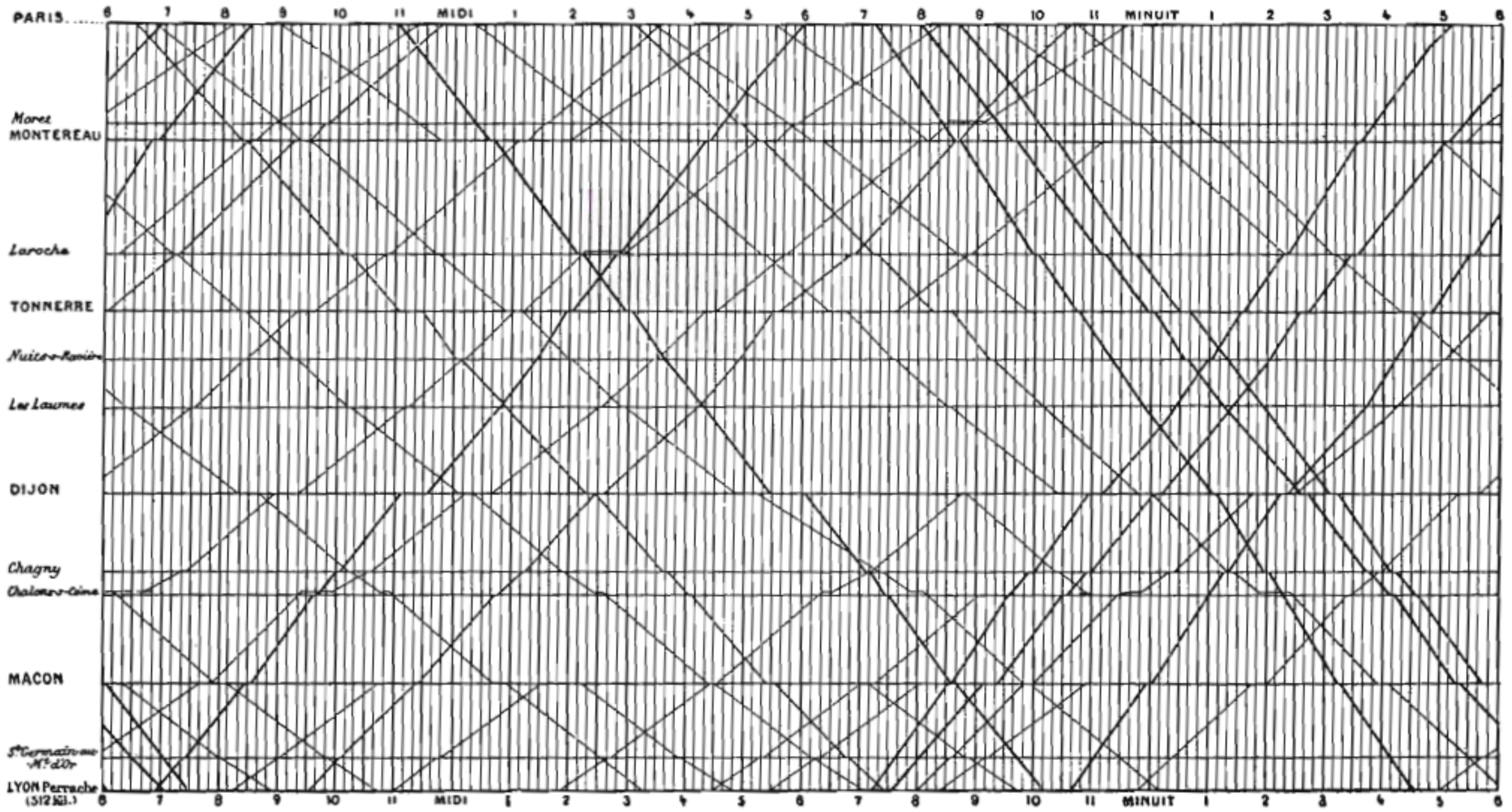

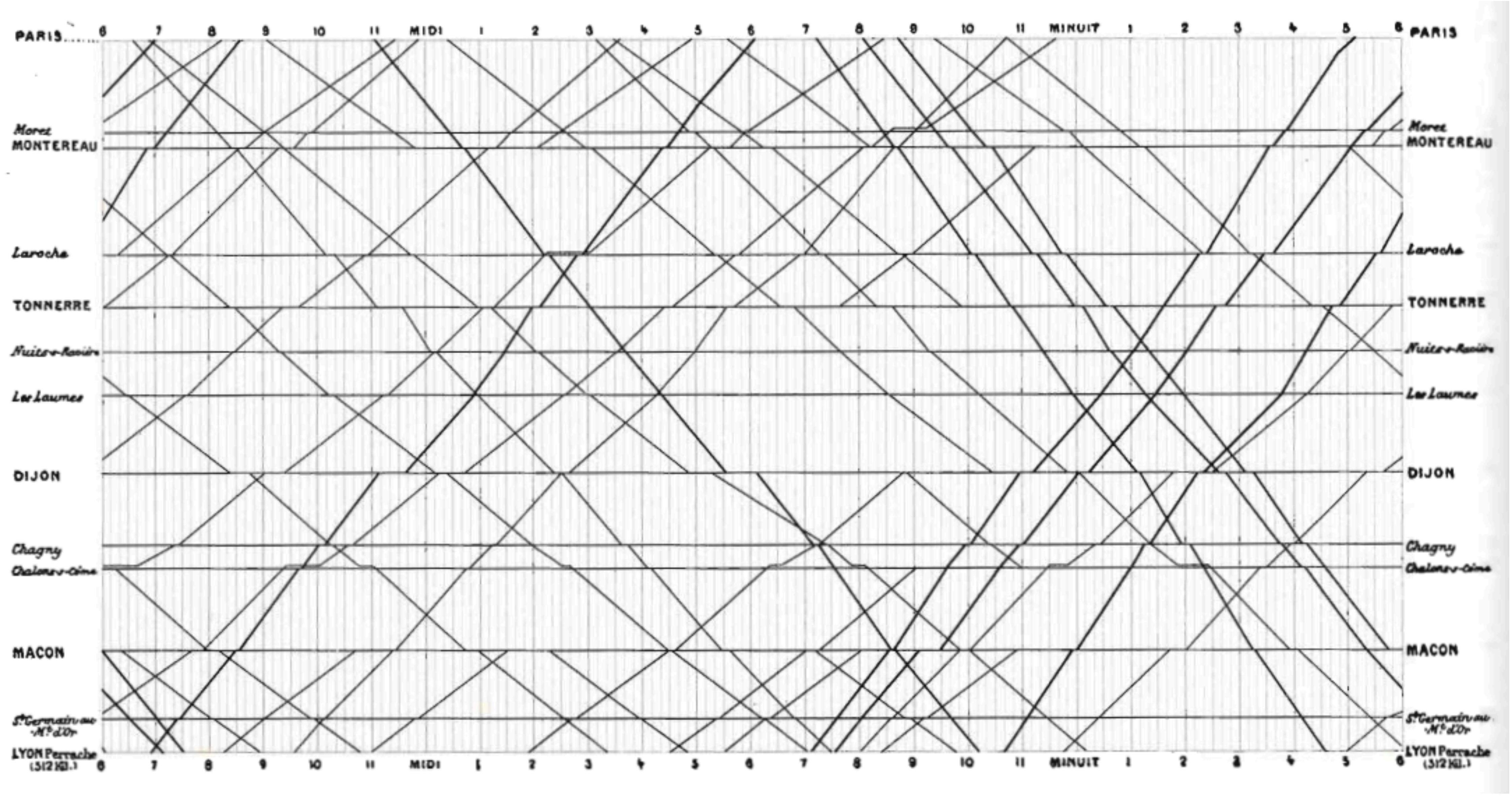

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Tabular Data

Country	Area	Density	Birthrate	Population	Mortality	GDP
Russia	17075200	8.37	99.6	142893540	15.39	8900.0
Mexico	1972550	54.47	92.2	107449525	20.91	9000.0
Japan	377835	337.35	99.0	127463611	3.26	28200.0
United Kingdom	244820	247.57	99.0	60609153	5.16	27700.0
New Zealand	268680	15.17	99.0	4076140	5.85	21600.0
Afghanistan	647500	47.96	36.0	31056997	163.07	700.0
Israel	20770	305.83	95.4	6352117	7.03	19800.0
United States	9631420	30.99	97.0	298444215	6.5	37800.0
China	9596960	136.92	90.9	1313973713	24.18	5000.0
Tajikistan	143100	51.16	99.4	7320815	110.76	1000.0
Burma	678500	69.83	85.3	47382633	67.24	1800.0
Tanzania	945087	39.62	78.2	37445392	98.54	600.0
Tonga	748	153.33	98.5	114689	12.62	2200.0
Germany	357021	230.86	99.0	82422299	4.16	27600.0
Australia	7686850	2.64	100.0	20264082	4.69	29000.0

What improvements can be made?

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Tabular Data

Country	Area	Density	Birthrate	Population	Mortality	GDP
Russia	17075200	8.37	99.6	142893540	15.39	8900.0
Mexico	1972550	54.47	92.2	107449525	20.91	9000.0
Japan	377835	337.35	99.0	127463611	3.26	28200.0
United Kingdom	244820	247.57	99.0	60609153	5.16	27700.0
New Zealand	268680	15.17	99.0	4076140	5.85	21600.0
Afghanistan	647500	47.96	36.0	31056997	163.07	700.0
Israel	20770	305.83	95.4	6352117	7.03	19800.0
United States	9631420	30.99	97.0	298444215	6.5	37800.0
China	9596960	136.92	90.9	1313973713	24.18	5000.0
Tajikistan	143100	51.16	99.4	7320815	110.76	1000.0
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Australia	7686850	2.64	100.0	20264082	4.69	29000.0

Country	Population	Area	Density	Mortality	GDP	Birth Rate
Afghanistan	31,056,997	647,500	47.96	**163.07**	700	36.0
Australia	20,264,082	7,686,850	2.64	4.69	29,000	**100.0**
Burma	47,382,633	678,500	69.83	67.24	1,800	85.3
China	**1,313,973,713**	9,596,960	136.92	24.18	5,000	90.9
Germany	82,422,299	357,021	230.86	4.16	27,600	99.0
Israel	6,352,117	20,770	305.83	7.03	19,800	95.4
Japan	127,463,611	377,835	**337.35**	3.26	28,200	99.0
Mexico	107,449,525	1,972,550	54.47	20.91	9,000	92.2
New Zealand	4,076,140	268,680	15.17	5.85	21,600	99.0
Russia	142,893,540	**17,075,200**	8.37	15.39	8,900	99.6
Tajikistan	7,320,815	143,100	51.16	110.76	1,000	99.4
Tanzania	37,445,392	945,087	39.62	98.54	600	78.2
Tonga	114,689	748	153.33	12.62	2,200	98.5
United Kingdom	60,609,153	244,820	247.57	5.16	27,700	99.0
United States	298,444,215	9,631,420	30.99	6.50	**37,800**	97.0

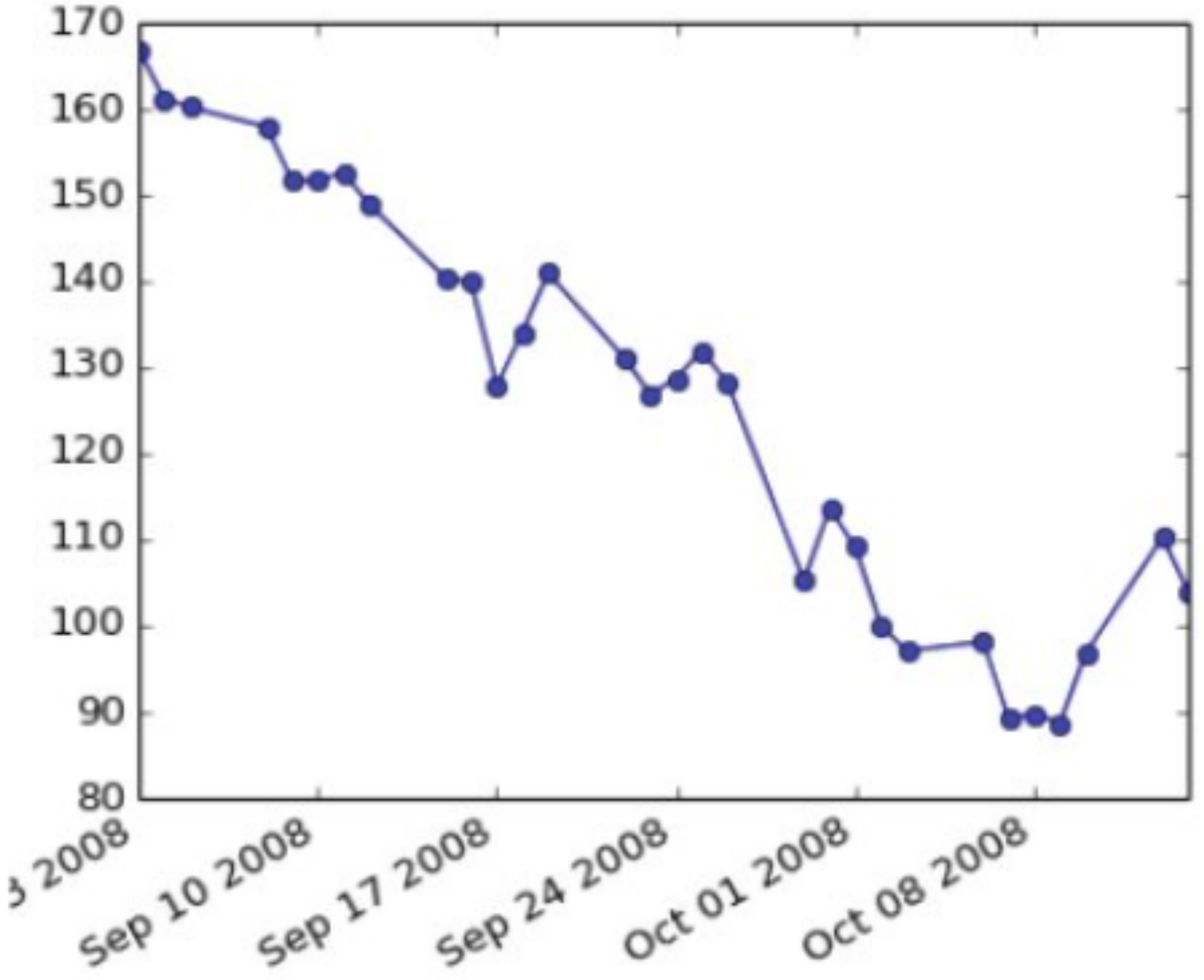
What improvements can be made?

- Order rows/columns
- Remove uninformative digits, right justify numbers, add commas
- Highlight biggest values

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Line Charts: Series of data points connected with lines.

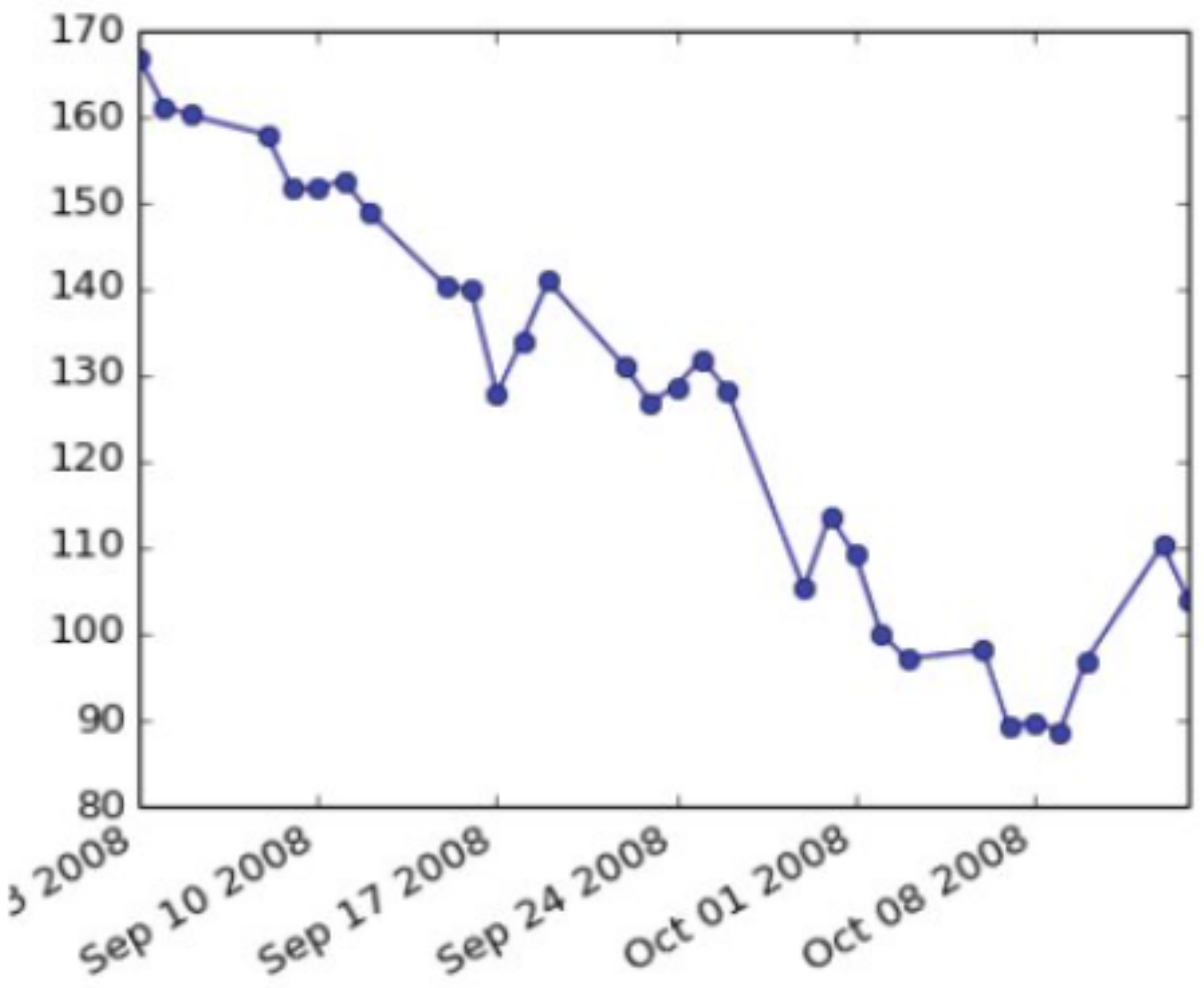


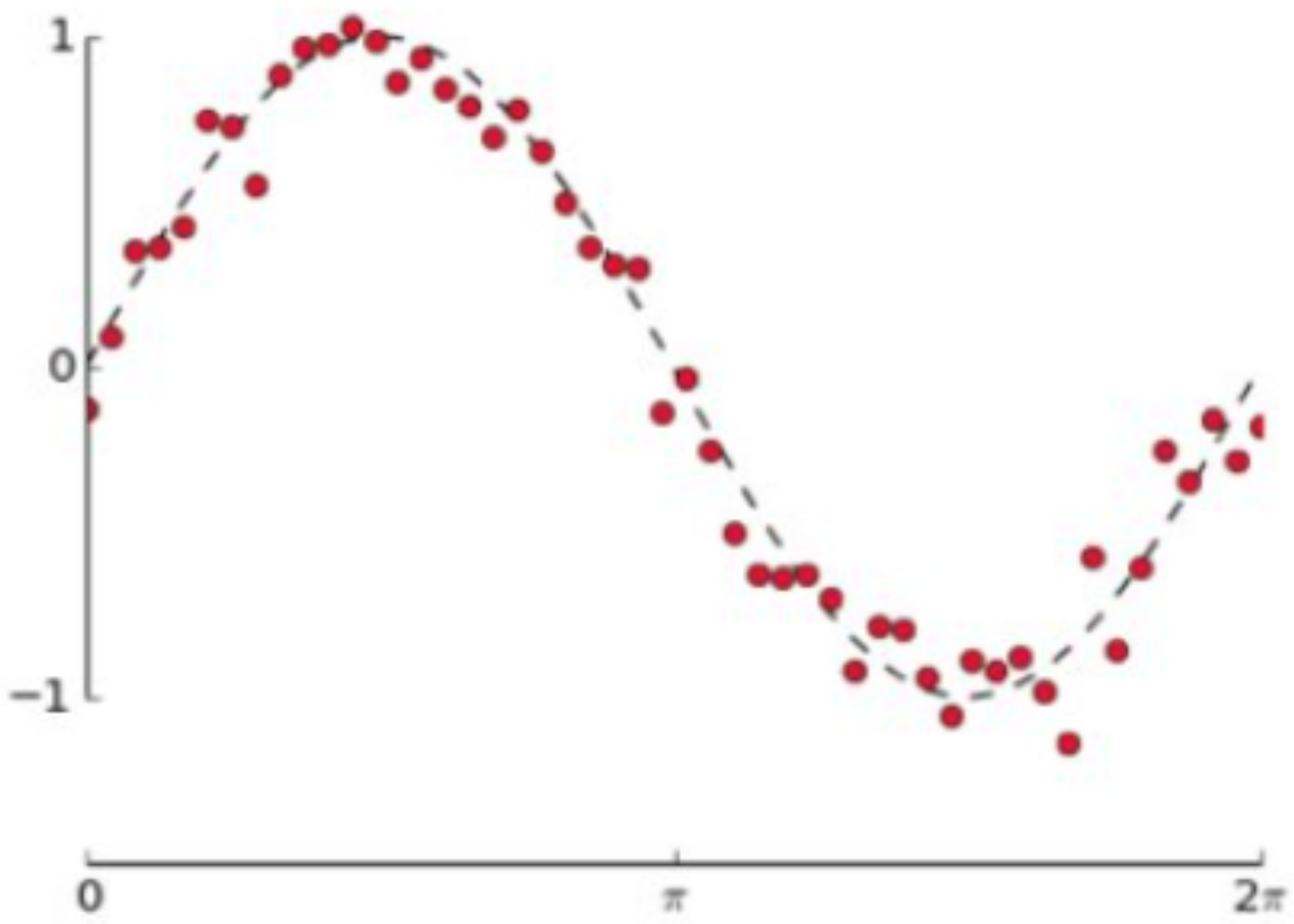
May make sense with time series data.

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Line Charts: Series of data points connected with lines.





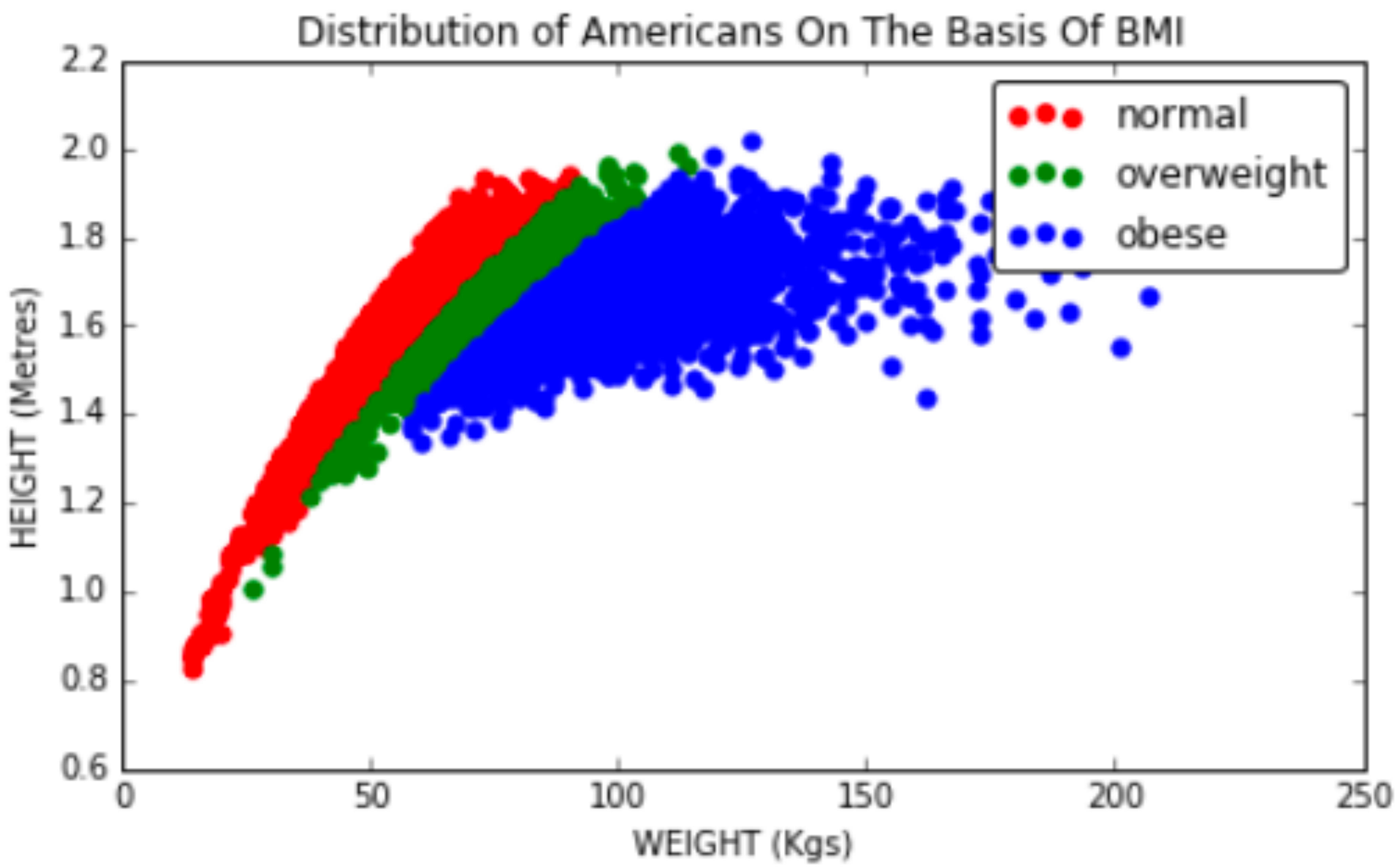
May make sense with time series data.

If no meaningful adjacency between data points, don’t us line chart, instead overlay fitted curve.

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Dot size

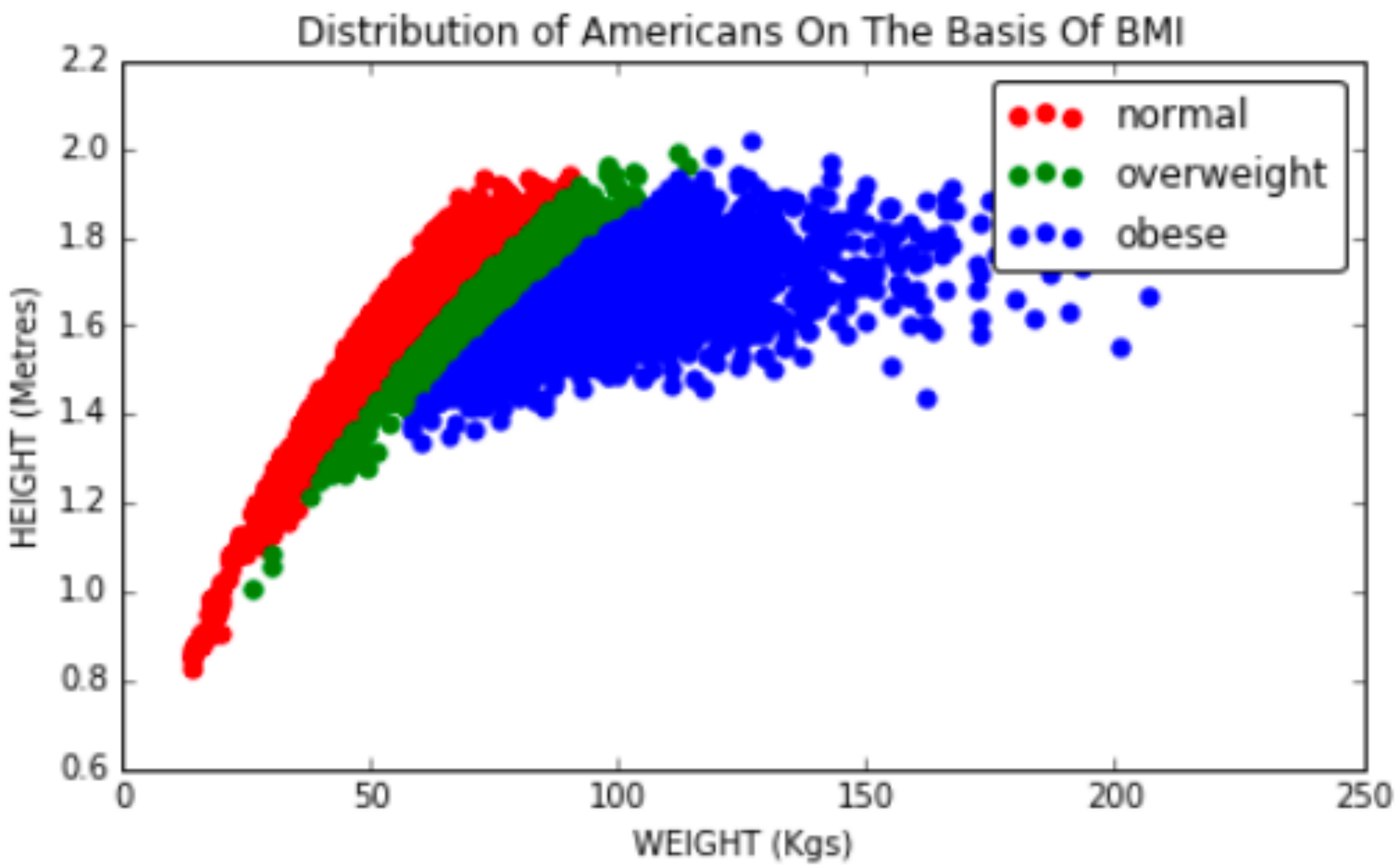


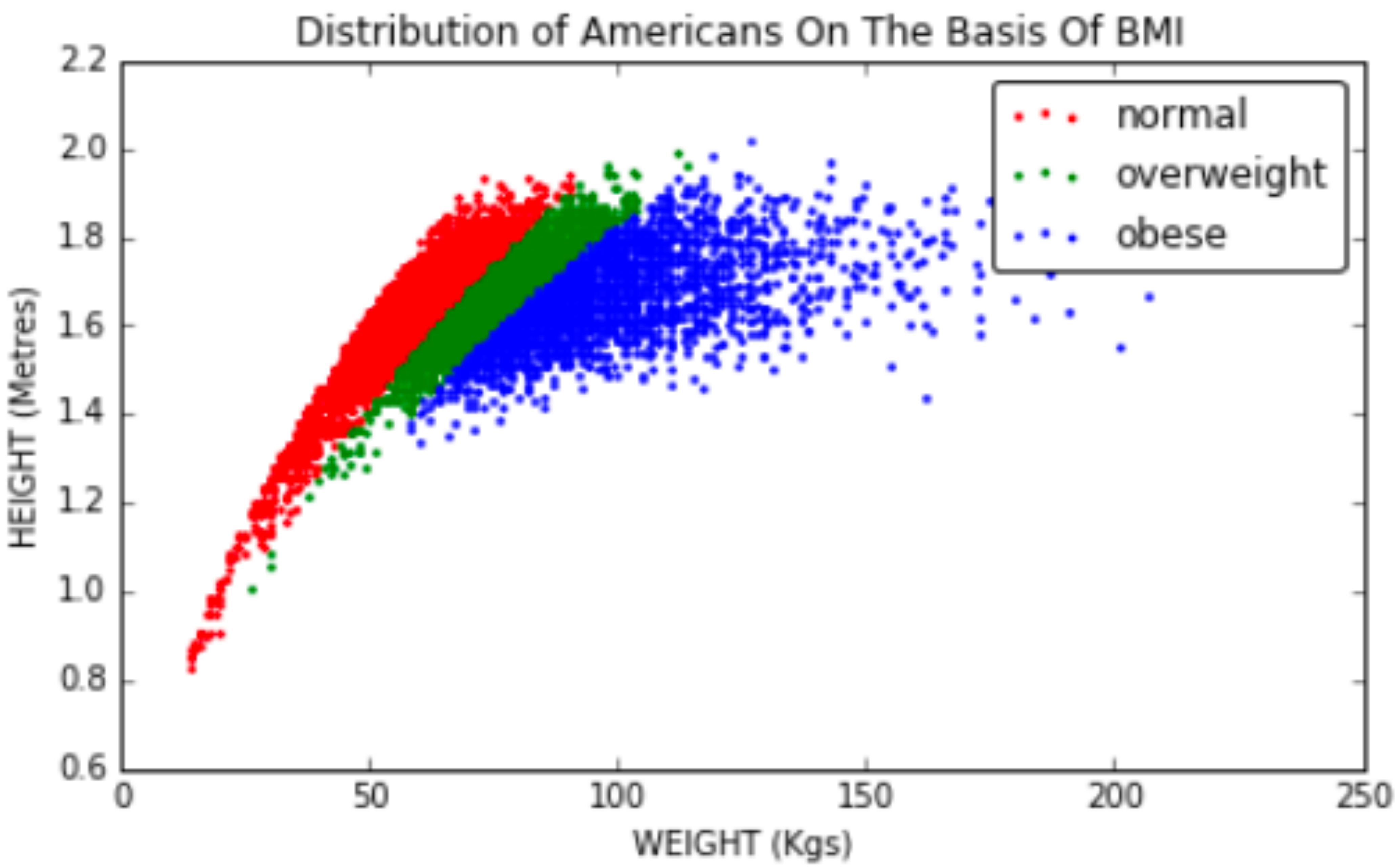
Large dot size: more overlaps and obscuring

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Dot size





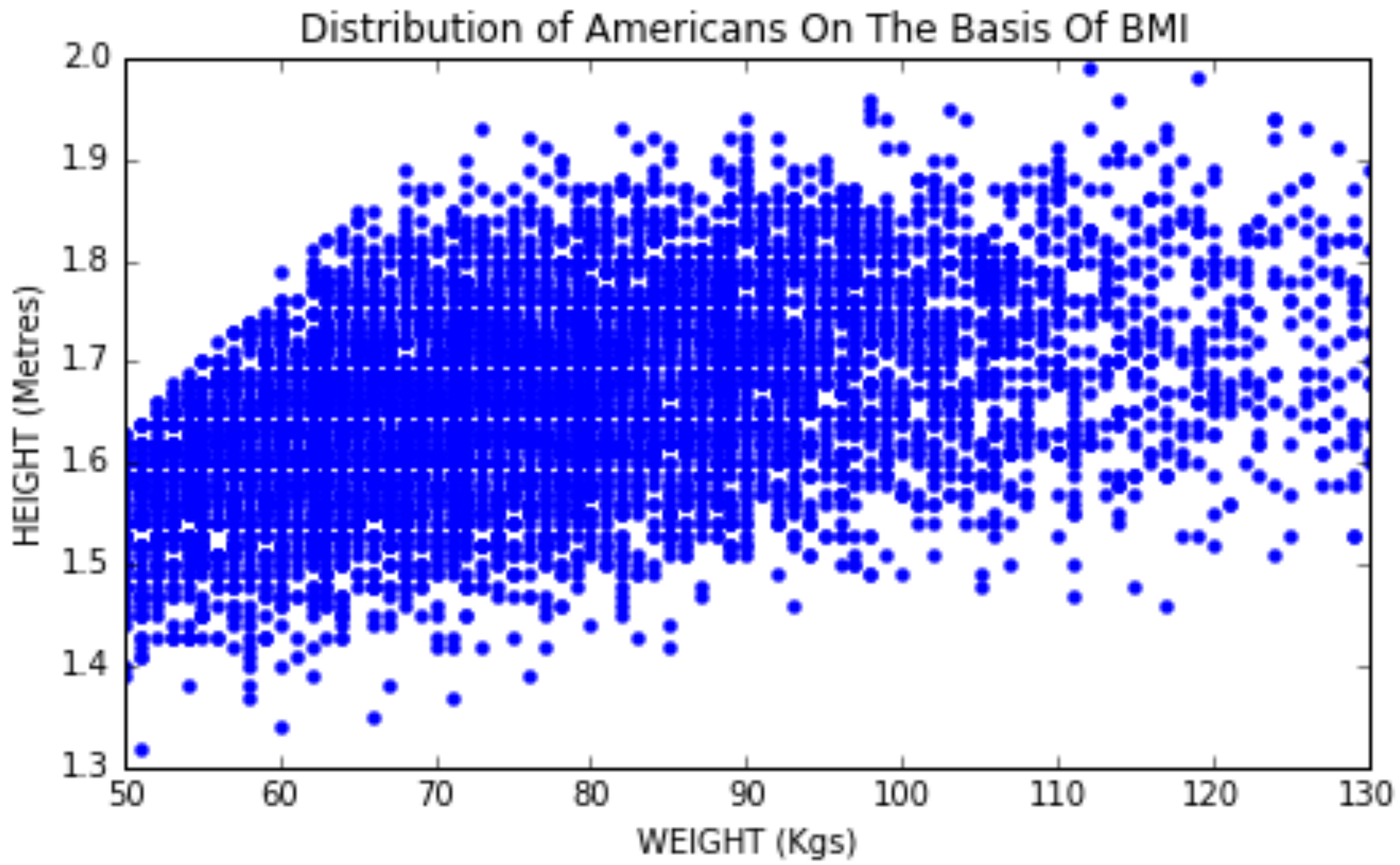
Large dot size: more overlaps and obscuring

Smaller dots for this size is better

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Heat map for better frequency visualization

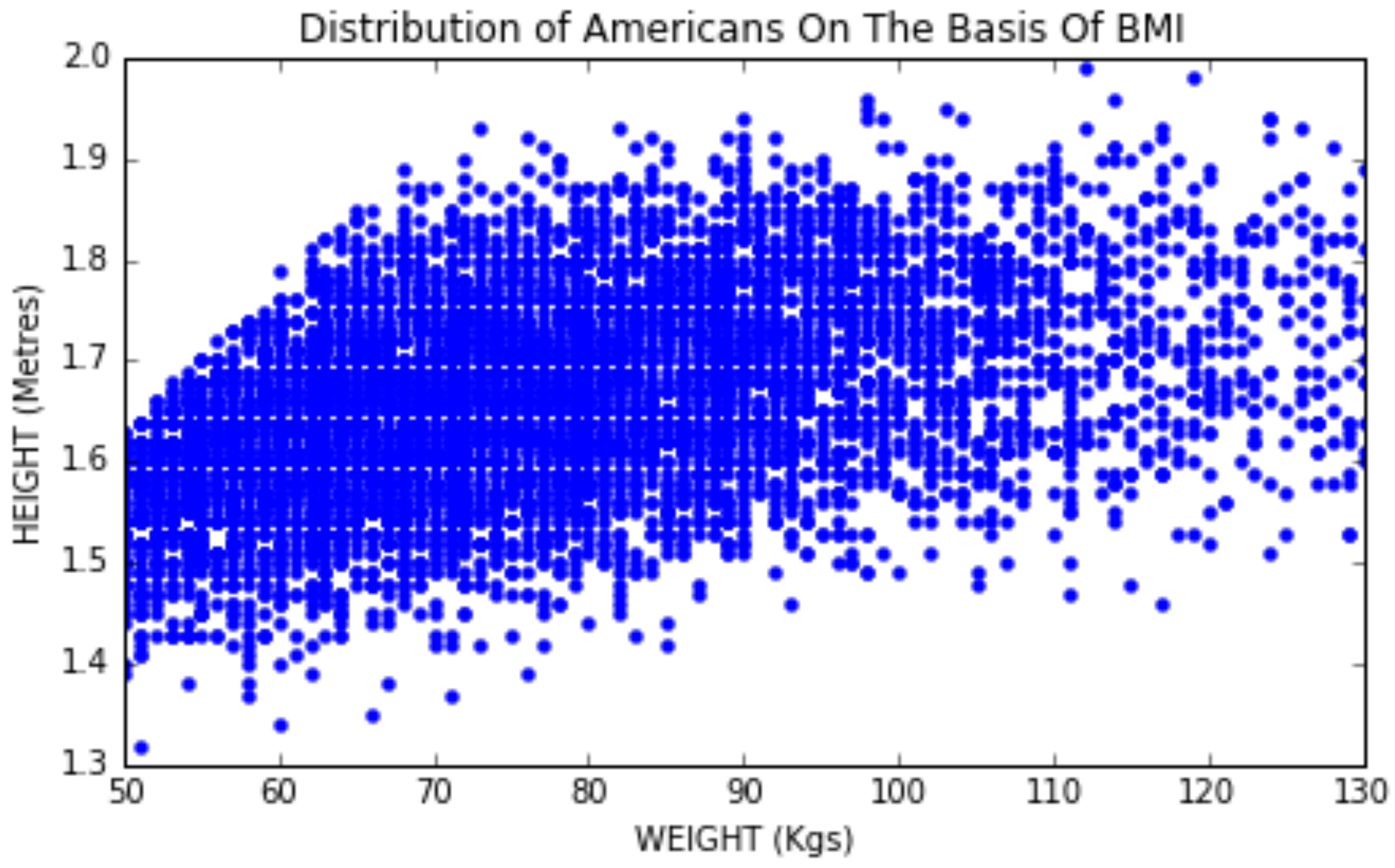


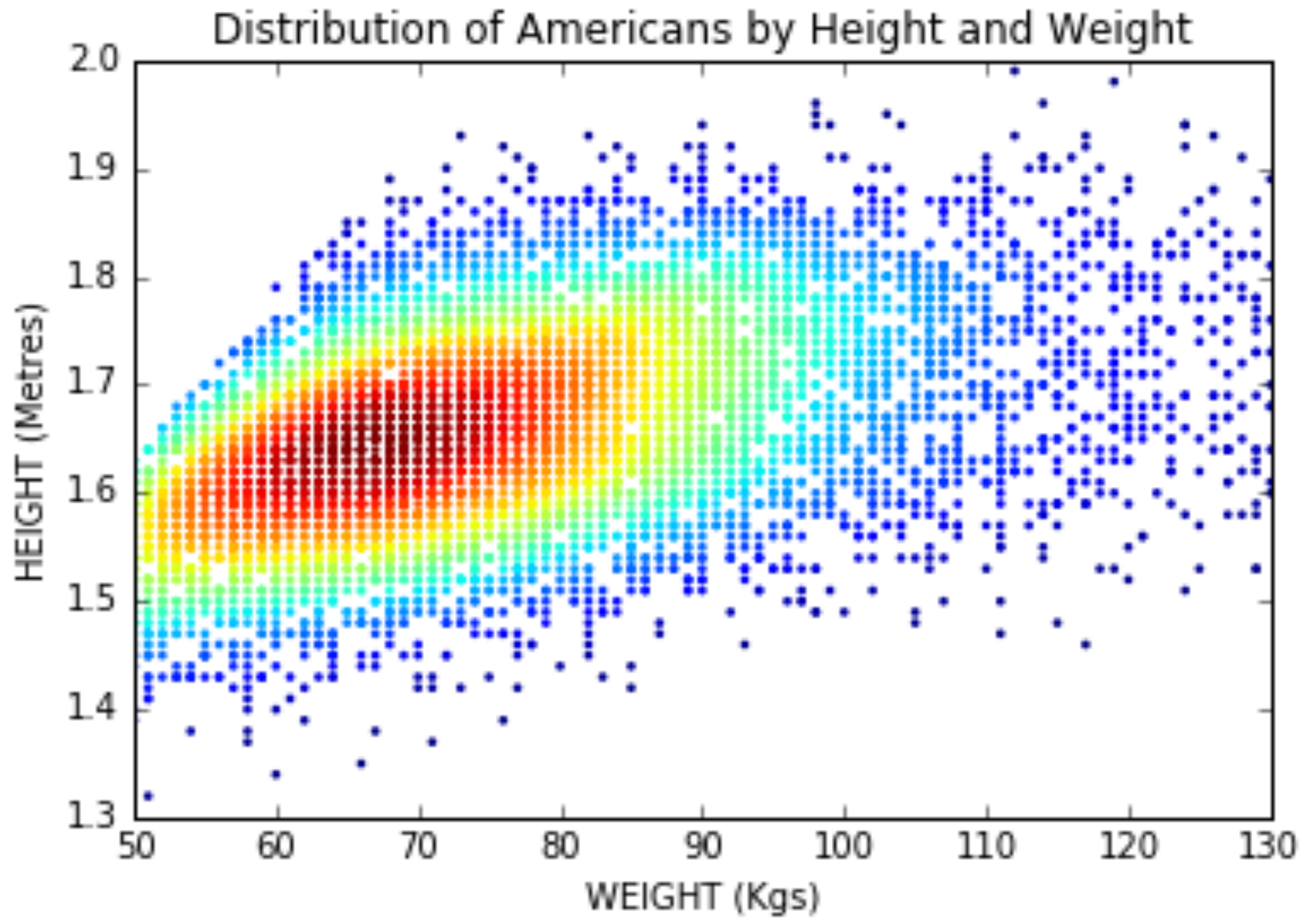
How to use color to
our advantage?

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Heat map for better frequency visualization





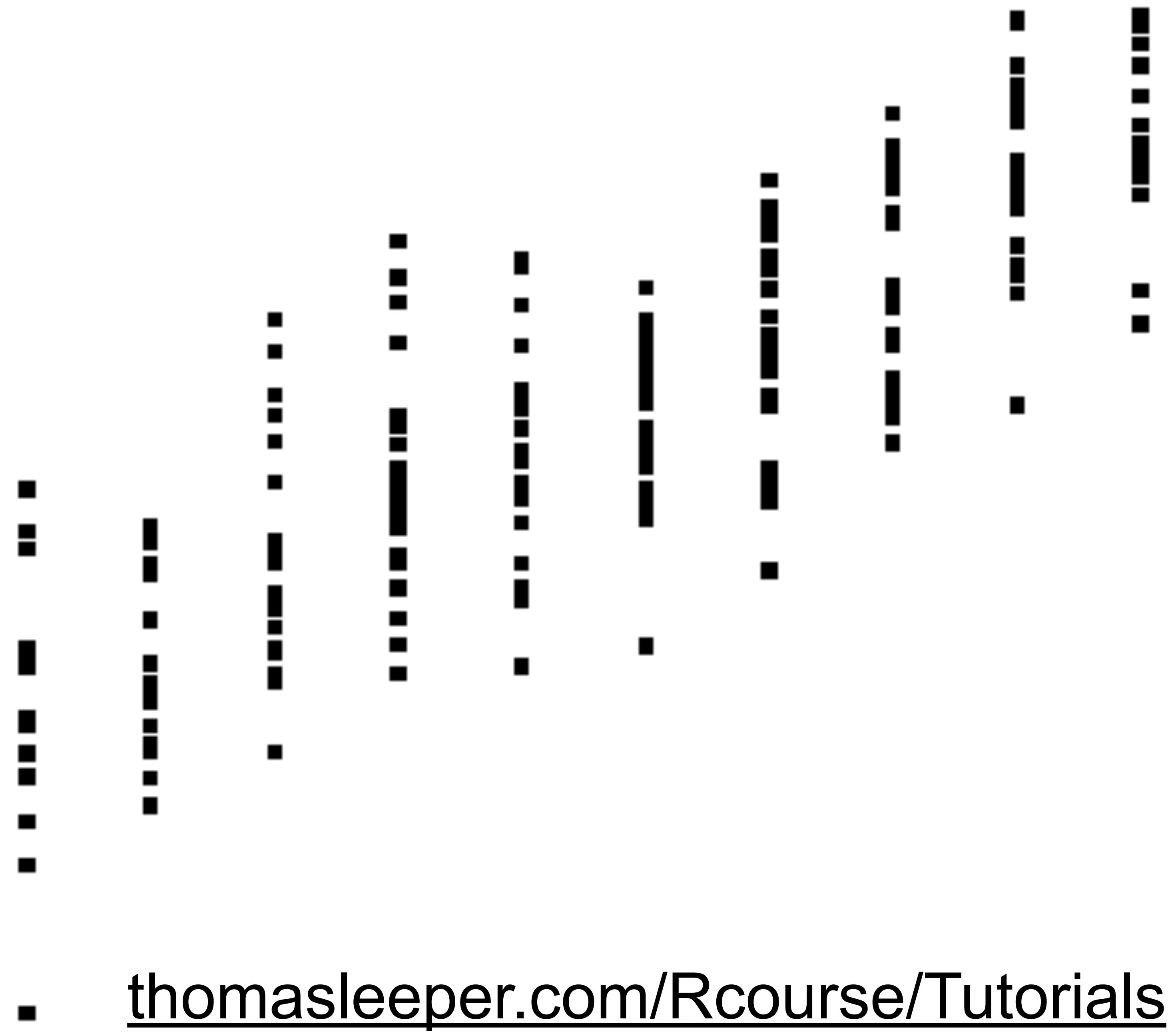
How to use color to our advantage?

Colored heat map reveals the distribution better.

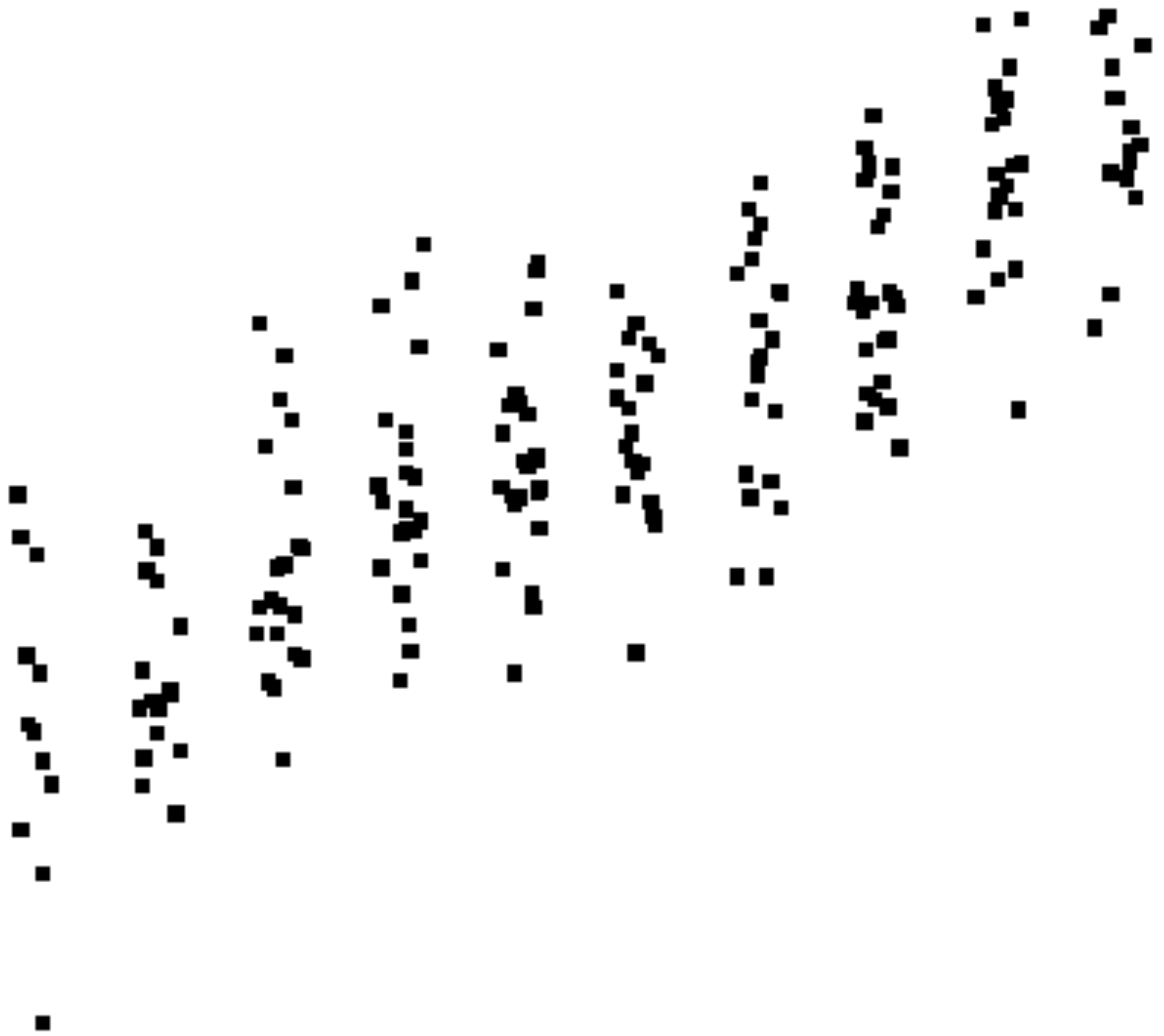
PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Jittering: add random noise ϵ to x and/or y coordinate.



thomasleeper.com/Rcourse/Tutorials



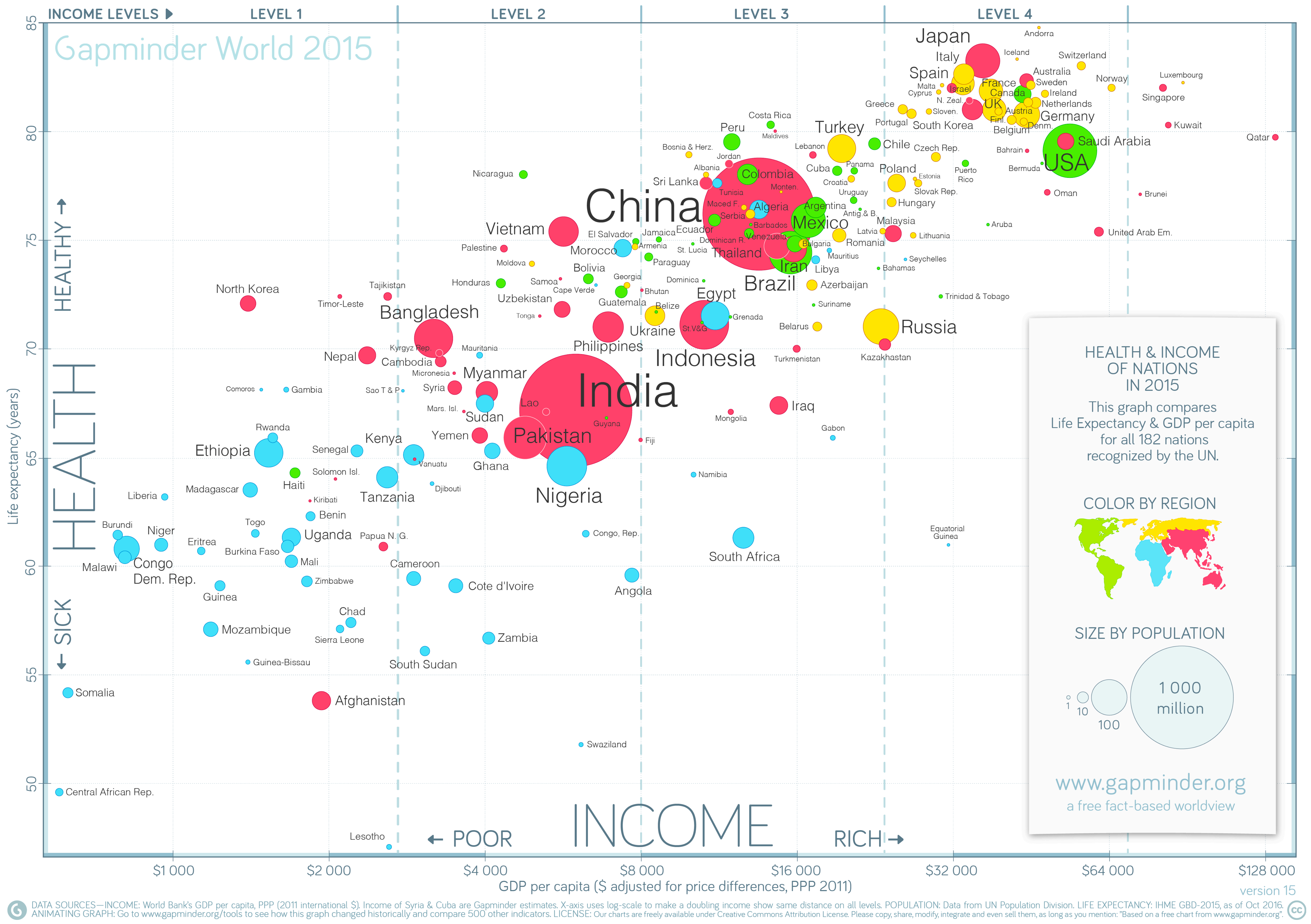
Difficult to see the distribution.

Much easier to see the distribution after jittering of x-coordinates.

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Visualize >2 variables: extra dimensions (size and color)

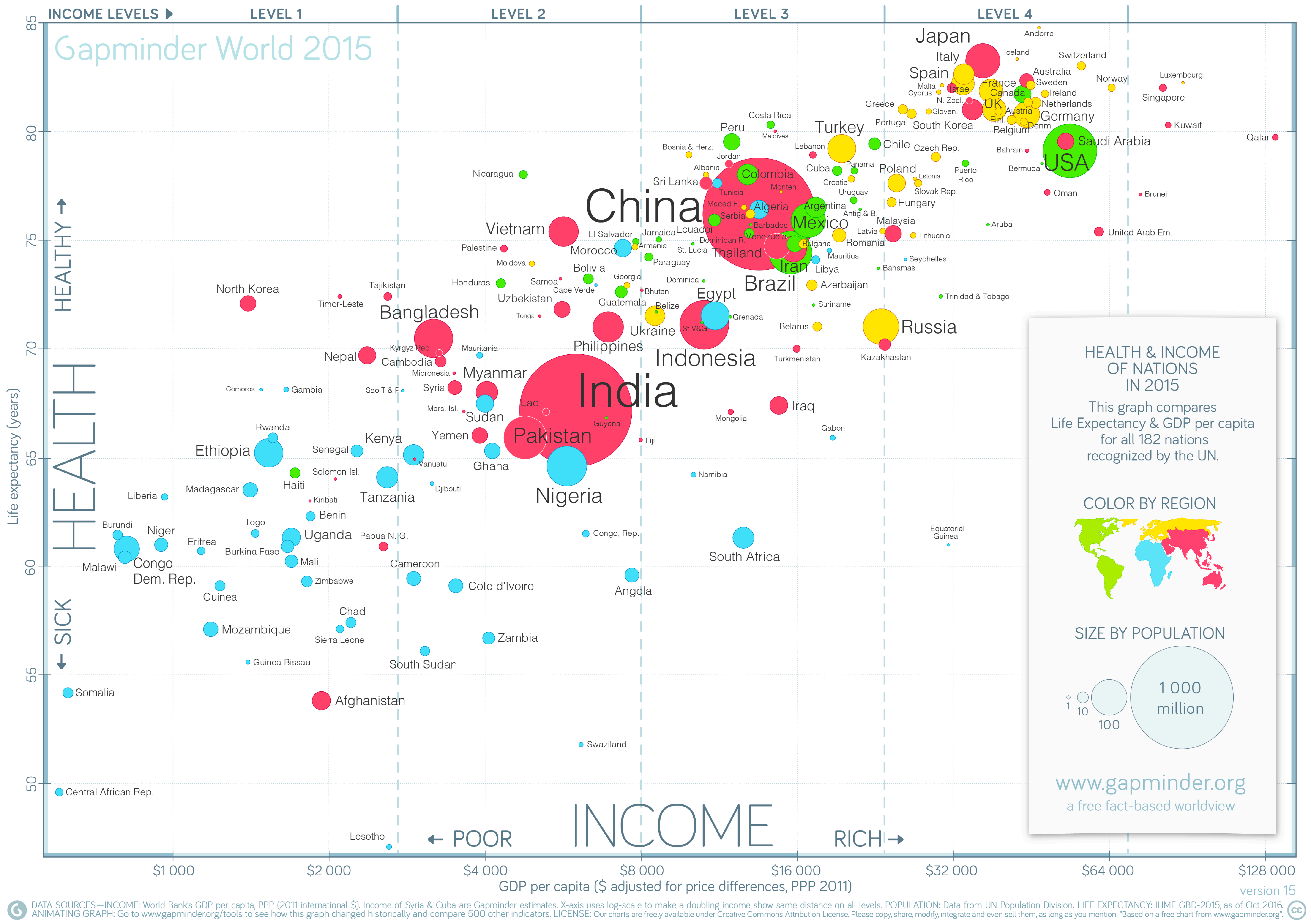


Gapminder World

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Scatter plots: Visualize >2 variables: extra dimensions (size and color)



Size: Population

Color: Region

y-coordinate: Life expectancy

x-coordinate: GDP per person (Log transformed)

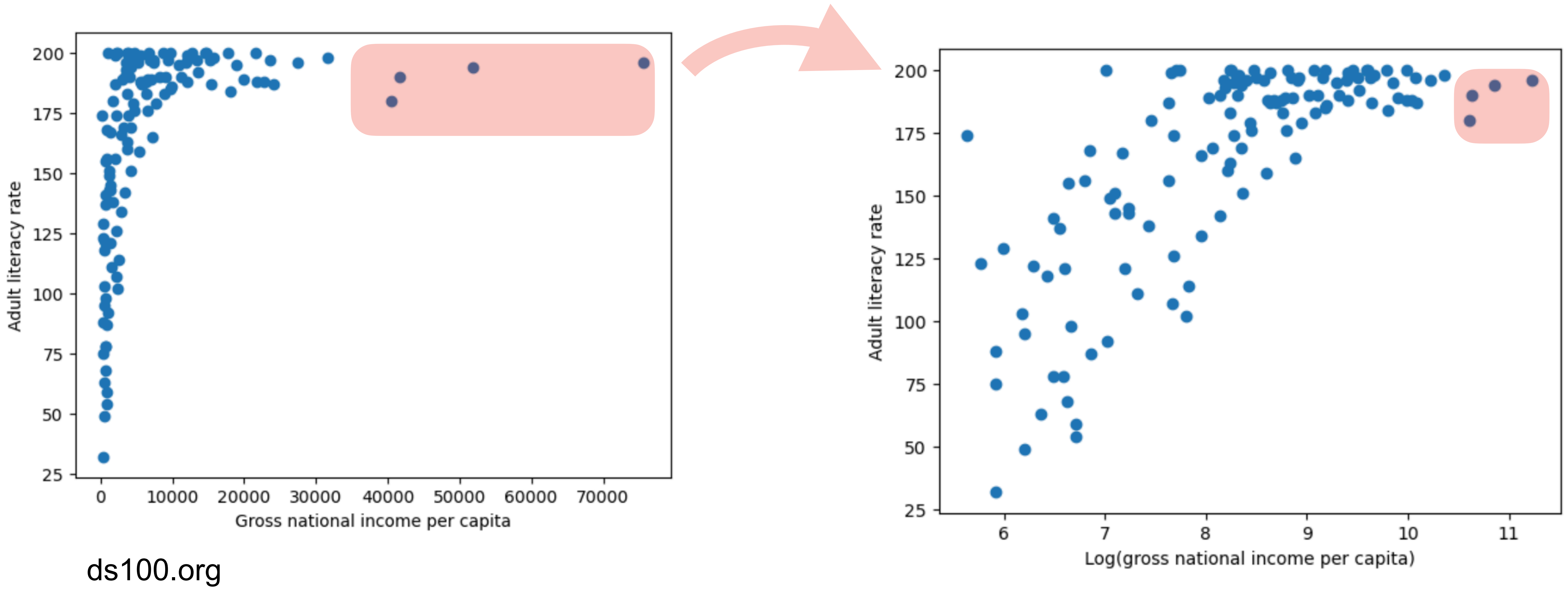
Why do we have transformations?

Gapminder World

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Transformations: better visualize/understand relationship bw variables



ds100.org

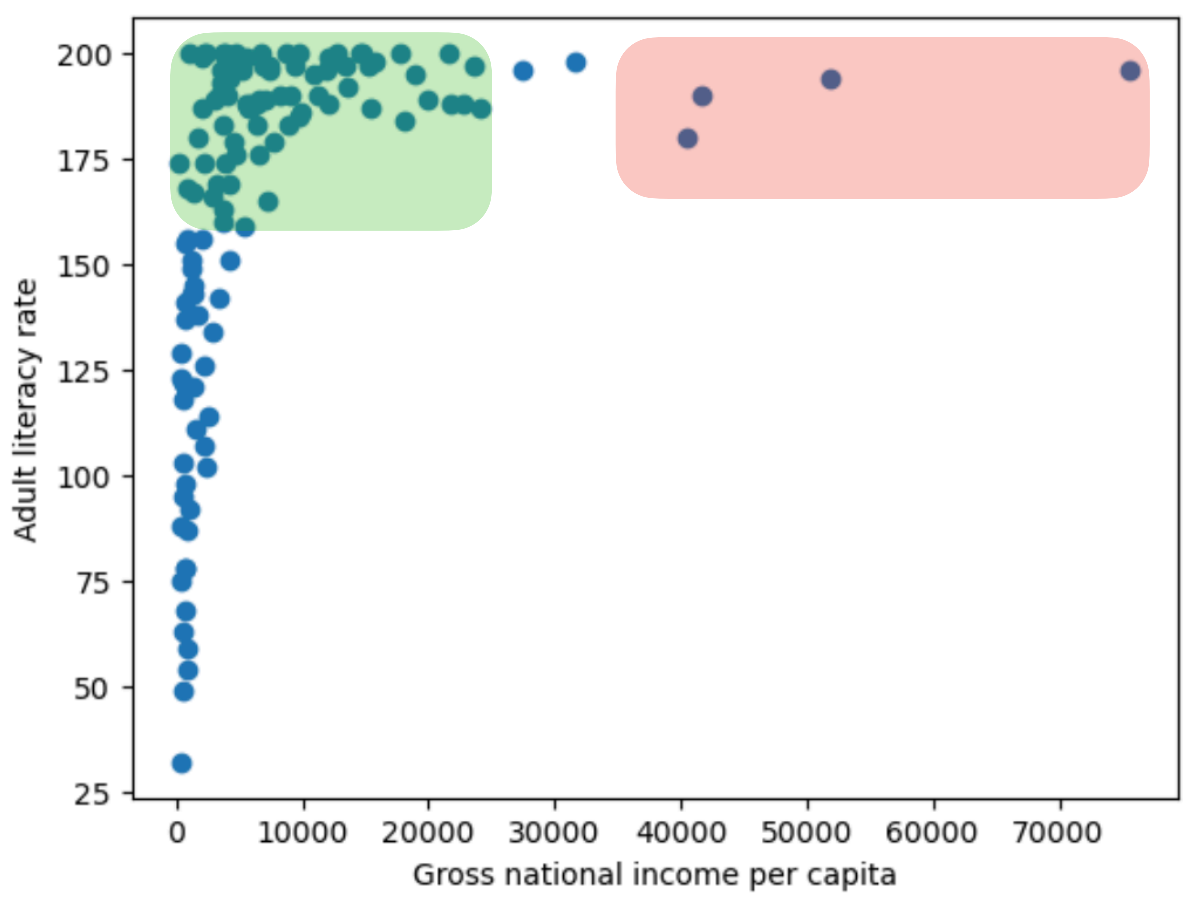
Few outliers at large x:

Log transform x-coordinates.

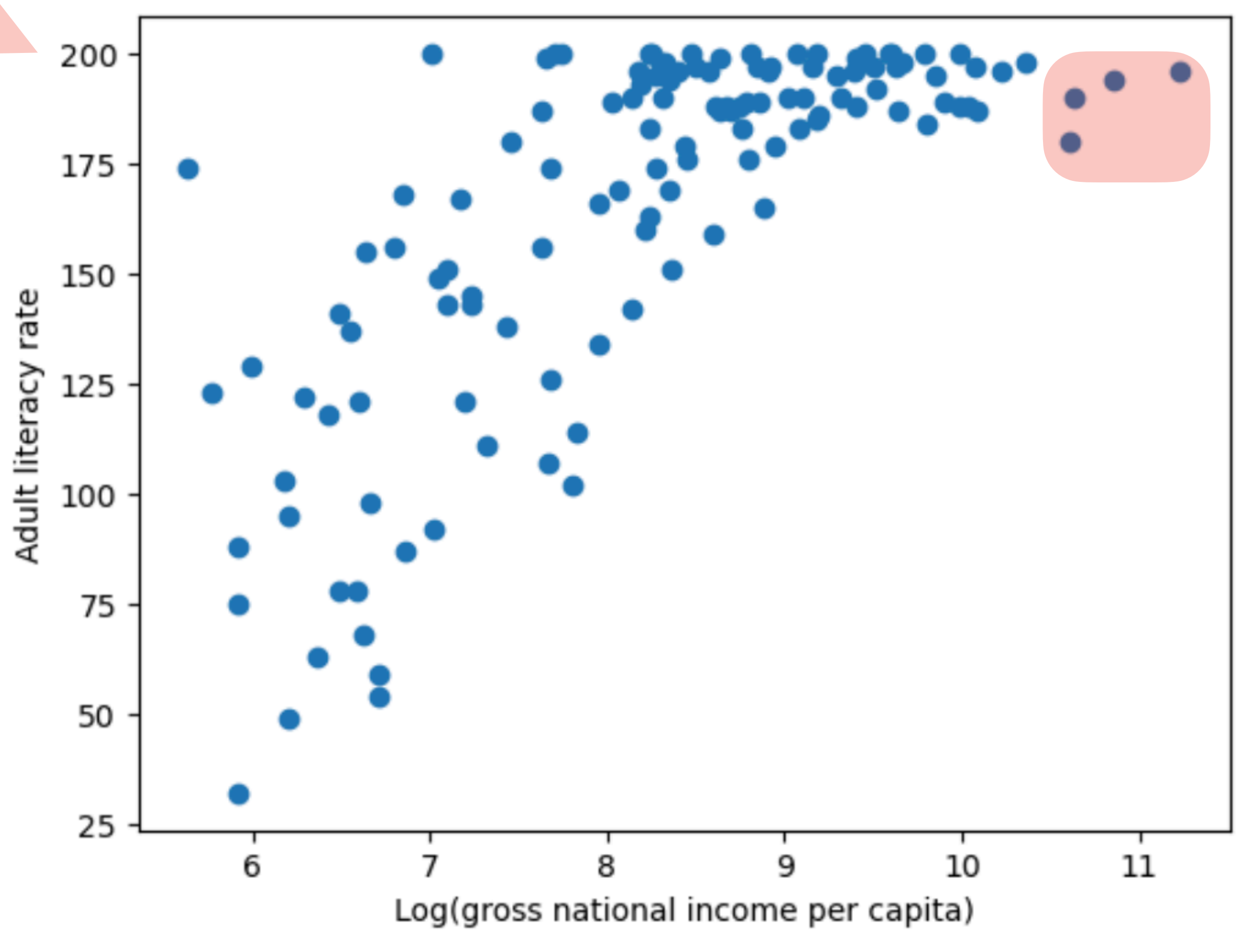
PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Transformations: better visualize/understand relationship bw variables



ds100.org



Few outliers at large x:

Log transform x-coordinates.

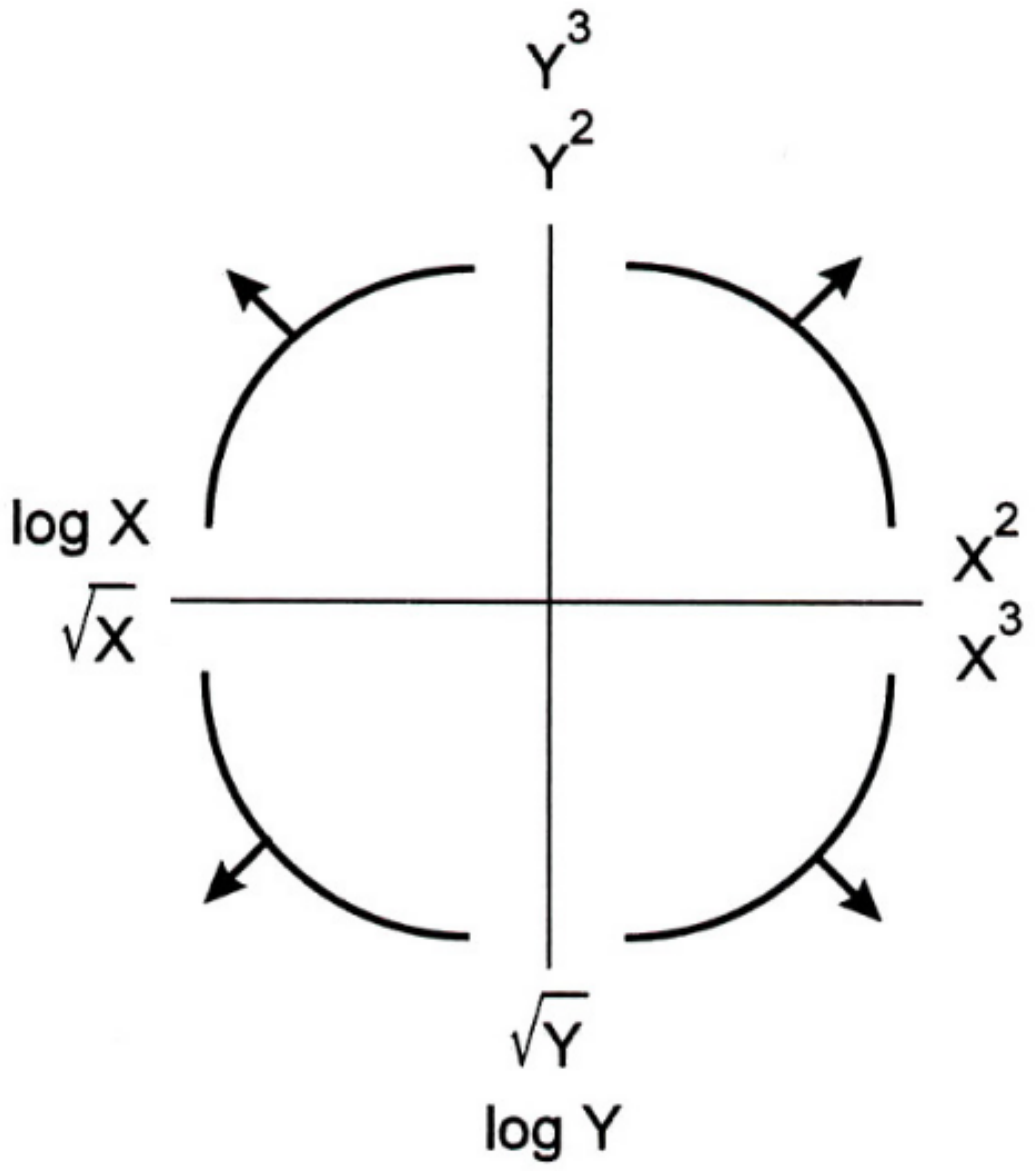
Heavy density at large y:

Power transform y-coordinates.

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Transformations: better visualize/understand relationship bw variables



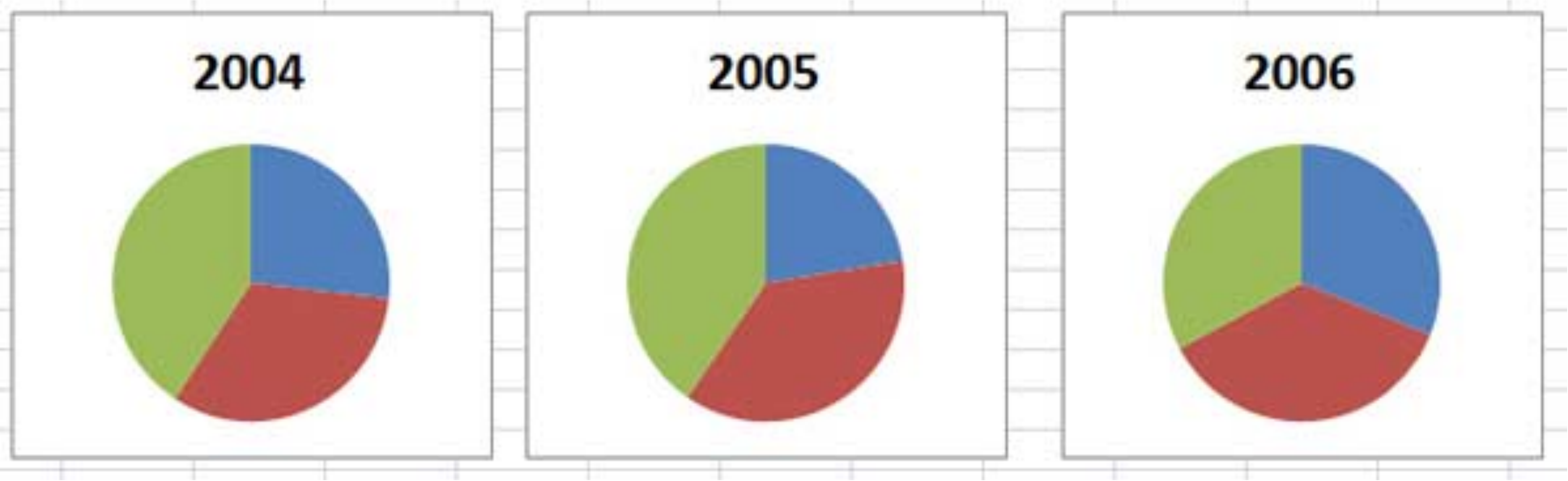
Use:

Tukey-Mosteller Bulge Diagram as a guide for finding suitable transformation.

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Pie charts vs bar plots:



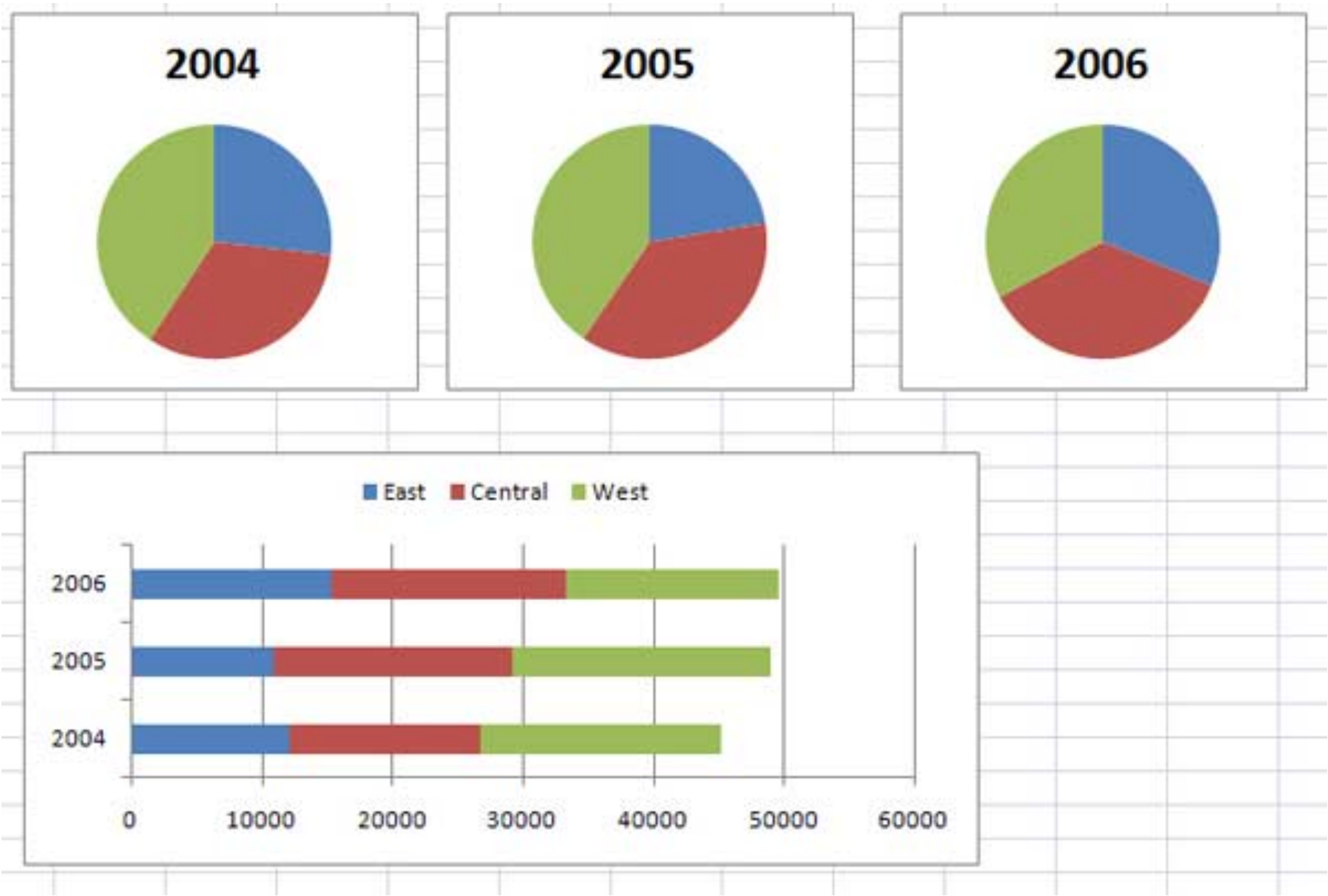
Hard to read & see differences

Maybe good for percentages

PRINCIPLES OF DATA VISUALIZATION

Things to Remember

Pie charts vs bar plots:



Hard to read & see differences

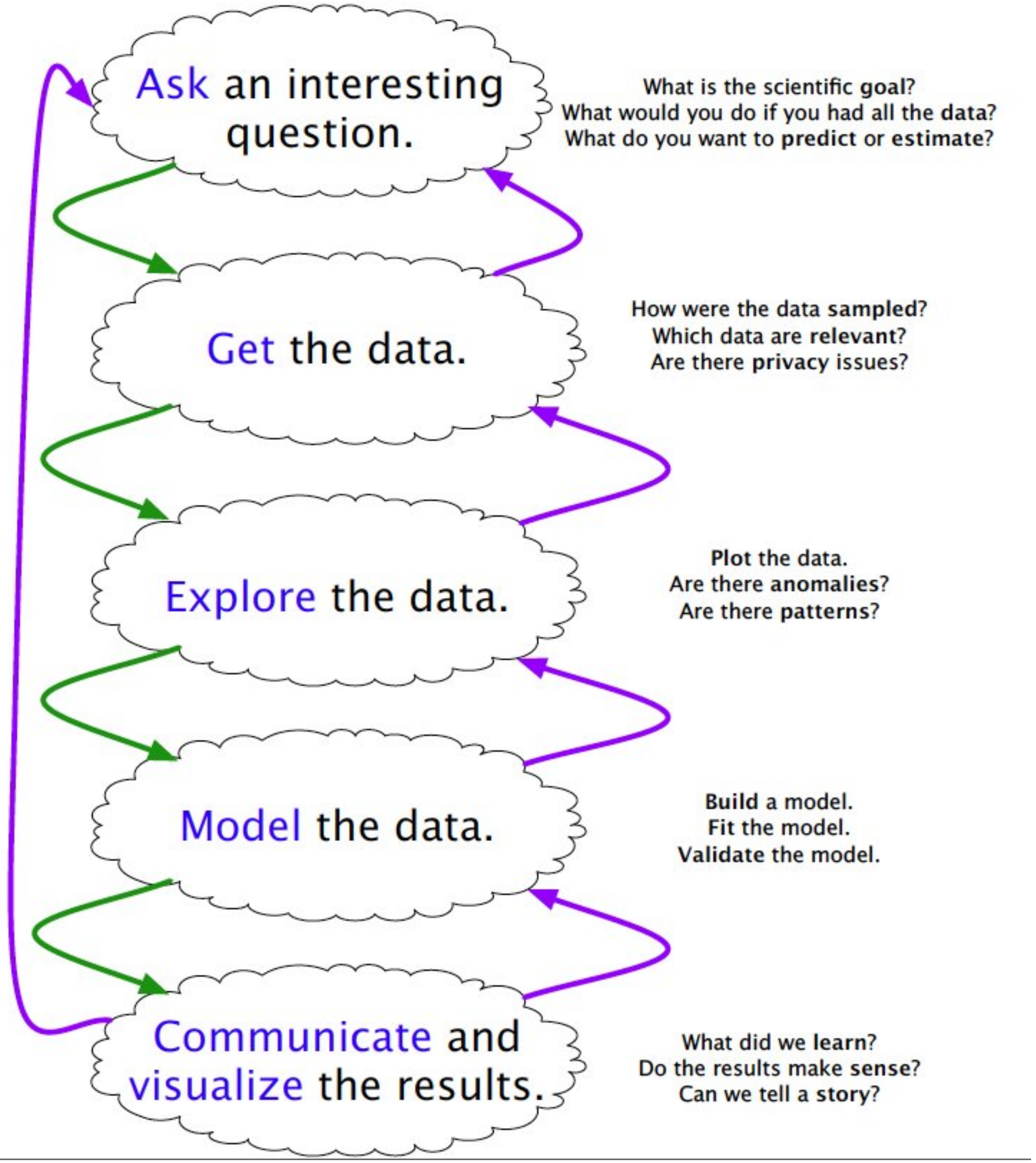
Maybe good for percentages

Lengths are easier to distinguish than angles.

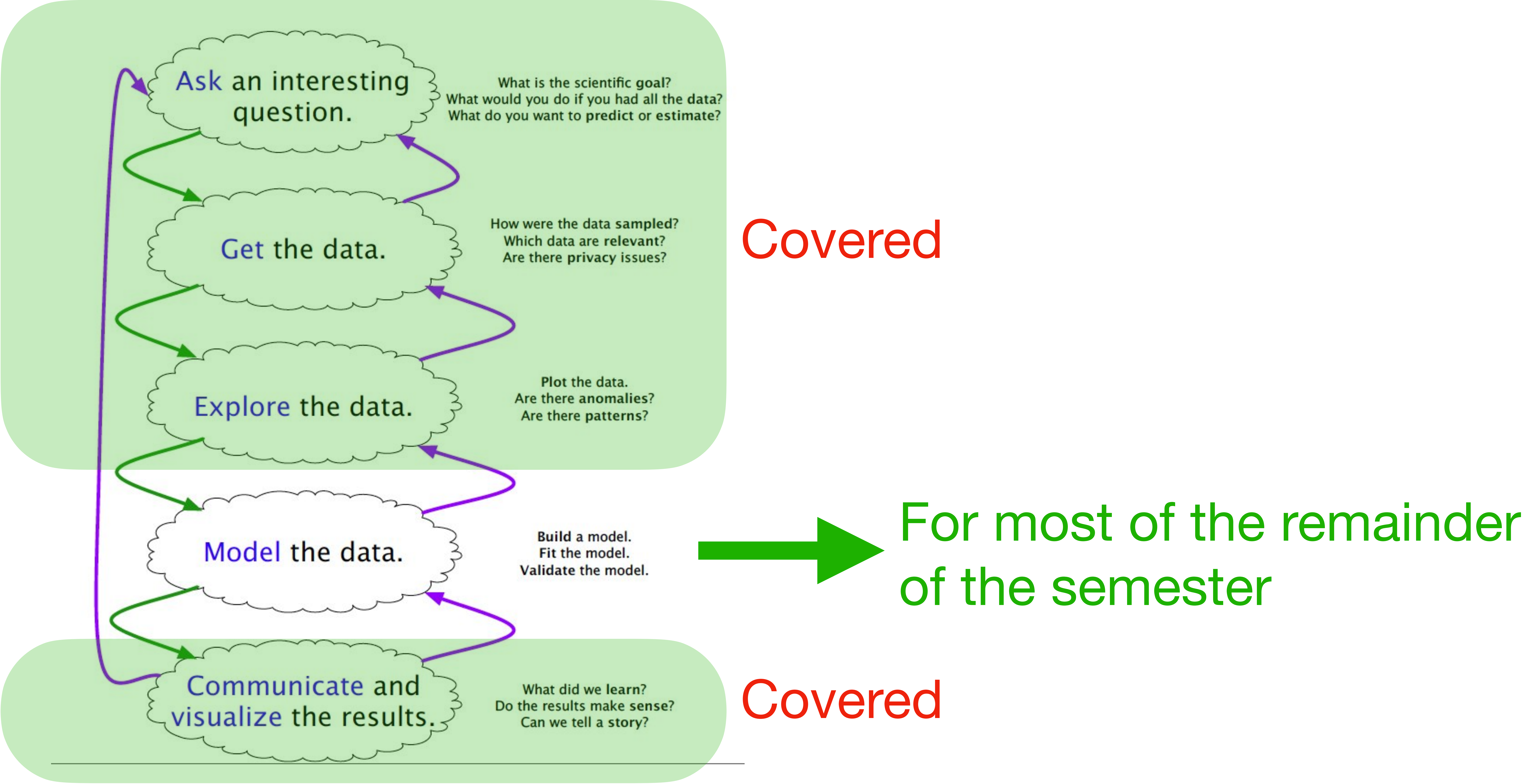
Bar plots better for comparisons, detecting differences.

MODELING

ROLE OF MODELING IN DATA SCIENCE



ROLE OF MODELING IN DATA SCIENCE



WHAT IS A MODEL?

Modeling: Process of encapsulating information into a tool which can make forecasts/predictions.

A model is an idealized representation of a system.

"All models are wrong, but some models are useful."
– George Box (1919-2013)

First-Principle vs. Data-Driven Models

Newton's Law of Gravitation vs predicting flight cancellations

PRINCIPLES OF MODELING

Occam's Razor: The simplest explanation is the best explanation

Minimize the number of model parameters.

Inherent trade-off between accuracy and simplicity

Bias-Variance Tradeoff:

Bias: Error from incorrect assumptions built into the model

E.g. interpolating function linear rather than a higher-order curve.

(Underfitting)

Variance: Error from sensitivity to small fluctuations in the training set

(Overfitting)

First-principle models may have bias, data-driven models may overfit.

MODELING PIPELINE

1. Choose a model	How should we represent the world?
2. Choose a loss function	How do we quantify prediction error?
3. Fit the model	How do we choose the best parameters of our model given our data?
4. Evaluate model performance	How do we evaluate whether this process gave rise to a good model?

CLASSIFICATION VS PREDICTION

$$\begin{array}{ccccc} & & \text{Features} & & \text{Output} \\ \hline x_{11} & x_{12} & \cdots & x_{1p} & y_1 \\ x_{21} & x_{22} & \cdots & x_{2p} & y_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} & y_n \end{array}$$

Classification:
Output is a categorical variable

Prediction:
Output is a numerical variable

CLASSIFICATION VS PREDICTION

Features				Output
x_{11}	x_{12}	$\cdots$	x_{1p}	y_1
x_{21}	x_{22}	$\cdots$	x_{2p}	y_2
$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
x_{n1}	x_{n2}	$\cdots$	x_{np}	y_n

Classification:
Output is a categorical variable

Prediction:
Output is a numerical variable

Other names:

Features	Output
Independent variables	Outcome
Explanatory variables	Response
Predictors	Dependent variable
Inputs	…
…	

PROJECT IS OUT: DUE DECEMBER 10

Suggestions:

Read the project guidelines carefully.

Find project partners soon (next few days).

Start with the “data” portion now (spend 2-3 weeks):

- Spend a day to see if other relevant datasets in CTDH (Data Hub) or somewhere else
- Get datasets, turn them into data frames, do EDA/visualizations
- Decide on interesting questions and tasks (prediction/classification)
- You may need to do a lot of data cleaning/preprocessing:
    - Merging datasets (unify location etc.), normalizations etc.
    - Retain only data you need, handle missing values etc.
    - If classification: form new relevant columns (e.g. crime index: 1-5)

Start with the “modeling” portion in 10-15 days (spend 2-3 weeks):

- Choose your algorithms (at least 2), Partition data (train/test split, CV etc.)
- Do the evaluations. Get resulting plots.

Start writing the report one week before the deadline:

- Include your findings from the “data” portion (statistics/correlations/nice visualizations etc.)
- Include your findings from modeling (tables/plots for presenting results, discussions etc.)